Cardiology II

Slides and condensed lecture notes

Julie Lorenzin MScN NP-PHC | AHAD I Module 6 | October 22 2026

Review the original slides with the relevant teaching directly underneath. The notes preserve explanations, clinical reasoning, case discussion and student questions while removing repetition, filler and meeting logistics.

All 178 slides remain in their original order. PDF bookmarks provide navigation by topic and slide. Timestamps refer to approximate locations in the supplied recording, not to the start time of the classroom session.

The recording begins during slide 6. Earlier slides are marked accordingly. Material skipped or left for self-study is labelled Slide reference and is summarized from the supplied deck; it is not presented as spoken teaching.

Clarification labels identify editorial corrections or qualifications separately from the lecturer's teaching. Numbered references for those additions appear at the end. The original slides themselves are reproduced unchanged.

Topic guide

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Cardiology II

Module 6
Advanced Health Assessment and Diagnosis I
Fall 2026

Julie Lorenzin MScN, NP-PHC
Western University
Primary Health Care Nurse Practitioner Program

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Cardiology II with Julie Lorenzin, MScN, NP-PHC. Module 6 of Advanced Health Assessment and Diagnosis I, Fall 2026.

The supplied recording begins partway through the Holter indications discussion on slide 6. No opening remarks are available for this slide.

Seminar Agenda

- Indications for a ECG monitoring
  - 12 Lead ECG
    - Holter and event monitors
- Sinus, atrial and ventricular arrhythmias
  - Sinus rhythms – normal, brady, tachy
- Palpitations
  - PVCs/PACs – benign causes
- Atrial Rhythms
  - Atrial fibrillation
  - Atrial flutter
  - Ventricular tachycardia
- Syncope
- Coronary artery disease
  - Stable angina
  - Acute coronary syndromes
    - STEMI, NSTEMI, UA (unstable angina)

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Use the agenda to organize revision into ECG monitoring and interpretation; sinus rhythms and ectopy; atrial arrhythmias; conduction blocks and syncope; ventricular arrhythmias; and coronary disease with diagnostic testing.

No corresponding spoken introduction is present in the supplied audio.

Indications for the 12 Lead ECG and Cardiac Rhythm Monitoring

- CNO does not outline an exhaustive list of treatments or diagnostic and/or laboratory tests NPs can provide
- Access to a broad range of diagnostic tests or procedures is a crucial component of NP practice to establish a diagnosis and provide access to quality client care, such as ordering of:
  - laboratory tests
  - imaging (X-ray, CT, MRI, Ultrasound)
  - miscellaneous services (pulmonary function test, *cardiac stress test, ECGs* and EEGs)
- 12 lead ECG
  - provides a brief sample of cardiac electrical activity over 10 seconds

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Slide reference: A standard 12-lead ECG samples about 10 seconds of electrical activity. It can identify an abnormality present during that brief window, but intermittent symptoms may require longer monitoring.

The ordering and scope discussion on this slide was not captured in the recording.

Indications for a 12-Lead ECG

- Acute Cardiac Ischemia / Infarction
  - Typical chest pain/pressure or atypical equivalents (jaw/arm radiation, shortness of breath, diaphoresis)
- Arrhythmias & Conduction Disturbances
  - Palpitations, unexplained syncope or dizziness, persistent tachycardia, or bradycardia
- Systemic & Metabolic Monitoring
  - Electrolyte imbalances (e.g., potassium/calcium), drug toxicities, and pacemaker/ICD evaluation
- Pre-Operative & Risk Assessment
  - Pre-procedural screening for major surgeries and risk evaluation in patients with cardiac risk factors

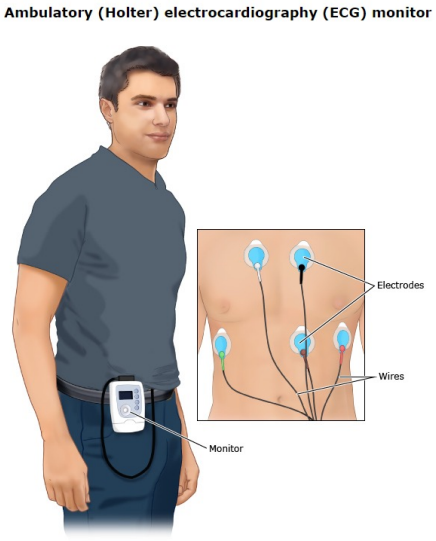
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Slide reference: ECG indications include suspected ischemia, palpitations, syncope or dizziness, persistent abnormal heart rate, metabolic or medication effects, and selected risk assessments.

Choose the test to answer the clinical question. A brief normal ECG cannot establish the cause of symptoms occurring at another time.

Holter Monitoring

- ambulatory ECG monitoring provides a view of ECG data over an extended period of time
- permits evaluation of dynamic and transient cardiac electrical phenomena
- most common ambulatory ECG application is in the diagnosis and assessment of cardiac arrhythmias or conduction abnormalities (symptomatic or asymptomatic) or the presence of potential arrhythmias (such as in patients with syncope or presyncope)
- short as 24-48 hours, 7-14 days or for as long as months to years (implantable loop recorders)



An ambulatory (Holter) monitor is a lightweight electrocardiographic (ECG) monitoring system worn on a belt or strap. The monitoring device is attached to surface electrodes on the chest to continuously record cardiac electrical

Not captured in recording

Slide reference: Ambulatory monitoring samples cardiac activity over a longer period to detect transient arrhythmias or conduction abnormalities. Options range from short Holters to longer external or implanted monitoring.

No separate explanation of this introductory slide is available in the recording.

Holter Monitoring Indications

- When to choose Holter monitoring
  - refer ambulatory ECG monitoring test for patients with daily or near daily symptoms, and for patients in whom a comprehensive/continuous assessment of all cardiac activity is required
- Clinical scenarios in which a Holter monitor is a good choice:
  - syncope, near syncope, or dizziness occurring on a daily or near daily basis
  - palpitations occurring on a daily or near daily basis.
  - atrial fibrillation (AF) for assessment of ventricular rate control
  - frequent ectopy PACs or PVCs, requiring quantification of the ectopy burden
  - recent acute coronary syndrome, hypertrophic cardiomyopathy, arrhythmogenic right ventricular cardiomyopathy, long QT syndrome, dilated or restrictive cardiomyopathy, congenital heart disease, or Brugada syndrome, screening for ventricular ectopy or non-sustained ventricular arrhythmias might alter prognosis and therapy (likely a Cardiology ordered and managed issue)

Lecture notes

The recording opens during the more specialized indications: prior ischemic events, cardiomyopathy, prolonged-QT concerns, congenital disease, and known arrhythmias requiring further characterization.

These situations are often investigated and managed by cardiology. Primary care still needs to understand why monitoring was ordered, what the findings mean, and which follow-up responsibilities remain with the NP.

The slide's general selection rule is daily or near-daily symptoms when continuous assessment is needed. Less frequent symptoms may need a longer monitoring strategy.

Holter Monitoring

- Typical Holter monitor report includes the following information:
  - Total heart beats
  - Average heart rate
  - Maximum and minimum heart rates
  - Number of premature beats (supraventricular and ventricular)
  - Episodes of tachyarrhythmia and the etiology of the arrhythmias (i.e. supraventricular or ventricular)
  - Longest R-R interval and any pauses greater than three seconds and etiology of the pauses (i.e. sinus pauses versus atrioventricular block)
  - ST segment changes
  - Patient-reported symptoms and any associated ECG findings (called symptom rhythm correlation)
  - Representative samples of ECG tracings (eg, hourly samples)

Lecture notes

A Holter report provides total beats, average/minimum/maximum rate, PAC and PVC counts, arrhythmia type, frequency and duration, pauses, ST changes, and sample tracings. Some systems can also flag important events remotely; this depends on the device and service.

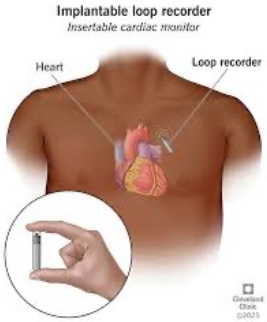
The symptom diary is especially useful. Ask the patient to record the exact date, time, activity and symptoms. The reader compares the rhythm before, during and after that event: symptom-rhythm correlation.

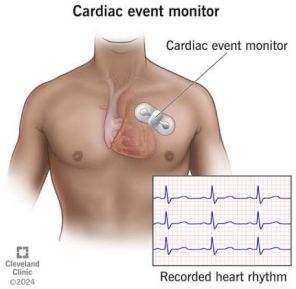
Symptoms without a corresponding rhythm abnormality can support broadening the differential to non-arrhythmic causes; interpret this within the limitations of the recording.

Clarification: The lecturer used a three-second pause threshold as an absolute symptom cutoff. Do not memorize that as a rule: clinical significance depends on symptom correlation and context, rather than one universal pause duration. [1]

Event Monitoring

- For patients with less frequent (ie, weekly to monthly) symptoms and for patients in whom a comprehensive assessment of all cardiac activity over a 24 to 48 hour interval is not required
- Looping memory monitor is a small device about the size of a pager that can be programmed to record your ECG for a period of time, such as five minutes
  - push a button to activate it, and it stores your ECG for the period before and during symptoms
- Symptom event monitor a hand-held device
  - place the monitor on your chest and activate a recording button
  - back of this device has small metal discs that function as the electrodes
  - stores your ECG in memory
  - won't store your before you activate it







Lecture notes

Longer external patches or event monitors can help when symptoms are too infrequent for a short Holter. A handheld recorder is placed against the chest and activated during symptoms; the metal contacts act as electrodes.

Distinguish a recorder that starts only when activated from a looping device that can preserve preceding rhythm. This affects whether the onset of the event is available for interpretation.

An implantable loop recorder is a very small subcutaneous device used for much longer surveillance. The lecturer described monitoring for a year or two when needed; selection depends on the clinical problem.

Ambulatory ECG monitor type	Indications/patient selection
Continuous ECG monitor (ie, Holter monitor)	▪ Daily or near daily symptoms (palpitations) ▪ Assessment for adequate rate control in AF ▪ Assessment of VPB or APB burden, NSVT in hypertrophic cardiomyopathy ▪ Heart rate assessment for inappropriate sinus tachycardia, postural orthostatic tachycardia syndrome, suspected chronotropic incompetence
Event (loop) monitor	▪ Less frequent symptoms (weekly or biweekly) ▪ Assessment for arrhythmic source of palpitations ▪ Assessment for arrhythmic source of dizziness or presyncope/syncope ▪ Assessment for AF (though asymptomatic episodes could be missed unless an auto-triggered event monitor is used)
Implantable loop recorder	▪ Infrequent (less than monthly) symptoms associated with syncope or other high-risk features ▪ Assessment for AF as a source of cryptogenic stroke ▪ Assessment for recurrence of AF (eg, postablation)

Clinical monitoring scenarios	Available monitoring options
Symptoms (palpitations) daily or near daily	▪ Continuous ECG monitor (Holter) ▪ Patch monitor
Symptoms (palpitations) weekly or biweekly	▪ Event (loop) monitor ▪ Patch monitor ▪ MCOT
Symptoms infrequently (monthly or less) associated with syncope or high-risk features	▪ Implantable loop recorder
Assessment for occult AF*	▪ Event (loop) monitor, ideally an auto-triggered event monitor ▪ Patch monitor ▪ MCOT ▪ Implantable loop recorder
Assessment for arrhythmia burden (eg, VPB burden) or average heart rate (eg, AF, inappropriate sinus tachycardia, etc)	▪ Continuous ECG monitor (Holter) ▪ Patch monitor ▪ MCOT

AF: atrial fibrillation; APB: atrial premature beats; ECG: electrocardiogram; MCOT: mobile cardiac outpatient telemetry; NSVT: nonsustained ventricular tachycardia; VPB: ventricular premature beats.

* Notably, the longer the monitoring duration, the higher the diagnostic yield.

Lecture notes

This comparison is a reference for choosing continuous ECG, a Holter, an external loop recorder or an implantable loop recorder. Match the monitoring window to symptom frequency and the information needed.

Implantable loops are inserted by cardiology or an arrhythmia specialist. Access and referral pathways vary by practice location.

Electrophysiology & Cardiac Conduction

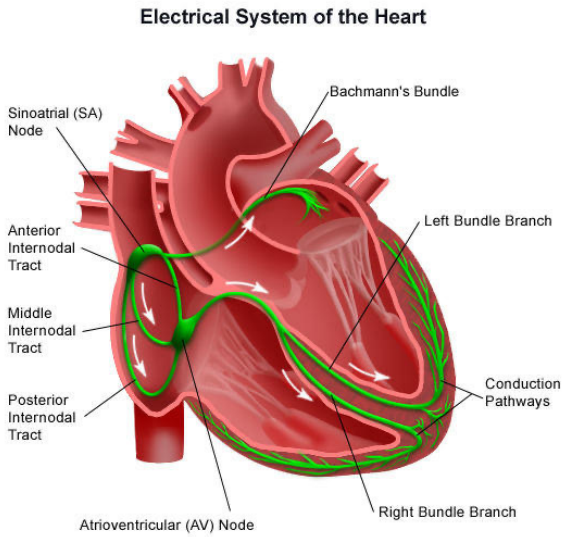
- **Electrophysiology** is the study of the electrical properties of biological cells and tissues
  - involves measurements of voltage change or electric current
- All cardiac cells have an electrical charge
- Stimulation of one cell initiates stimulation of adjacent cells, ultimately leading to cell contraction (you remember action potentials right!)
- There are 2 types of cardiac cells:
  - **Mechanical (Myocardial)**
    - Responsible for muscular contraction and relaxation of the heart
    - Form the middle (muscle) layer of the atria and ventricles
    - Are able to stretch and contract
  - **Electrical Cells**
    - Specialized cells responsible for impulse formation and conduction

Lecture notes

The opening ECG section was mainly provided for independent review. Electrical cells generate and transmit impulses; myocardial cells produce the mechanical contraction.

Keep the electrical event distinct from its mechanical consequence when relating the tracing to cardiac physiology.

Electrophysiology & Cardiac Conduction



- The electrical impulse begins in the SA node and travels through the internodal tracts and Bachmann's bundle to the AV node
- From the AV node, the impulse travels down the bundle of His, along the bundle branches, and through the Purkinje fibers
- Intrinsic firing rates
  - SA node — 60 to 100/minute
  - AV junction — 40 to 60/minute
  - Purkinje fibers — 20 to 40/minute

Lecture notes

Follow the impulse from the SA node across the atria to the AV node, then through the His-Purkinje system to the ventricles. The intended sequence is atrial contraction followed by ventricular contraction.

Intrinsic pacing slows farther down the conduction system: SA node 60-100/min, AV junction 40-60/min, and Purkinje tissue 20-40/min. When a lower escape focus takes over, the resulting rate is usually slower.

This relationship helps explain why more distal conduction disease can produce a markedly slow ventricular rate.

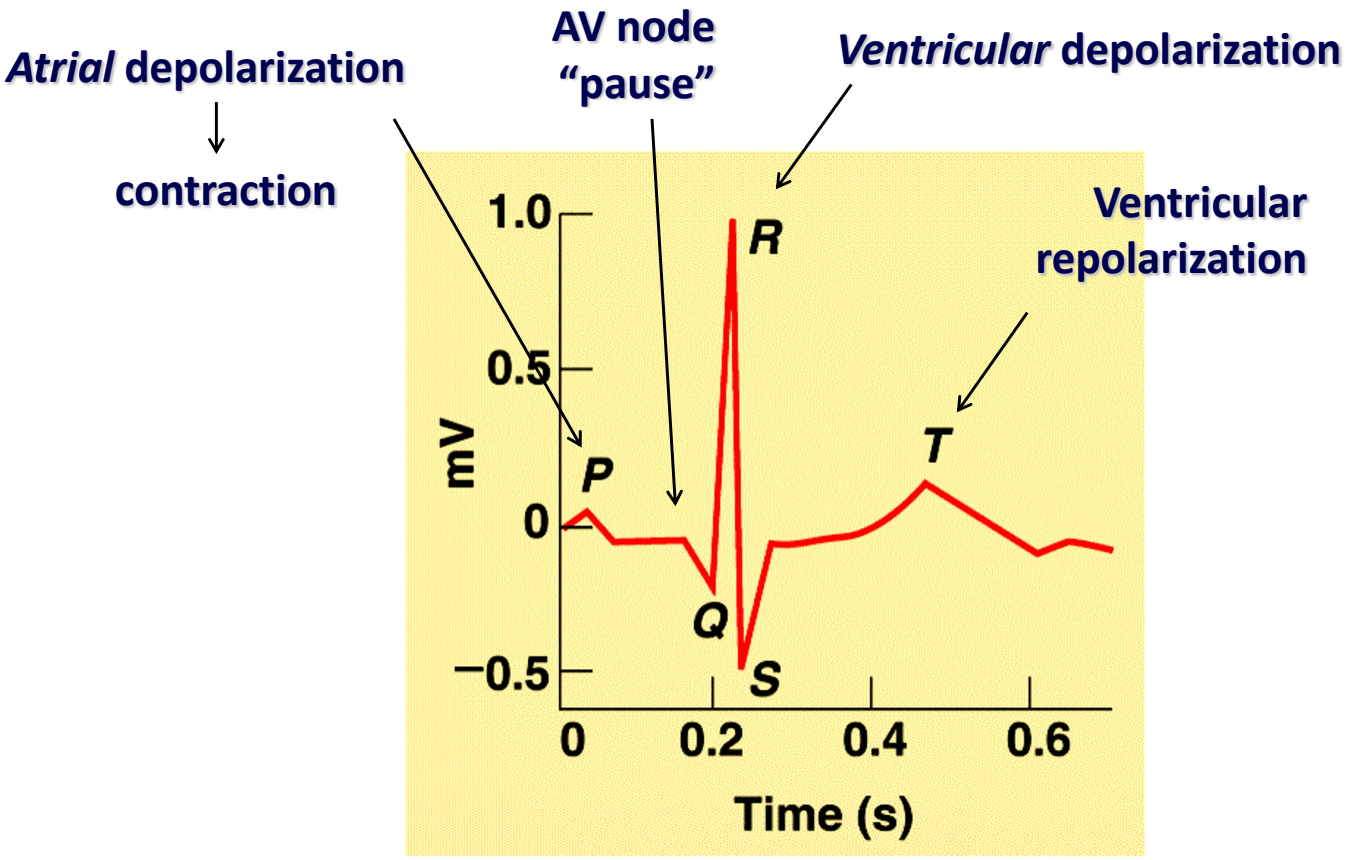
A Word About RE and DE ‘polarization’

- The electrical cells are able to generate and conduct electrical impulses resulting in depolarization (contraction) and repolarization (relaxation and filling) of myocardial cells
- These impulses are a result of a flow of positively charged ions back and forth across the semi-permeable cardiac cell membrane
- **R** in repolarization as standing for **REST**… remember that repolarization is the resting phase of the cardiac cycle

Slide reference

This slide was passed over as part of the ECG review section. Depolarization is electrical activation; repolarization is electrical recovery.

Use the slide to review the movement of ions and the relationship between electrical activation and the cardiac cycle.

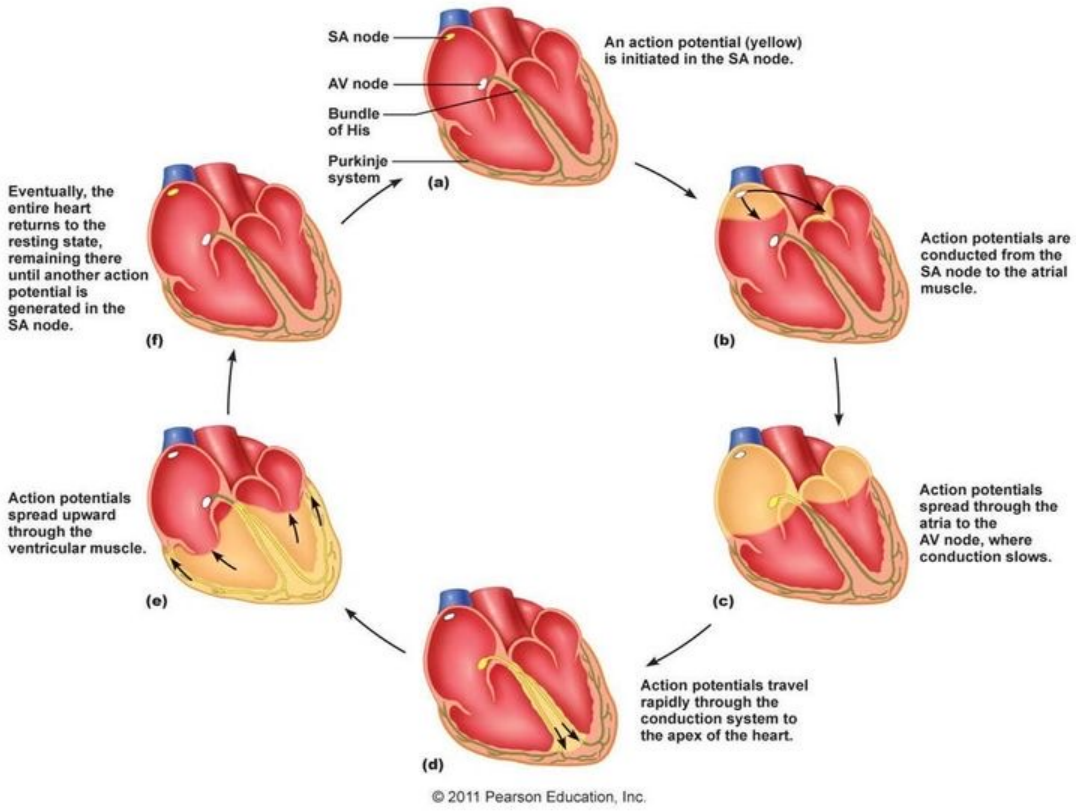


Slide reference

Relate the P wave to atrial depolarization, the QRS to ventricular depolarization, and the T wave to ventricular repolarization. The PR interval includes conduction from atrial activation to ventricular activation.

The figure was not discussed individually in the recording.

Cardiac Conduction



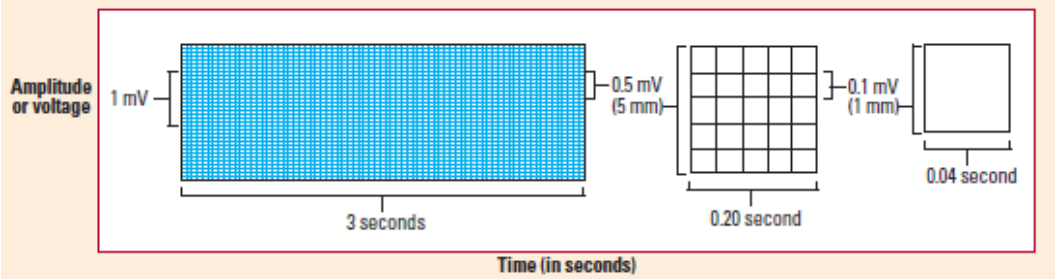
Slide reference

Use this conduction illustration to trace the sequence from the SA node through the AV node and His-Purkinje system. Relate the pathway to the waveforms on the previous slide.

No separate spoken explanation was given for this figure.

The 12 Lead ECG

- horizontal axis represents time
- vertical axis represents voltage
- grid is divided into large squares 5mm X 5mm
  - 1 small square = 40 ms (0.04 sec)
  - 1 large square (5 small squares) = 200 ms (0.2 sec)
  - 5 large squares (25 small squares) = 1000 ms (1 sec)

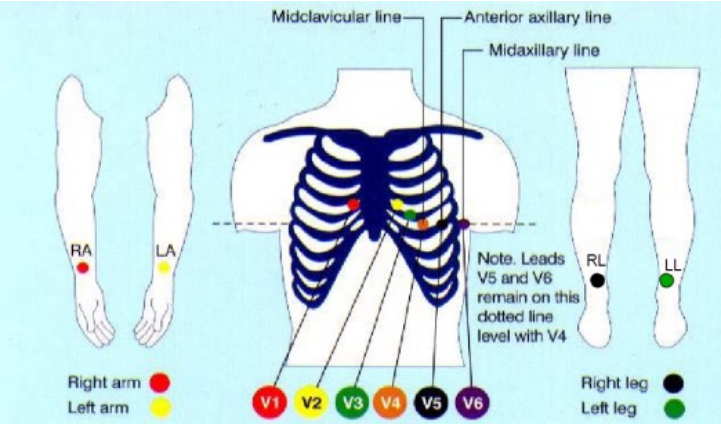


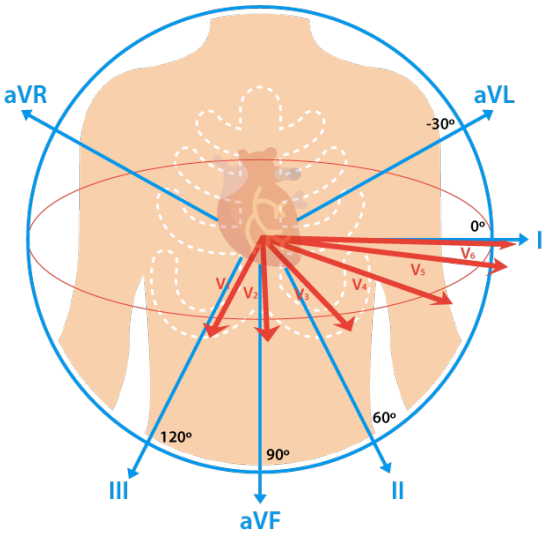
Slide reference

At standard paper speed, one small horizontal square is 0.04 seconds, one large square is 0.20 seconds, and five large squares equal one second.

The lecturer emphasized remembering the box durations before calculating rates or measuring intervals. The horizontal axis measures time; the vertical axis measures voltage.

The 12 Lead ECG



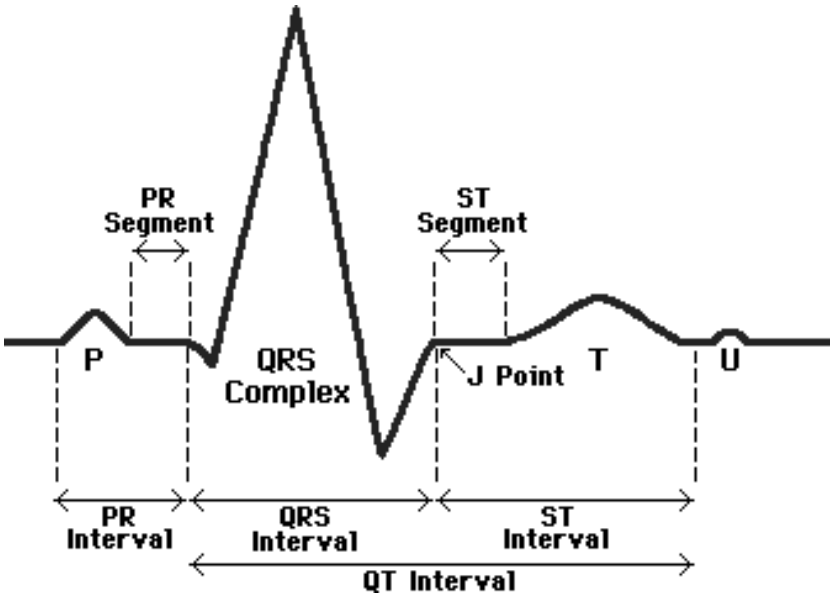


Slide reference

Use the lead-placement illustration to relate the limb and chest leads to their views of cardiac electrical activity. A 12-lead ECG records different perspectives of the same cardiac cycle.

This was included for review rather than a separate live teaching segment.

The Normal ECG

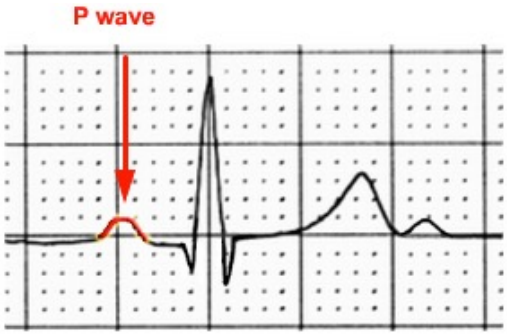


Slide reference

Identify the normal waveform sequence before comparing abnormal examples. Locate the P wave, PR interval, QRS complex, ST segment and T wave.

The lecturer's main instruction was to return to this introductory material independently if a refresher is needed.

The P Wave



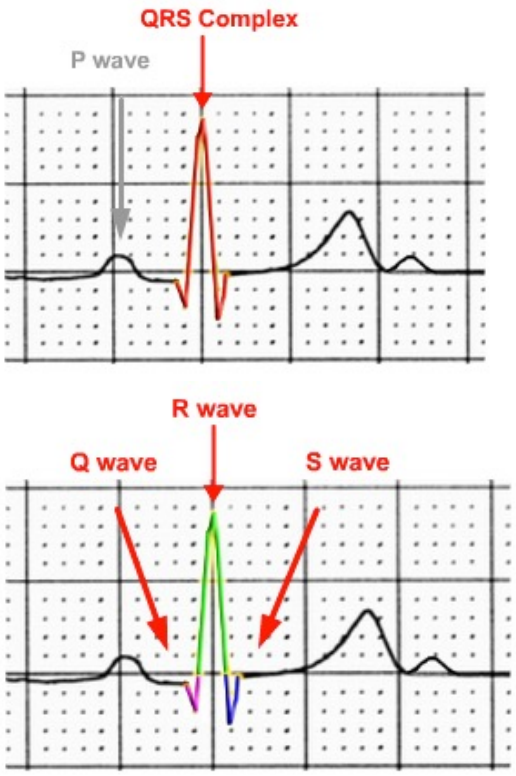
- first deflection representing the depolarization of the atria
- location - precedes the QRS complex
- amplitude - 2 to 3 mm high
- duration - 0.06 to 0.12 second
- configuration - rounded and upright
- peaked, notched, or enlarged P waves - atrial hypertrophy or enlargement associated with COPD, pulmonary emboli, valvular disease, or heart failure

Slide reference

The P wave represents atrial depolarization. Assess its presence, shape and relationship to the QRS; compare waves across the strip rather than identifying just one.

The slide lists size and duration features and examples of abnormal morphology. These details were not expanded on in the recording.

The QRS Complex



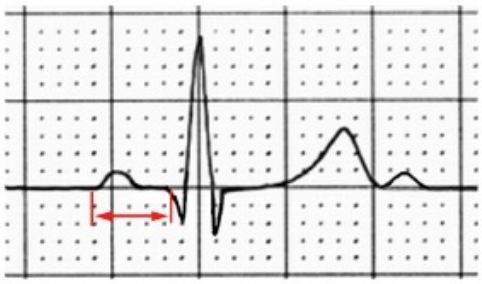
- The QRS complex consists of one or multiple waves representing the depolarization of the ventricles.
- The Q Wave is the initial negative (downward) deflection.
- The R Wave is the first positive (upright) deflection.
- The S Wave is a negative deflection following the R wave.

Slide reference

The QRS represents ventricular depolarization. Q is the initial negative deflection, R the first positive deflection, and S a negative deflection after R.

Not every complex contains all three visible components. Use this terminology when describing the tracing.

PR Interval



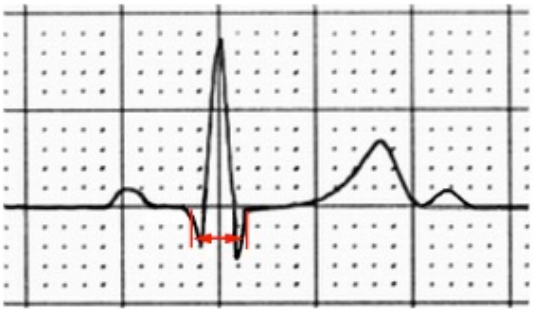
- PR interval measured from the beginning of the P wave to the beginning the QRS complex
- duration - 0.12-0.20 seconds
- includes the time when the impulse travels the atria, A-V node, bundle of His, bundle branches, Purkinje fibres

Slide reference

Measure PR from the start of the P wave to the start of the QRS. The teaching range is 0.12-0.20 seconds.

Check both its duration and whether it stays constant. Later AV-block examples depend on recognizing a prolonged, progressively changing or unrelated PR interval.

QRS Duration



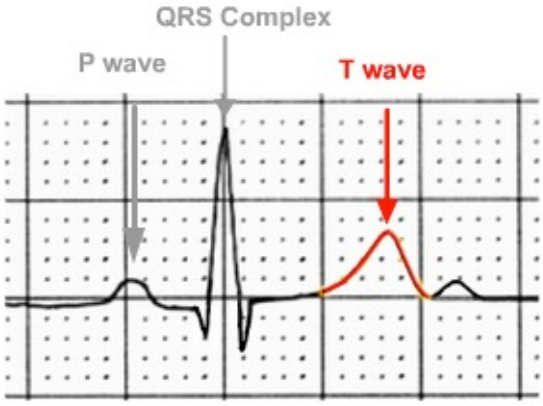
- QRS duration measured from the beginning to the end of the QRS complex (J point)
- represents the time required for the ventricles to depolarize
- Location - follows the PR interval
- Amplitude - 5 to 30 mm high but differs for each lead used
- Duration - 0.06 to 0.10 second, or half of the PR interval
- may not always see all three waves; ventricles depolarize quickly, minimizing contact time between the stylus and the ECG paper

Slide reference

Measure QRS from the beginning to the end of ventricular depolarization. Compare width and shape across the strip.

The deck gives slightly different normal ranges in different places. For the later rhythm exercises, the practical narrow-versus-wide boundary is 0.12 seconds. [2]

The T Wave



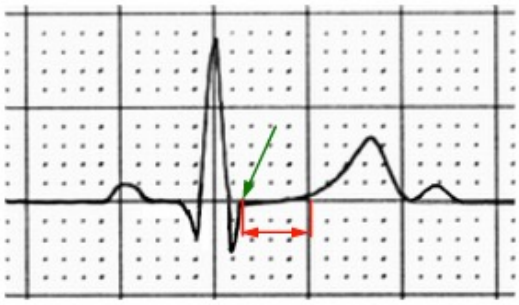
- The T wave represents the repolarization of the ventricles
- Location - follows the S wave
- Amplitude - 0.5 mm in leads I, II, and III and up to 10 mm in the precordial leads
- Configuration - typically round and smooth

Slide reference

The T wave represents ventricular repolarization. Assess its contour and direction in the relevant leads and in relation to the rest of the ECG.

T-wave changes return later in the discussion of ischemia; this slide itself was not explained separately.

ST Segment



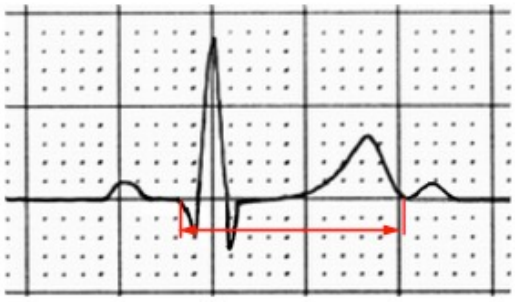
- ST segment: the portion of the ECG tracing from the end of the QRS complex (J point) to the onset of the T wave
- Location - extends from the S wave to the beginning of the T wave
- Deflection - usually isoelectric (neither positive nor negative); may vary from - 0.5 to + 1 mm in some leads

Slide reference

The ST segment extends from the end of the QRS, at the J point, to the onset of the T wave. Assess it relative to the baseline.

Elevation or depression must be interpreted across the relevant leads and in the clinical context, rather than from an isolated shape alone.

The QTc Duration



- The QT interval is measured from the beginning of the QRS complex to the end of the T wave
- Location - extends from the beginning of the QRS complex to the end of the T wave
- Duration - varies according to age, sex, and heart rate
  - usually lasts from 0.36 to 0.44 second
  - shouldn’t be greater than half the distance between consecutive R waves when the rhythm is regular

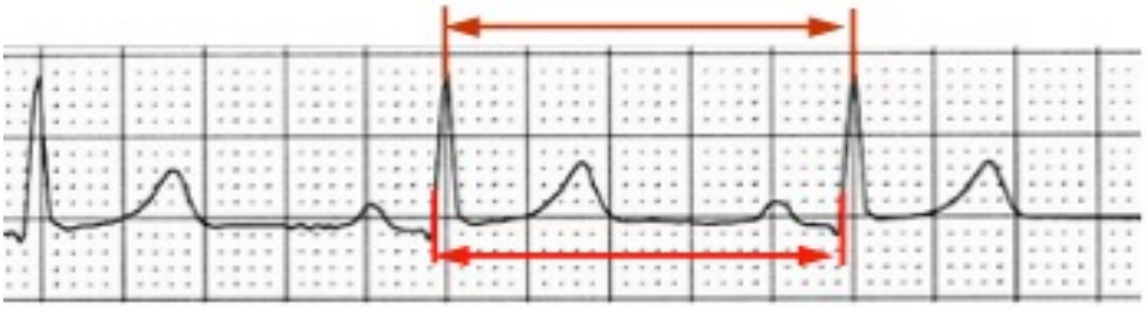
Slide reference

QT extends from the start of QRS to the end of T and includes ventricular depolarization and repolarization. Its expected duration changes with heart rate.

Clarification: QT and QTc are different: QTc is QT corrected for heart rate. The half-R-R comparison is only a rough screen, not a substitute for assessing QTc. [2]

RR Interval

- measured from the beginning of an R wave to the beginning of the next R wave
- represents the period of one cardiac cycle
- used to calculate the heart rate
- any wave in the QRS complex (or part of a wave, such as the peak of the R wave) can be used to determine the RR interval as long as you are consistent with the wave you choose

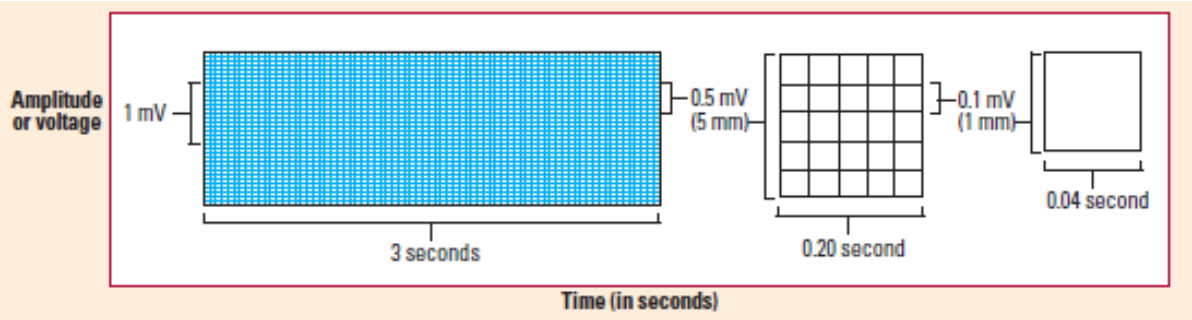


Slide reference

R-R is the interval between consecutive ventricular complexes. Use the same reference point on each complex, usually the R-wave peak.

Compare several intervals to assess regularity. The R-R calculation of rate works best when the rhythm is regular.

Calculating Heart Rate





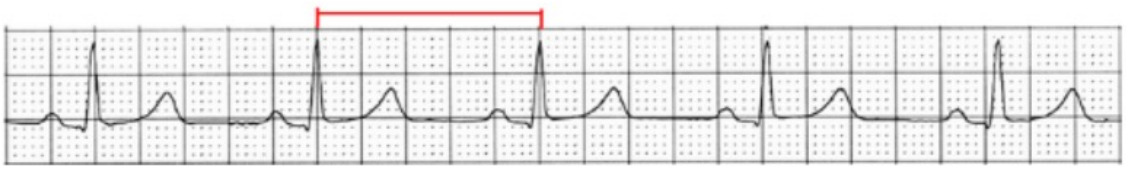
Lecture notes

The lecturer reviewed two approaches: count complexes over a known duration and scale to one minute, or use the R-R method for a regular rhythm.

For a 10-second strip, multiply the number of QRS complexes by six. The following slide examples use six seconds, so multiply those counts by ten. Confirm the duration before choosing the multiplier.

Calculating Heart Rate

R to R Method



There are 300 large squares in 1 minute (300 x 200 msec). To calculate the rate, divide 300 by the number of large squares between 2 P waves (for the atrial rate) or 2 QRS complexes (for the ventricular rate).

$$\text{Rate (beats/min)} = \frac{300}{\text{\# of large squares}}$$

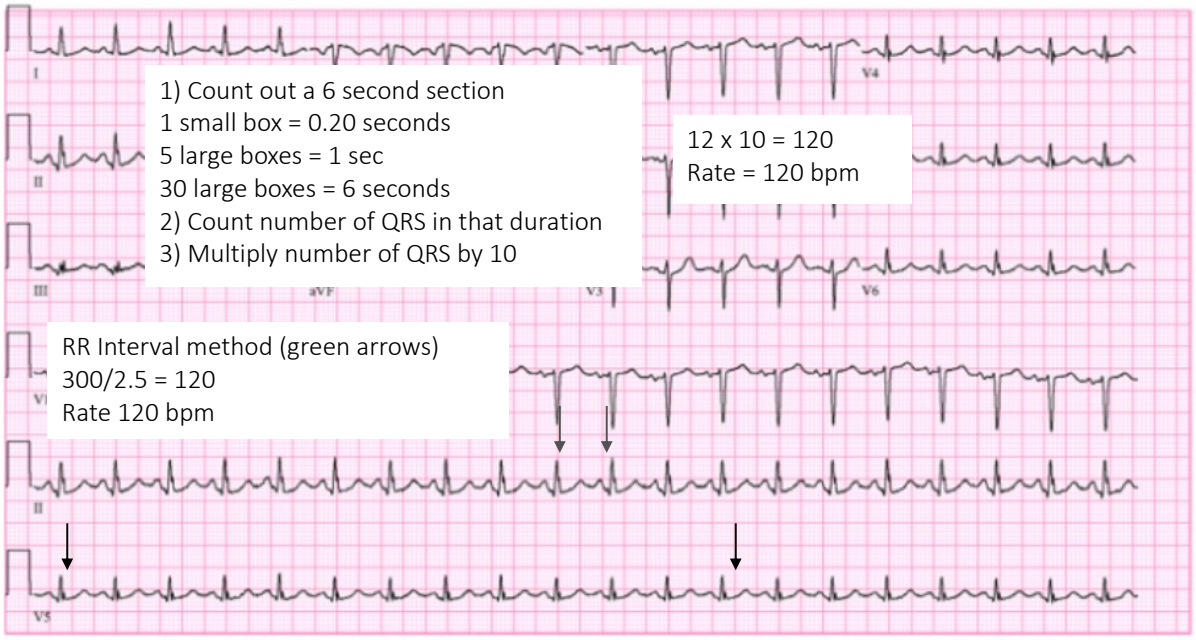
# large squares	Rate (beats/min)
1	300
2	150
3	100
4	75
5	60
6	50
7	43
8	37
9	33
10	30

Lecture notes

For a regular rhythm, heart rate is approximately 300 divided by the number of large boxes between two successive R waves.

Example: two large boxes gives 150/min; two and a half gives 120/min. If the R-R intervals vary, use a timed-strip estimate rather than relying on one interval.

Let’s Calculate the Heart Rate



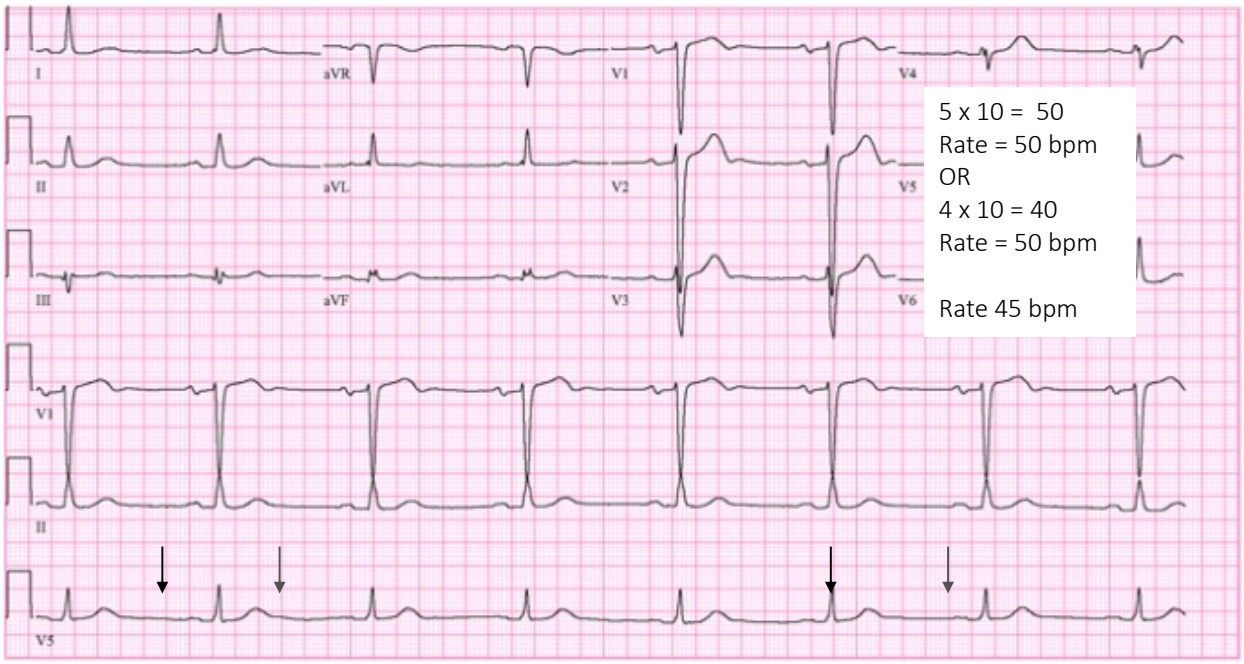
Slide reference

Six-second method: count 30 large squares, count QRS complexes within that window, then multiply by ten. Twelve complexes give 120/min.

The R-R example reaches the same answer: 300 divided by 2.5 large squares equals 120/min.

Clarification: The printed statement that one small box is 0.20 seconds is a typo. One small box is 0.04 seconds; one large box is 0.20 seconds.

Let’s Calculate the Heart Rate

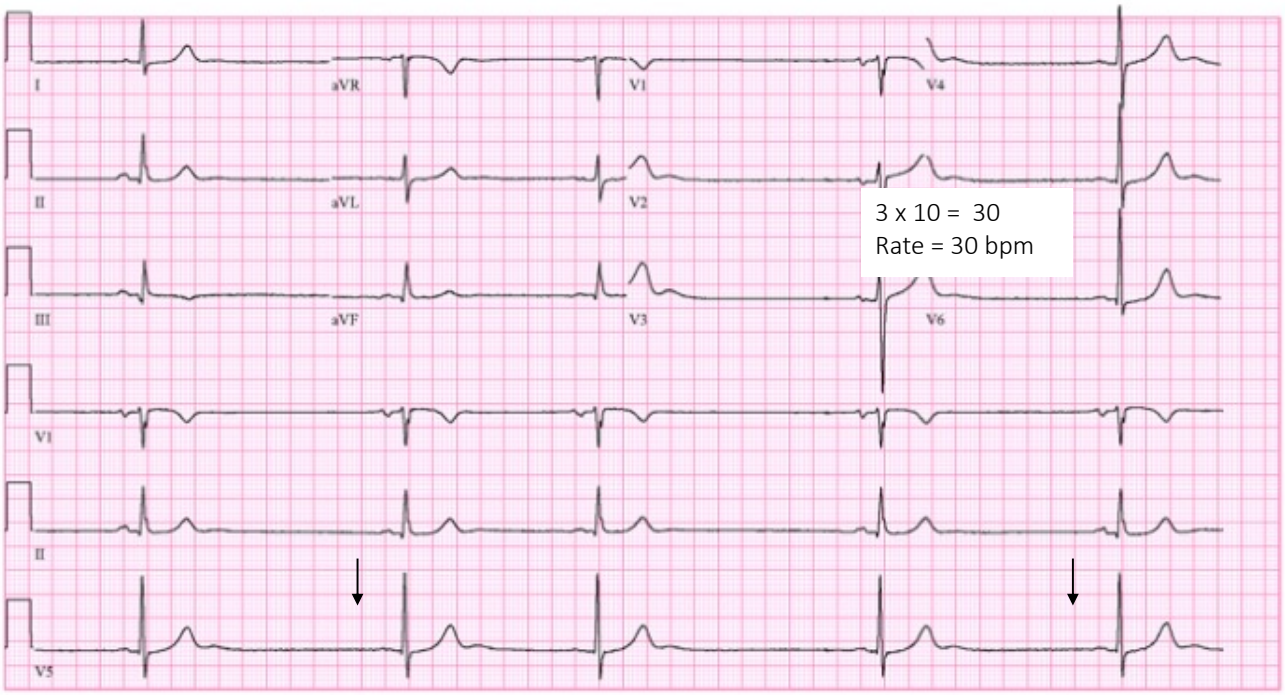


Slide reference

The example illustrates the imprecision of counting a short strip, particularly when a complex lies at the edge of the window. Four complexes over six seconds estimate 40/min; five estimate 50/min.

Clarification: The slide's line '4 x 10 = 40' is followed by a printed rate of 50/min. The arithmetic result is 40/min. A longer sample or regular R-R measurement can refine the estimate.

Let’s Calculate the Heart Rate

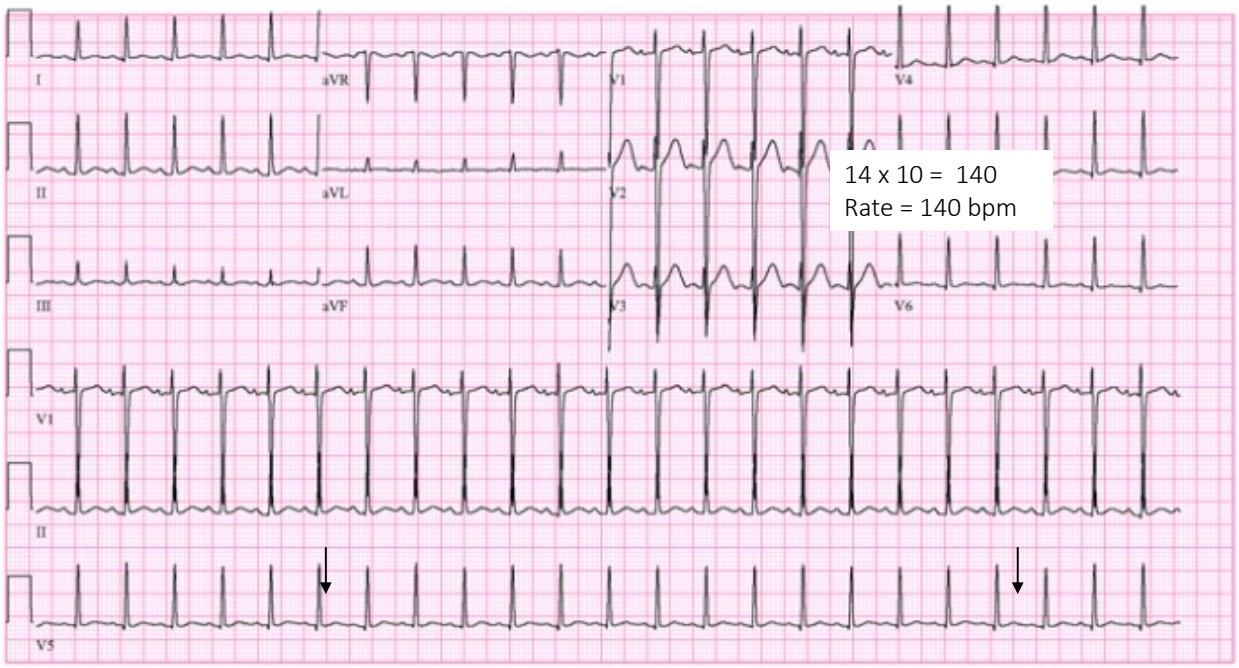


Slide reference

Practice the timed-strip method: three QRS complexes in six seconds give an estimated ventricular rate of 30/min.

The number identifies marked bradycardia; the rest of the tracing and the patient's perfusion determine its clinical meaning.

Let’s Calculate the Heart Rate

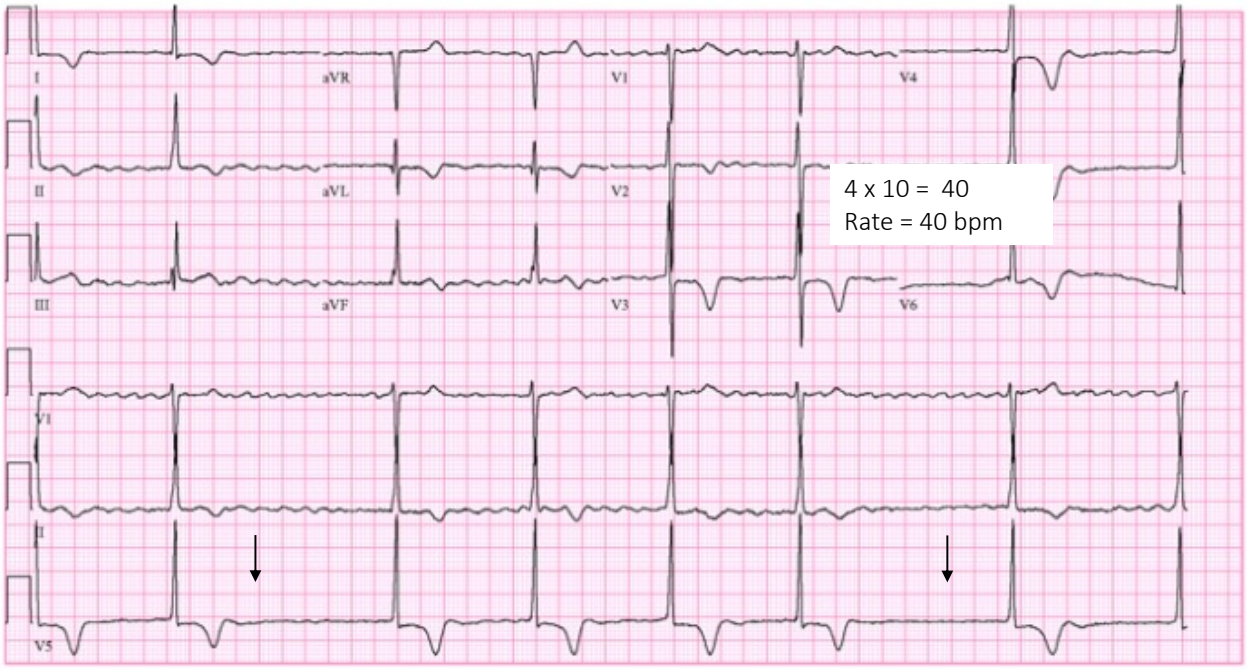


Slide reference

Fourteen QRS complexes in six seconds give approximately 140/min.

After calculating the rate, assess regularity, P waves and the P-QRS relationship before naming the rhythm. Rate alone does not identify its origin.

Let’s Calculate the Heart Rate



Slide reference

Four QRS complexes in six seconds give approximately 40/min.

Use the same sequence for every tracing: rate, rhythm, atrial activity, conduction and clinical context.

A Stepwise Approach to ECG Interpretation

1) Determine the rhythm - atrial and ventricular, regular or irregular?
2) Determine the rate
3) Evaluate the P wave
4) Determine the duration of the PR interval
5) Duration of the QRS duration
6) Evaluate the T waves
7) Determine the duration of the QT interval
8) Evaluate any other components

Lecture notes

The lecturer supplied a stepwise approach for later reference rather than working through every step live. Use one consistent sequence so that an obvious abnormality does not distract you from the remaining components.

Assess rhythm and rate, P waves, PR, QRS, T waves and QT, then other features such as ST changes.

A Stepwise Approach to ECG Interpretation

- **Determine the rhythm - atrial and ventricular, regular or irregular?**
  - Starting at the left side of the rhythm strip, measure the distance between 2 R waves
  - Measure from R wave to R wave across the rhythm strip
  - Can be regular or irregular
- Sometimes this is very obvious and you do not need to measure the R-R interval to determine regularity

Slide reference

Compare R-R intervals across the whole strip, not just a single pair. Assess atrial regularity separately if P waves can be identified.

Some patterns are clearly irregular by inspection; measurement is helpful when the variation is subtle.

A Stepwise Approach to ECG Interpretation

- **Determine the rate**
  - Six-second method
    - Look at six second strip (30 squares)
    - Count the number of QRS complexes in the period of six seconds and multiply by 10 (ventricular rate) & count the number of P waves for that same period to determine the atrial rate
    - Can be used for regular or irregular rhythms
    - Simplest method

Slide reference

In six seconds, QRS count multiplied by ten estimates ventricular rate. P-wave count multiplied by ten estimates atrial rate.

Keeping the two rates separate becomes important when some atrial impulses are not conducted to the ventricles.

A Stepwise Approach to ECG Interpretation

- **Evaluate the P wave**
  - What does it look like?
    - Normal in appearance?
    - Are they identical?
  - Is there a P wave for every QRS complex?
  - What is the ratio of P waves to QRS complexes?
    - Is there a QRS for every P?

- ECG hint… P for every QRS, QRS for every P?
  - SINUS RHYTHM

Slide reference

Ask whether P waves are present, look alike, precede each QRS, and are followed by a QRS. Count the P-to-QRS relationship.

Clarification: A one-to-one relationship supports organized conduction but does not alone prove sinus origin. Assess compatible P-wave morphology and the rest of the tracing as well. [2]

A Stepwise Approach to ECG Interpretation

- **Determine the duration of the PR interval**
  - Measured from the beginning of the P wave to the beginning of the QRS complex
  - Should be between 0.12-0.20 seconds
  - Is the PR Interval measurement the same for each complex?

Slide reference

Measure PR from the start of P to the start of QRS and compare it from beat to beat. The teaching range is 0.12-0.20 seconds.

A long but fixed PR, a progressively lengthening PR, and P waves unrelated to QRS represent different patterns in the later block examples.

A Stepwise Approach to ECG Interpretation

- **Duration of the QRS duration**
  - All QRS complexes should look alike
  - Measure from where the QRS leaves the baseline to where the ST segment begins
  - Normal duration is 0.04-0.12 seconds
    - Considered ‘narrow’ if less than 0.12 seconds
    - Considered ‘wide’ if less than 0.12 seconds

Slide reference

Compare QRS width and morphology throughout the tracing. A wide complex directs attention to altered ventricular conduction or ventricular origin, depending on the full pattern.

Clarification: The final bullet is a typo: narrow QRS is less than 0.12 seconds; wide QRS is 0.12 seconds or greater, not less than 0.12 seconds. [2]

A Stepwise Approach to ECG Interpretation

- **Evaluate the T waves (measure QT)**
  - QT Interval measured from the beginning of the QRS complex to the end of the T wave
  - Normal duration is based on heart rate
  - T wave is electrical only, there is no simultaneous mechanical action
  - Inversion or flattening of the T wave may indicate myocardial ischemia or injury

Slide reference

Assess T waves and measure QT from QRS onset to the end of T. Compare findings with heart rate, other leads and the clinical presentation.

The lecturer left this measurement section for independent review; the later coronary discussion revisits T-wave abnormalities.

A Stepwise Approach to ECG Interpretation

- **Evaluate any other components**
  - **ST segment**
  - Measure the ST segment from the end of the QRS complex to the beginning of the T wave
  - Should be flat
  - Elevation or depression may indicate myocardial ischemia or injury

Slide reference

Assess ST segments after reviewing the basic rhythm. Look for deviations from the baseline and whether they occur in a meaningful group of leads.

ST changes may support ischemia or injury but require clinical interpretation; the deck returns to this in the ACS section.

The ‘Sinus’ Rhythms

Lecture notes

Recognizing the normal pattern is the foundation for recognizing an abnormal rhythm. The lecturer introduced the sinus examples by asking students to establish the expected P-QRS sequence, regularity and rate first.

Then describe what has changed: speed, regularity, premature beats or conduction.

Recognizing Normal Sinus Rhythm

- Before you can recognize an arrhythmia, you must first be able to recognize normal sinus rhythm!
- Characteristics of normal sinus rhythm:
  - Regular rhythm
  - Normal rate (60-100 bpm)
  - A P wave for every QRS complex; all P waves similar in size and shape
  - All QRS complexes similar in size and shape
  - Normal PR and QT intervals
  - Normal (upright and round) T waves

Lecture notes

Normal sinus rhythm has a regular atrial and ventricular pattern, a rate of 60-100/min, consistent P waves, and one P associated with each QRS. Assess intervals and QRS morphology as well.

The lecturer's quick visual approach was to look for the repeating P-QRS pattern and regular R-R spacing before deciding which sinus category applies.

Keep the full set of criteria in mind rather than using the one-to-one relationship by itself.

Sinus Rhythm

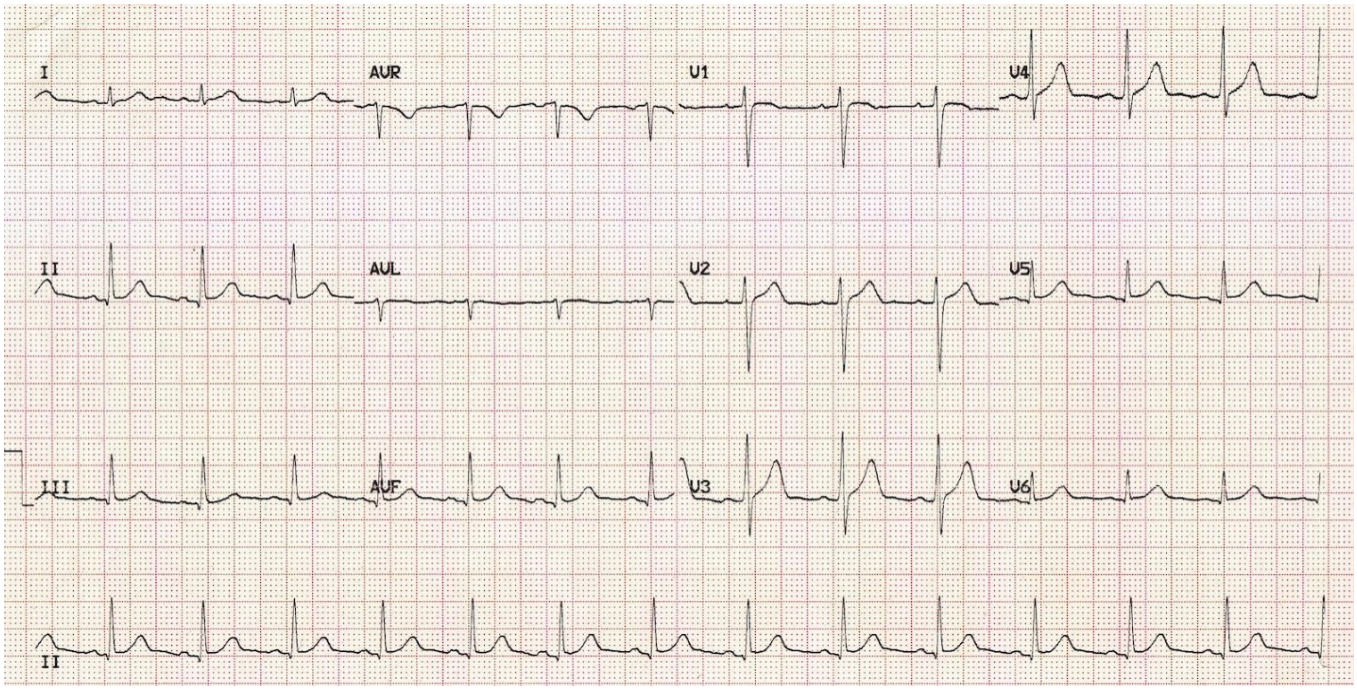
- Rate: 60 to 100 bpm
- Rhythm: Regular, P-P constant - atrial regular, R-R constant - ventricular regular
- P waves: Uniform
- P/QRS ratio: 1:1
- PR interval: 0.12-0.20 seconds
- QRS Complex: 0.04-0.12 seconds

Lecture notes

Check that P-P spacing is consistent and R-R spacing is consistent. This means the atrial and ventricular rhythms are both regular.

In the uncomplicated sinus example, P waves have a consistent appearance, P:QRS is 1:1 and PR is within the stated range. Once the rhythm is established, use the rate to distinguish normal sinus rhythm from sinus tachycardia or bradycardia.

Sinus Rhythm



Lecture notes

On the 12-lead example, the lecturer used the continuous strip at the bottom to follow the repeating P-QRS relationship. Different leads view the same electrical activity from different directions.

Scan the rhythm strip for continuity, then review the individual leads. The bottom strip is commonly lead II; some printouts use a different lead.

Sinus Tachycardia

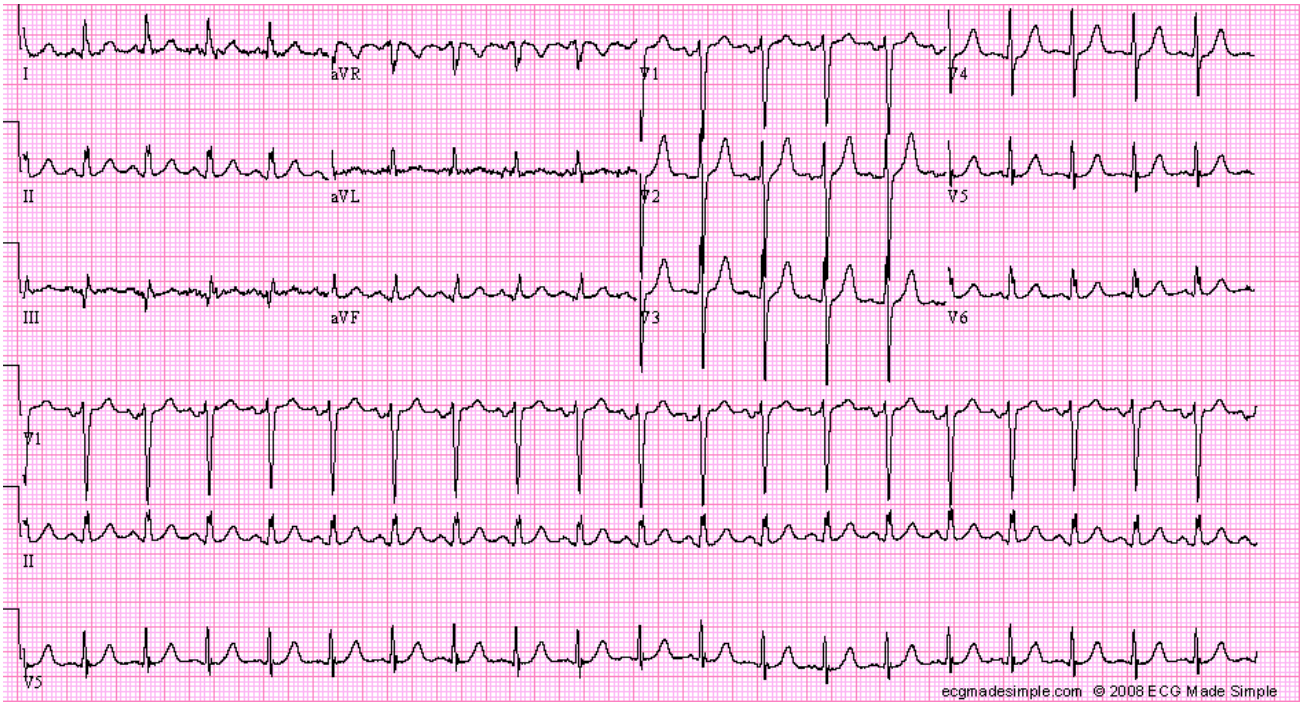
- Rate: 100 to 160 bpm
- Rhythm: Regular, P-P constant-atrial regular, R-R constant-ventricular regular
- P waves: Uniform
- P/QRS ratio: 1:1
- PR interval: 0.12-0.20 seconds
- QRS Complex: < 0.12 seconds
- Often is a response to another clinical event - treatment is usually directed toward the underlying cause (i.e. infection, hypovolemia, pain)

Lecture notes

Sinus tachycardia retains the sinus pattern but is faster. Faster rates reduce ventricular filling time; consider the effect on cardiac output, especially when heart function is already impaired.

Usually look for a driver such as infection or sepsis, hypovolemia, or pain, and direct management toward that cause. The lecturer also mentioned inappropriate sinus tachycardia and POTS as distinct diagnoses to consider in the appropriate presentation.

Sinus Tachycardia



Lecture notes and discussion

Student question: can AI interpret ECGs? The lecturer emphasized that machine-generated ECG interpretations use algorithms and can be wrong. A clinician must review the actual tracing and confirm or correct the interpretation.

Her workplace uses cardiologist review and sign-off. An automated label does not remove the NP's responsibility to recognize the basic pattern and findings that need attention.

Clarification: Do not use a narrow or wide QRS as a simple measure of how fast the heart is going. Measure ventricular conduction separately; QRS width is not determined by rate alone. [2]

Sinus Bradycardia

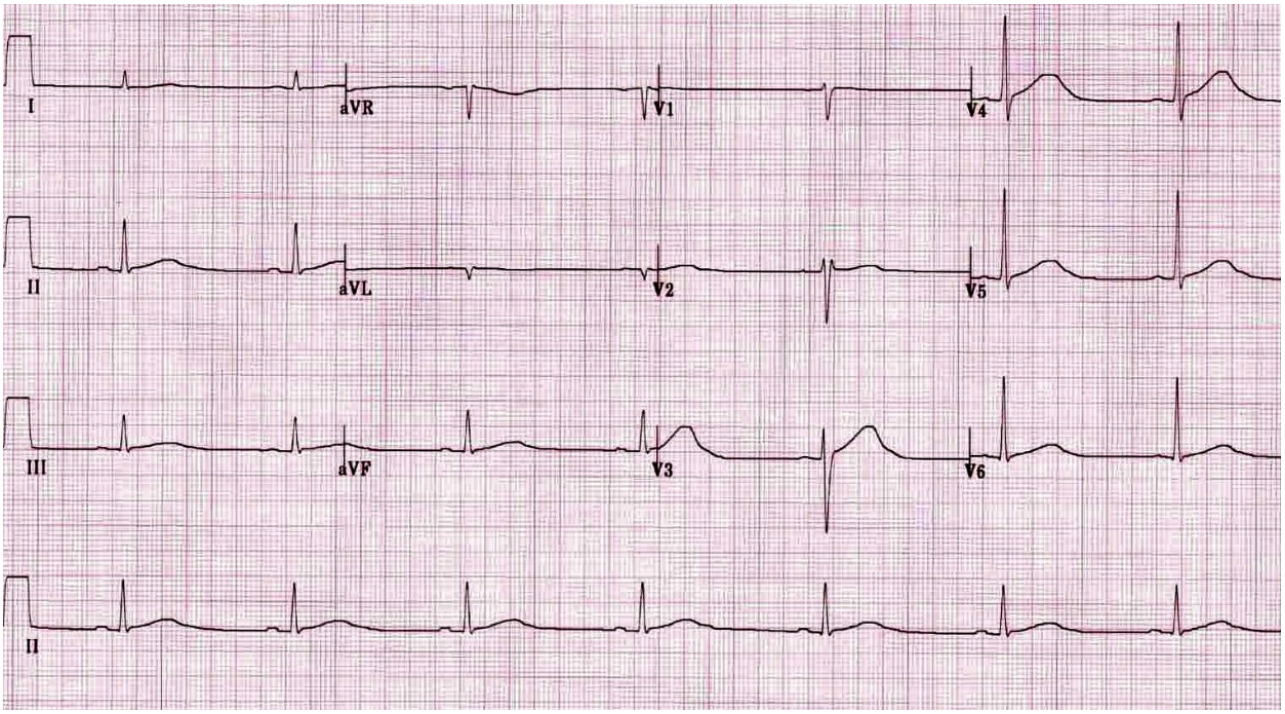
- Rate: 40 to 60 bpm
- Rhythm: Regular, P-P constant-atrial regular, R-R constant-ventricular regular
- P waves: Uniform
- P/QRS ratio: 1:1
- PR interval: 0.12-0.20 seconds
- QRS Complex: <0.12 seconds
- Decreased rate is usually from a reduction in sympathetic output or excessive vagal tone
- May be normal & patient may be asymptomatic

Lecture notes

Sinus bradycardia retains the sinus pattern at a slower rate. It can be a normal finding when the person is asymptomatic and the rhythm truly is sinus.

The practical question is whether the rate is tolerated. A slow number alone and a slow rhythm with evidence of poor perfusion are different presentations.

Sinus Bradycardia



Lecture notes

Use the example to confirm the sinus P-QRS relationship and slow rate rather than assuming every slow rhythm is sinus bradycardia.

Clarification: The spoken suggestion that QRS simply widens because the heart is slower is an oversimplification. QRS duration reflects ventricular depolarization and conduction; a widened complex needs its own assessment. [2]

Sinus Bradycardia…When to Worry (and we will get into this more with syncope!)

- Some patients are asymptomatic… don't worry!
- At slower rates some patients may show signs and symptoms consistent with decreased perfusion
- If a patient can't tolerate bradycardia, they may develop these signs and symptoms:
  - Hypotension
  - Cool, clammy skin
  - Altered mental status
  - Dizziness
  - Blurred vision
  - Heart failure symptoms (crackles, dyspnea, and an S3 heart sound)
  - Chest pain
  - Presyncope (faint, lightheaded, say they feel like they are going to pass out
  - Syncope (overt LOC)

Lecture notes

Look for evidence that bradycardia is impairing perfusion: hypotension, cool clammy skin, confusion, dizziness, visual symptoms, chest pain, presyncope or syncope.

Some patients tolerate slow rates; others, particularly with underlying disease, develop symptoms or heart failure. The lecturer described prolonged unrecognized complete heart block presenting with heart failure.

Distinguish feeling faint from actual loss of consciousness. Both matter when deciding whether a slow rhythm is clinically significant.

Sinus Arrhythmia

- Impulse originates in SA node in an irregular pattern
- Arrhythmia is characterized by an increase in rate during inspiration and a decrease during expiration
- Each beat is conducted normally

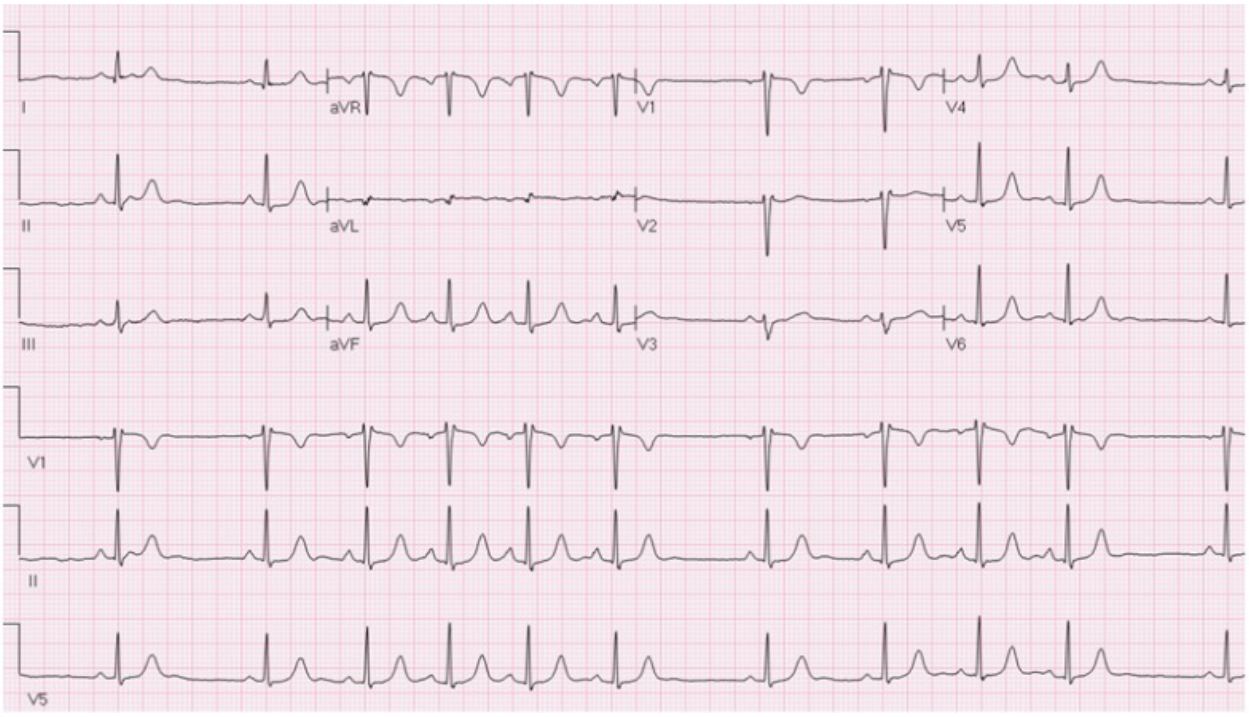
- Rate: Normal (60-100) or sometimes slow (less than 60)
- Rhythm: Irregular
- P waves: Uniform
- P/QRS ratio: 1:1
- PR interval: 0.12-0.20 seconds
- QRS Complex: < 0.12 seconds
- Very common in children and young adults

Lecture notes

Sinus arrhythmia meets the other sinus criteria but has variable R-R spacing. Respiratory variation is common in children and young adults.

The rhythm can therefore be irregular without being AF. Look for consistent sinus P waves and a P-QRS relationship before interpreting the irregularity.

Sinus Arrhythmia



Lecture notes

Follow each P wave into its QRS. The complexes maintain the sinus pattern even though the distance between successive R waves changes.

The lecturer contrasted intervals that look similar with intervals that are visibly shorter or longer. This is the key visual difference from the completely regular sinus examples; the teaching example is benign respiratory sinus variation.

AHAD I Case Study #3

- Julie, 38 years old, presents to the clinic reporting that she has felt her heart "skipping beats" over the past two months. Her family history is positive for paternal death from an MI at age 40. She smokes two packs of cigarettes and drinks two pots of coffee per day. Her cardiac examination reveals: HR 84, regular; BP 124/74; normal S1 and S2; no ectopic beats; no S3 or S4.

    - 1. List the differential diagnoses relevant to Julie's symptoms.
    - 2. For your top three differential diagnoses, review the typical clinical presentation (history and physical examination) associated with each.
    - 3. What diagnostic test(s) would the NP order for Julie at this time?

Lecture notes and case discussion

Julie is 38, with two months of skipped-beat sensations, a father who died young from MI, heavy smoking and two pots of coffee daily. Her examination and vital signs are reassuring between episodes.

Caffeine and nicotine are plausible contributors to ectopy or anxiety symptoms, but do not dismiss a cardiac cause solely because stimulants are present. The family history requires attention.

The lecturer returned to this case during the PAC/PVC and investigation discussion. A normal examination does not capture what the rhythm was doing during an earlier symptom episode.

Palpitations

- one of the most common symptoms of patients presenting to primary care providers and cardiologists
- although the cause is usually benign, palpitations are occasionally a manifestation of a concerning or potentially life-threatening arrhythmia
- subjective symptom
  - defined as an unpleasant awareness of the forceful, rapid, or irregular beating of the heart
  - may describe sensation as a rapid fluttering or a flip-flopping in the chest, or a pounding sensation in the chest or neck, ‘butterflies’
- most common causes of palpitations include cardiac disorders, medical conditions, psychiatric disorders, medication effects, and drug or other substance use effects

Lecture notes

Palpitations are a subjective awareness of heartbeat, not a specific rhythm diagnosis. Patients may say fluttering, flip-flopping, pounding, butterflies or skipped beats; some feel the sensation in the neck or across the chest.

Most causes are benign, but the same complaint can accompany a concerning arrhythmia. Clarify the patient's description, associated symptoms, triggers and background risk rather than treating the word 'palpitations' as the diagnosis.

Preatrial Contractions (PACs)

- An early beat that originates in an ectopic pacemaker in the atria
- Occurs before the next expected normal complex
- Single beat (not a rhythm)
- Every other beat called bigeminy), every 3rd called trigeminy
- Shape of P wave is different (i.e. notched or flattened)
- Underlying rhythm must always be identified
- May originate from a single ectopic pacemaker site or from multiple sites in the atria

Lecture notes

A premature atrial contraction is an early beat from an atrial site outside the SA node. It arrives before the next expected sinus complex. A PAC is an individual ectopic beat, not the name of the underlying rhythm.

The early P wave may be notched, flattened or otherwise unlike the usual sinus P waves. Describe the baseline rhythm first, then the ectopy.

Bigeminy means every second beat is premature; trigeminy means every third. Different atrial origins can produce different P-wave shapes.

Clarification: Use the standard term 'premature atrial contraction'; the slide's 'preatrial' wording is nonstandard.

Preatrial Contractions (PACs)

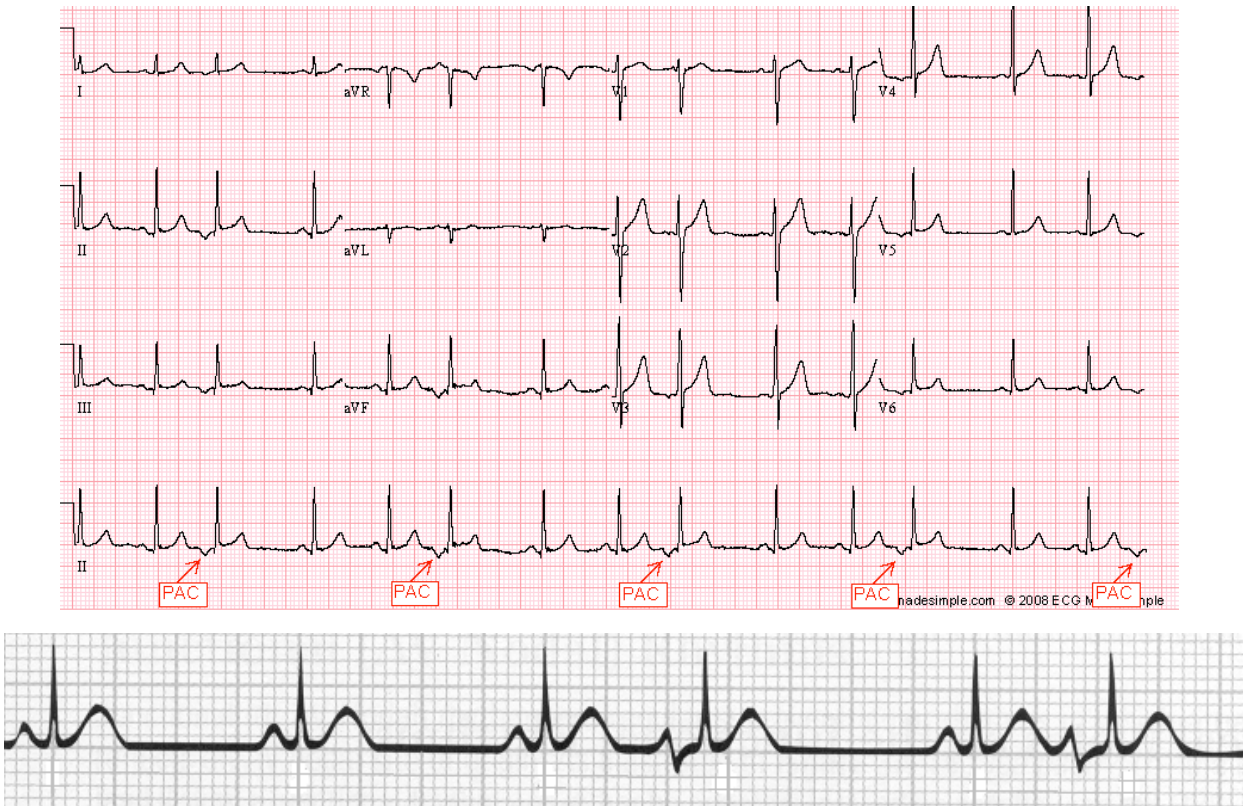
- Rate: Normal for age, dependent on normal rhythm
- Rhythm: Irregular, due to premature ectopic beat
- P waves: P waves associated with PACs are premature and abnormal in size, shape (sometimes peaked), direction (can be inverted), often found hidden in preceding T wave distorting T wave
- P/QRS Ratio: 1:1
- PRI: 0.11-0.20, may be prolonged
- QRS: < 0.12 seconds

Lecture notes

The overall rate is determined by the underlying rhythm. The premature beat makes the strip appear irregular, but the remaining beats still need ordinary rhythm interpretation.

The important change is the timing and morphology of the ectopic P wave. It may partly overlap the previous T wave, making that T wave look different.

Preatrial Contractions (PACs)



Lecture notes

Compare the patient's usual P-QRS complex with the early complex: the ectopic P looks different, arrives early and is followed by a QRS. The tracing then returns to its underlying pattern.

The lecturer pointed out a repeating every-third-beat pattern in one example. Patients may perceive the premature beat as a palpitation even when the overall underlying rhythm is sinus.

Preventricular Contractions (PVCs)

- Common in hearts that are diseased or injured
- Impulse arises from an irritable focus in the ventricle
- Ectopic ventricular beat that occurs before the sinus conducted QRS
- QRS is wide and bizarre
- Usually compensatory pause
- May be unifocal or multifocal
- Impulse of the PVC doesn’t follow the usual bundle branch conduction system
- Results in slow conduction and a wide QRS

Lecture notes

A PVC comes from an irritable ventricular focus before the next expected sinus-conducted beat. It can occur in a normal heart but deserves more attention in a diseased or injured heart.

The teaching visual is a premature, wide, unusually shaped QRS that does not match the surrounding sinus complexes. Ventricular activation does not follow the usual rapid conduction pathway, helping explain the broad appearance.

Clarification: The lecturer initially described four or more beats as a VT run, then later corrected this: three or more consecutive ventricular beats at a tachycardic rate meet the teaching definition of VT. Use slide 126 for the consistent definition.

Premature Ventricular Complexes

Possible Causes

- Aging
- Electrolyte imbalances
- Medications: aminophylline, ventolin, caffeine
- Toxicities
- Stress
- Pregnancy
- And other bad stuff… like cardiac ischemia, hypoxia, acidosis, hypothermia, trauma, hyper/hypokalemia

Signs and Symptoms

- Generally asymptomatic
- Describe feeling a ‘skipped beat’, palpitations
- Observe for increasing frequency/duration
- Hemodynamic instability

Lecture notes and case discussion

For Julie, heavy nicotine and caffeine exposure are plausible contributors in the absence of other reported symptoms. The lecturer favored starting with this explanation while continuing an appropriate assessment.

Review aging, medications such as salbutamol, other stimulants, drug toxicity, stress, pregnancy, oxygenation, acid-base issues and potassium abnormalities. Ischemia is one possible serious cause in the right context.

Clarification: Absence of chest pain or a normal examination alone does not exclude ischemia. The case discussion was a probability judgment about this presentation, not a rule-out test.

Preventricular Contractions (PVCs)

- **Rate:** Depends on the underlying rhythm
- **Rhythm:** Atrial is irregular due to PVC; ventricular can be regular or irregular
- **P Waves:** P waves associated with PVCs
- **P/QRS:** PVC is not proceeded by a P wave
- **PRI:** No P waves associated with PVCs
- **QRS:** <0.12 seconds or more, wide and bizzare (I like to say… ugly!)

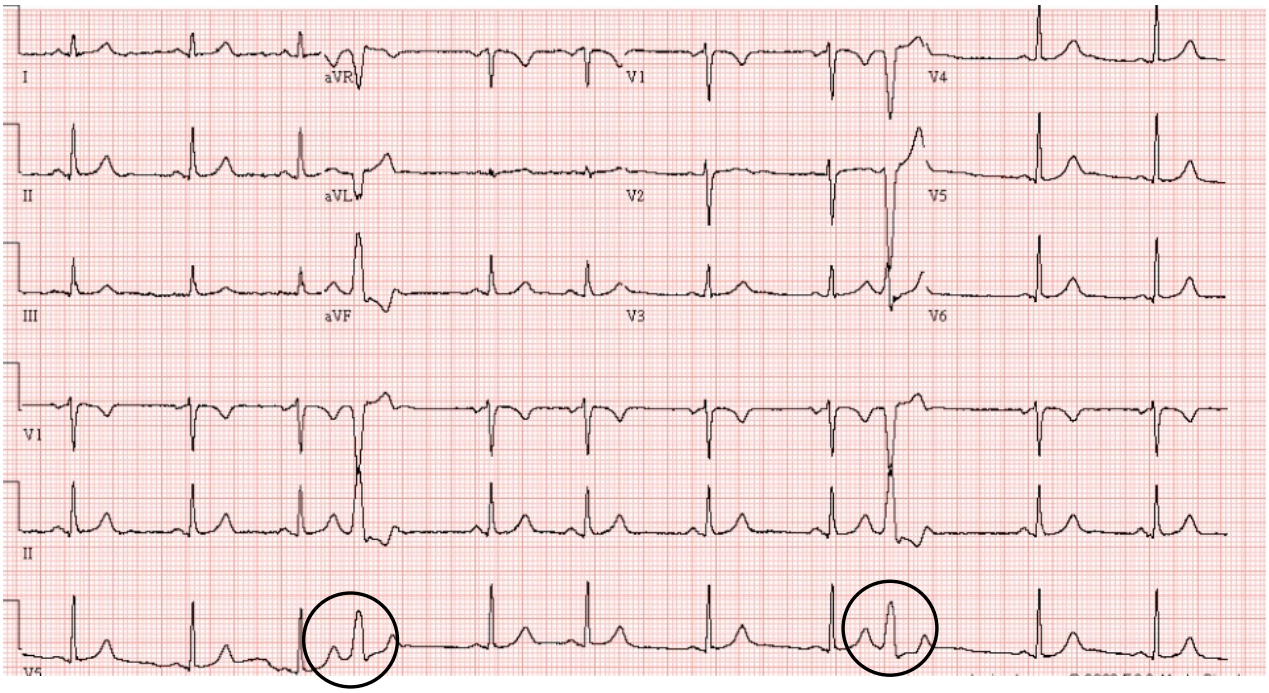
Lecture notes

A PVC has no preceding atrial P wave responsible for that ventricular beat. Start by naming the underlying rhythm, then describe the ventricular ectopy.

The lecturer's visual shorthand was an 'ugly' QRS: it looks different from the patient's usual complexes. Use measurable width, timing and morphology rather than the adjective alone.

Clarification: The printed QRS inequality is wrong. The typical PVC described here is wide, usually 0.12 seconds or greater; the slide's less-than sign should not be memorized. [2]

Preventricular Contractions (PVCs)

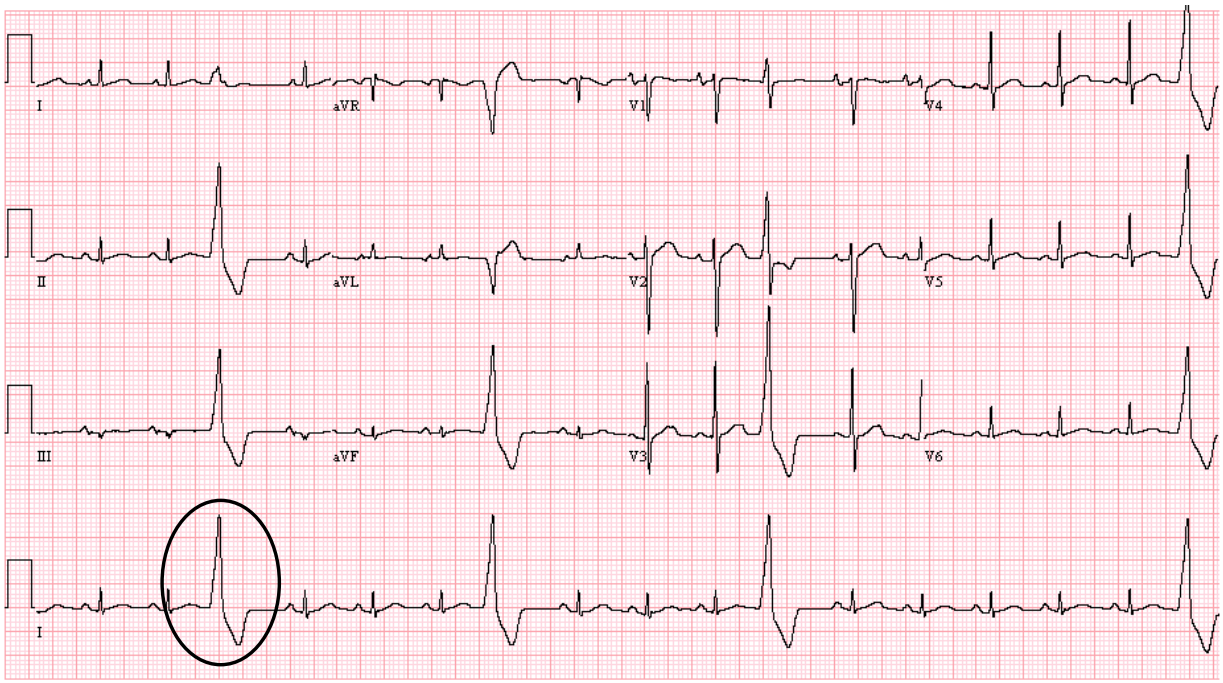


Lecture notes

Follow the normal P-QRS complexes, identify the early broad complex with no preceding P responsible for it, then observe the return to the baseline rhythm.

The surrounding R-R intervals may also vary; do not let that distract from identifying the individual PVC. The underlying rhythm and the ectopic beat are separate parts of the description.

Preventricular Contractions (PVCs)



Lecture notes

The lecturer contrasted a tracing with occasional PVCs with one containing several PVCs in a short sample, which she described as frequent.

Use the description 'sinus rhythm with PVCs' when appropriate, then qualify frequency and pattern. A short ECG gives an impression; a Holter provides the actual burden over the monitoring period.

Causes of palpitations

Cardiac

- Arrhythmias (tachyarrhythmias, bradyarrhythmias, and ectopic beats) due to:
  - Structural heart disease
  - Underlying conduction system abnormality
  - Medical comorbidity (eg, COPD, pulmonary embolism)
  - Idiopathic
- Mitral valve prolapse
- Pacemaker syndrome
- Atrial myxoma
- Intracardiac shunt

High-output states

- Normal pregnancy
- Anemia
- Paget disease of bone
- Fever

Metabolic and endocrine

- Hypoglycemia
- Hyperthyroidism
- Pheochromocytoma

Catecholamine excess

- Stress
- Exercise

Catecholamine excess

- Stress
- Exercise

Substance use

- Cocaine
- Caffeine
- Alcohol
- Amphetamines
- Nicotine

Medications

- Sympathomimetic agents
- Vasodilators
- Anticholinergics
- Beta blocker withdrawal

Psychiatric disorders

- Generalized anxiety
- Panic disorder
- Somatization disorder

Copyrights apply

Lecture notes

This reference table groups causes of palpitations. The lecturer used it to reinforce that the symptom has a broad differential, including cardiac, systemic, psychiatric, medication and substance-related causes.

Use the history to narrow those categories rather than ordering the same investigations for every patient.

Common Causes of PVCs

- Cardiac Causes
  - Ischemic heart disease – e.g., prior myocardial infarction or active ischemia
  - Cardiomyopathy – dilated or hypertrophic cardiomyopathy
  - Heart failure – structural remodeling increases irritability
  - Valvular heart disease – especially aortic or mitral valve
  - Myocarditis – inflammation increases ventricular excitability
- Electrolyte and Metabolic Disturbances
  - Hypokalemia or hypomagnesemia – lower threshold for ectopic activity
  - Hypercalcemia – can increase myocardial excitability
  - Thyrotoxicosis (hyperthyroidism) – increases sympathetic tone
  - Hypoxia or acidosis – respiratory failure or shock
- Stimulants and Substances
  - Caffeine – excessive intake can trigger PVCs
  - Nicotine – sympathetic stimulation
  - Alcohol – especially binge drinking
  - Illicit drugs – cocaine, amphetamines
  - Certain medications – digoxin toxicity, some sympathomimetics, theophylline
- Autonomic or Lifestyle Factors
  - Stress or anxiety – elevated catecholamines increase ectopy
  - Physical exertion – particularly if excessive or sudden
  - Sleep deprivation – may increase ectopic activity
  - Postoperative state – especially after cardiac surgery
- Idiopathic (Benign) PVCs
  - Occur in otherwise healthy individuals with no identifiable cause
  - Often originate from the right ventricular outflow tract (RVOT)
  - Typically benign but may require monitoring if frequent
- Clinical note:
  - Occasional PVCs are usually benign and asymptomatic.
  - Frequent PVCs (>10% of total beats) or symptomatic PVCs (palpitations, presyncope) may warrant evaluation for underlying disease or treatment.
  - PVCs in the setting of structural heart disease or reduced ejection fraction carry a higher risk of malignant arrhythmias

Lecture notes

Consider structural or ischemic disease, cardiomyopathy, heart failure, valve disease and myocardial inflammation. Also review potassium and magnesium, calcium, thyroid function when indicated, hypoxia and acid-base disturbance.

Ask about caffeine, nicotine, alcohol, stimulants, relevant medications, stress, sleep and exertion. Some PVCs are idiopathic, with no identified underlying disease.

A PVC burden above about 10% on Holter was the lecturer's trigger for further assessment, commonly beginning with echocardiography. PVCs in a heart with prior infarction, reduced function or structural disease carry greater concern for dangerous ventricular arrhythmias.

Palpitations
History Taking

- Age of onset
- Duration of palpitations
- Heart rate and rhythm regularity
- Additional descriptive sensations - rapid fluttering, regularity or irregularity, sensation of "flip-flopping" in the chest, pounding, or very strong heart beat followed by "stopping" sensation, irregular, pounding feeling in the neck
- Associated presyncope or syncope
- Abruptness of palpitation onset and resolution
- Patient self-termination of palpitations
- Effect of positional changes
- Temporal association with exercise or emotional stress
- Personal and family history of cardiac disease
  - MVP
  - Structural heart disease
  - Hx of cardiac conduction system disease
- Coexisting medical conditions
  - High-output cardiac state
  - Pregnancy related
  - Metabolic and endocrine causes
    - Hypoglycemia
    - Hyperthyroiodism
    - Pheochromocytoma
  - Chronic obstructive pulmonary disease
- Coexisting psychiatric disorders
- Medications
- Caffeine, nicotine, and other substance use

Lecture notes

Characterize the event: onset, duration, frequency, abrupt start or stop, perceived rate and regularity, position, exertion and emotional stress. Ask directly about presyncope or actual syncope.

Review personal and family cardiac history and all comorbid conditions. Poorly controlled disease, medication changes or difficulty following a treatment plan may explain a change in symptoms.

Include prescribed drugs, OTC products, caffeine, nicotine, alcohol and other substances. The lecturer emphasized a robust history because a brief, normal examination may miss intermittent rhythm events.

What Kind Of PVCs Should I Worry About?

Feature	**Benign PVCs**	**Pathologic PVCs**
Patient profile	Often healthy, no known cardiac disease	Known heart disease (e.g., MI, cardiomyopathy)
Symptoms	Usually asymptomatic or mild palpitations	Palpitations with syncope, dizziness, or presyncope
Occurrence	Occasional or infrequent (<10% of total beats)	Frequent (>10–20% of beats) or in runs
Triggers	Often associated with caffeine, stress, stimulants, or exercise	Occur at rest or during sleep, independent of triggers
Physical exam	Normal heart sounds, no signs of heart failure	May show signs of heart failure or structural disease
Response to activity	Often decrease or disappear with exercise	May persist or worsen with exertion

If PVC burden >10% on a 24-hour Holter, consider further evaluation (echocardiogram, stress test, etc.).

Lecture notes

Known cardiac disease, syncope or presyncope, more frequent ectopy, and impaired heart function should increase concern. An otherwise healthy person with occasional, well-tolerated symptoms offers more room for a staged outpatient assessment.

Quantify burden rather than relying only on how dramatic the sensation sounds. The lecturer recommended an echo as an early next investigation when Holter PVC burden exceeds 10%, to assess cardiac structure and function.

Read the table as a set of risk clues, not as a guarantee that a particular trigger or timing makes PVCs benign.

Palpitations
Physical Exam

- Focused physical exam
  - Vital signs
  - Cardiovascular exam
    - Close attention to presence of cardiac murmurs
    - Findings of heart failure
      - Displaced PMI
      - Peipheral edema
      - Elevated JVP, +ve HJR
      - Presence of an S3
  - Respiratory exam
    - Think COPD… prolonged expiratory phase, wheezing, decreased a/e
  - General
    - Thyroid disease – hyper
      - Sweating
      - Tremor
      - Exopthalmos
      - Risk tendon relfexes

Lecture notes

Perform a comprehensive cardiac examination, including careful assessment for a murmur and signs of heart failure. The lecturer highlighted the aortic and mitral areas, JVP, hepatojugular reflux and S3 in addition to the usual examination.

Assess the lungs and consider respiratory disease or hypoxia as contributors. Look for hyperthyroid features such as sweating, tremor, agitation, eye signs and brisk reflexes.

The lecturer explicitly corrected the slide wording: it should say 'brisk tendon reflexes.' Combine signs with the history rather than diagnosing a cause from one finding.

Palpitations
Laboratory and Diagnostics

- ECG
- Echocardiogram (if appropriate, see next slide)
- Limited laboratory testing – based on history or physical exam findings
  - rule out anemia and hyperthyroidism
  - history or examination suggests substance use or withdrawal, perform toxicology testing

Lecture notes and discussion

Begin with ECG, recognizing that a 10-second sample often misses an intermittent symptom. Daily or near-daily events are a reason to consider Holter monitoring.

Laboratory testing should answer questions raised by the history and examination, such as anemia or hyperthyroidism; it is not automatically an extensive panel. Consider toxicology only when clinically indicated.

Student question: how do you discuss substance testing? Explain the possible causes, why the test might help and what the patient agrees to. If they decline, document that the cause remains unassessed and continue evaluating alternatives. Refusal is not evidence that substances caused the symptoms.

Palpitations

- Echocardiogram in persons whose history, physical examination, or ECG raises concern for structural heart disease
  - family history, symptoms, or ECG concerning for hypertrophic cardiomyopathy
  - ECG with Q waves suggesting prior myocardial infarction, a left bundle branch block, or changes consistent with ventricular hypertrophy
  - murmur on physical examination that suggests valvular heart disease (i.e. aortic stenosis or mitral regurgitation)
- Patients at low risk for concerning arrhythmias
  - palpitations are unsustained and well tolerated and who do not have evidence of heart disease by history, physical examination, or 12-lead ECG
  - generally do not obtain ambulatory rhythm monitoring
  - counsel the patient regarding the low risk of a serious cardiac arrhythmia and the limited utility of further testing
  - if the patient remains very concerned about their symptoms despite counseling, we perform two weeks of event monitoring
- Patients at high risk for concerning arrhythmias
  - ambulatory cardiac rhythm monitoring for all persons whose initial diagnostic evaluation suggests a high risk for concerning arrhythmias, specifically, a disorder that is associated with the development of ventricular tachycardia
  - palpitations that are sustained, poorly tolerated, or associated with syncope or presyncope
  - organic heart disease (eg, scar formation from myocardial infarction, dilated cardiomyopathy of any cause, clinically significant valvular heart disease, hypertrophic cardiomyopathy)
  - personal or family history of arrhythmia, syncope, sudden death, cardiomyopathy, or long QT syndrome
  - patients with frequent symptoms and associated syncope often benefit from expedited evaluation in an inpatient setting.

Lecture notes and case discussion

For Julie, choose the next steps according to symptom frequency and risk. The lecturer favored Holter monitoring for daily symptoms, quantifying PVC burden, and echo when structural disease is suspected or burden is significant. Reduce nicotine and excessive caffeine while investigating appropriately.

Complete reasonable primary-care investigations before a routine cardiology referral when the presentation allows; urgent risk features change that approach. Reassuring, infrequent, well-tolerated symptoms may need less testing.

Ask about family sudden death, including death during sleep, abrupt unexplained collapse, or unexpected drowning in a competent swimmer. These stories may reveal an inherited arrhythmia risk that 'family history of heart disease' alone misses.

Left Bundle Branch Block (LBBB)

- Normal cardiac conduction down both anterior and posterior left fascicles of the His-Purkinje system is compromised
- LBBB is often associated with significant heart disease it can also be seen in patients without any clinical disease
  - i.e. CAD, HTN, valvular heart disease, cardiomyopathy (dilated), ACS, myocarditis, medication
- In isolation the presence of LBBB does not lend itself to any specific clinical concern, nor does it affect prognosis
- In the proper clinical context, LBBB can of great consequence and importance, especially in patients presenting with chest pain, pre/syncope and heart failure with reduced ejection fraction (HFrEF)
- New onset LBBB in the proper setting of concerning clinical symptoms should always be considered a sign of pathology
- ***Patients with a new LBBB require cardiac evaluation***

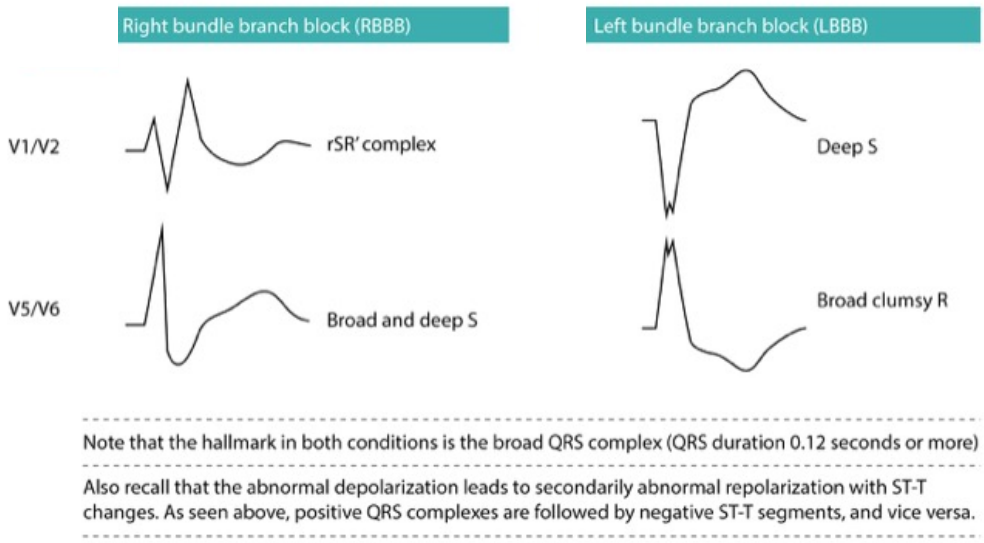
Lecture notes

A new LBBB warrants further cardiac evaluation. It can accompany coronary disease, hypertension, valvular disease, cardiomyopathy, myocarditis or other pathology, even before overt heart-failure symptoms appear.

The lecturer emphasized context: new LBBB with chest pain, presyncope or syncope, or heart-failure features is especially concerning. Do not dismiss an abnormal conduction finding because the patient currently looks comfortable.

Use it as a reason to investigate possible underlying disease, not as a stand-alone diagnosis of the cause.

- QRS duration greater than 120 milliseconds
- Absence of Q wave in leads I, V5 and V6
- Monomorphic R wave in I, V5 and V6
- ST and T wave displacement opposite to the major deflection of the QRS complex



Slide reference

The slide lists a broad QRS, absent lateral Q waves, characteristic lateral R-wave morphology and secondary ST-T changes as LBBB features.

The recording concentrated on the importance of a new LBBB rather than walking through this tracing criterion by criterion.

Clarification: A new LBBB alone is not a STEMI equivalent. Suspected ischemia in this setting requires assessment of symptoms and appropriate ECG criteria and urgent pathways. [5]

The Atrial Rhythms

Lecture notes

The lecture now moves from sinus rhythms and premature beats to atrial fibrillation and atrial flutter. AF was the main focus because it is common and frequently encountered in primary care.

Keep rhythm recognition, symptom burden, structural assessment and stroke prevention as separate parts of the evaluation.

Atrial Fibrillation (AF)

- Atrial fibrillation (AF) is the most commonly treated cardiac arrhythmia
- State of disorganized electrical activity from multiple locations within the atria
- Rapid atria depolarizations are not uniform
    - No discernible P waves
    - Small irregularly shaped fibrillatory (F waves) are seen
- Ventricular response is irregular
- Hemodynamic status may become compromised due to loss of “atrial kick”
    - Contributes about 30% of cardiac output
    - Decreased ventricular filling time
    - Less filling time means less blood will be ejected during ventricular systole and less will be sent through the circulation

Lecture notes

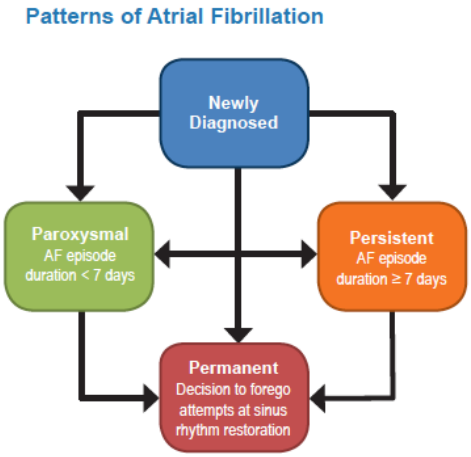
In AF, disorganized atrial activation replaces the coordinated sinus impulse. Multiple electrical signals reach the AV node; variable conduction through that node produces an irregular ventricular response. The lecturer described the AV node as a gatekeeper that cannot pass every impulse through.

There are no discrete sinus P waves. Instead, the baseline may show fibrillatory activity, with irregular R-R intervals.

Loss of coordinated atrial contraction removes the atrial contribution to ventricular filling, or atrial kick. Combined with a fast rate and less filling time, this can reduce output, particularly in an already impaired heart. The lecture cited an atrial contribution of up to roughly 30%.

Atrial Fibrillation

- Paroxysmal (self-terminating or intermittent) AF
  - terminates spontaneously or with intervention within seven days of onset
  - episodes may recur with variable frequency
- Persistent AF
  - fails to self-terminate within seven days
  - episodes often require pharmacologic or electrical cardioversion to restore sinus rhythm
  - a patient who has had persistent AF can have later episodes of paroxysmal AF, AF is generally considered a progressive disease.
- Long-standing persistent AF
  - refers to AF that has lasted for more than 12 months
- Permanent AF
  - term used to identify persistent AF for which a joint decision by the patient and clinician has been made to no longer pursue a rhythm control strategy



Lecture notes

Paroxysmal AF ends within seven days; persistent AF lasts beyond seven days and may require an intervention to restore sinus rhythm. Symptoms vary greatly: some people can identify the exact onset and end, while others do not know they are in AF.

Classification concerns duration and the ongoing treatment strategy, not a different ECG appearance. Some patients benefit from restoring sinus rhythm; in others, the patient and clinician decide to stop pursuing it.

Clarification: AF lasting more than 12 months is long-standing persistent AF. Duration alone does not make it permanent. Permanent AF reflects the shared decision to accept the rhythm, as the slide itself explains.

Atrial Fibrillation

- **Rate:**
  - Atrial rate: cannot be measured (400-700)
  - Ventricular rate: varies with number of impulses conducted through AV node (60-200)
- **Rhythm:** Unable to measure atrial, ventricular is irregular
- **P Waves:** Atria are fibrillating, no P waves are produced
- **PRI:** No P waves
- **QRS:** <0.12 seconds

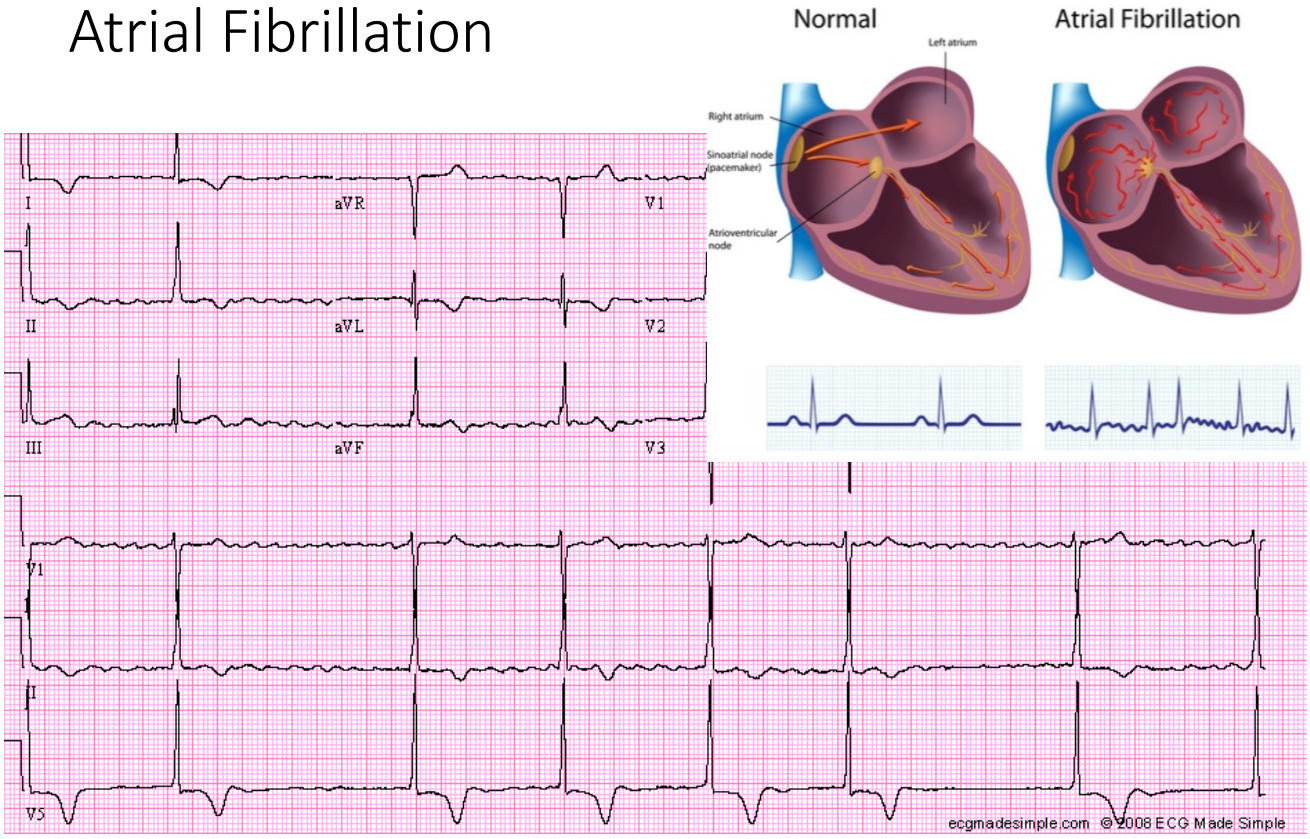
Lecture notes

Recognize AF by absent discrete P waves and an irregular ventricular rhythm. A PR interval cannot be measured when there are no identifiable P waves.

The atrial activity is too rapid and disorganized to count conventionally; the slide gives 400-700/min. Ventricular rate depends on AV conduction and can be controlled or rapid. The lecturer used over about 110/min as a practical rapid-response discussion point.

Describe the ventricular rate separately from the rhythm diagnosis.

Atrial Fibrillation

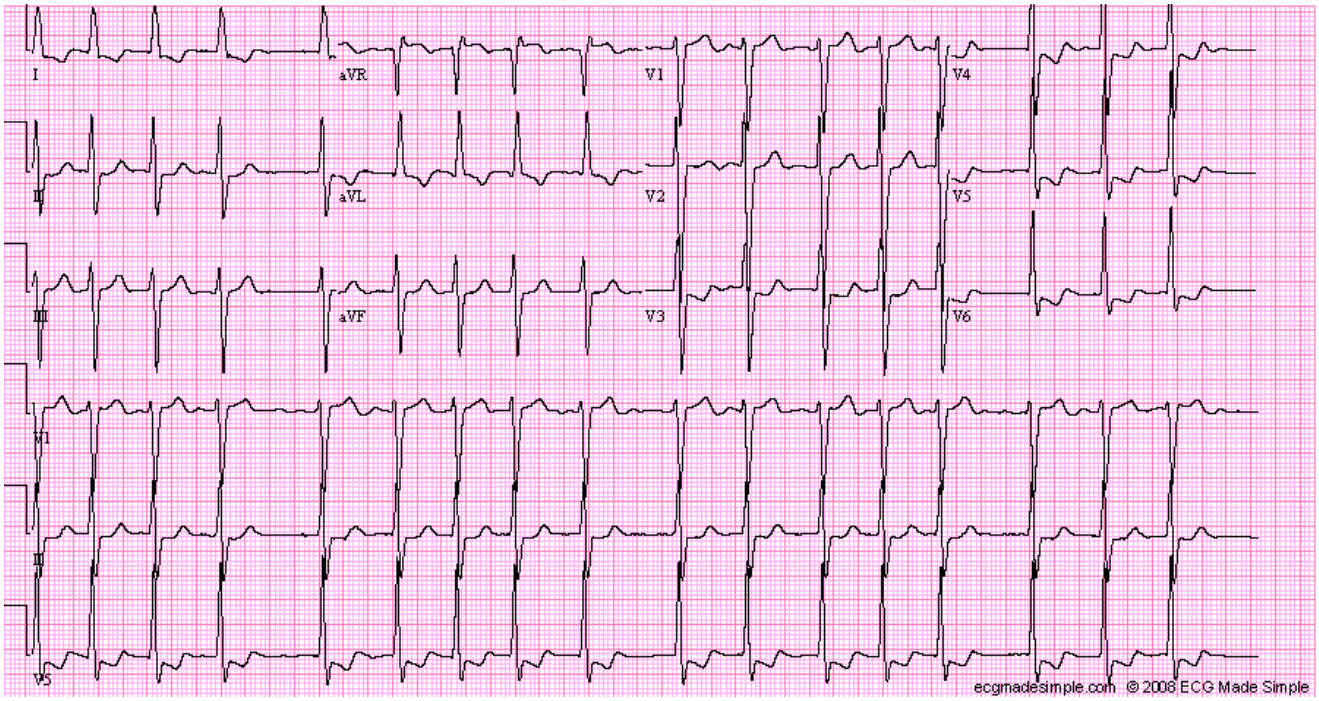


Lecture notes

Compare organized sinus activity with the AF tracing. Look for the loss of discrete P waves and the uneven R-R intervals.

Do not confuse a noisy-looking baseline alone with AF. The lecturer used the combination of atrial activity and ventricular irregularity to explain the pattern.

Atrial Fibrillation



Lecture notes

This second example demonstrates the same AF pattern at a faster ventricular rate. Changing the speed does not change the underlying rhythm diagnosis.

Use the tracing to practice separating the questions 'what rhythm is this?' and 'how fast are the ventricles responding?' Then consider how well the patient is tolerating it.

Atrial Flutter

- Characterized by an atrial rate of 250 to 350 beats/minute (generally around 300 beats/minute)
- Originates from a single atrial focus
- On an ECG, the P waves lose their distinction due to the rapid atrial rate
- P waves blend together in a saw-toothed appearance and are called *flutter waves*, or *ƒ waves*… these waves are the hallmark of atrial flutter
- Not all flutter waves are conducted
- Ventricular rate depends on the number of flutter waves that are blocked
- Ventricular response is regular (block can be variable, resulting in irregular ventricular response)
- 2:1/ 3:1 ratio
- PRI not measured

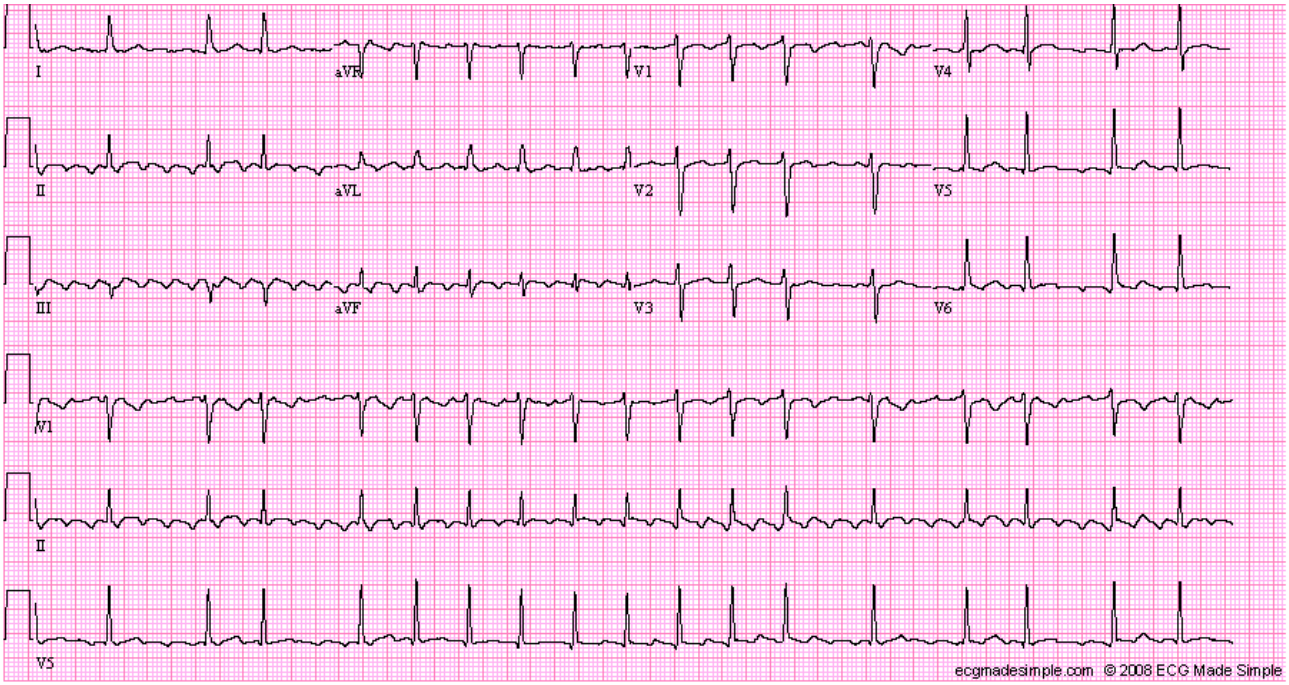
Lecture notes

Flutter is more organized than AF. Its rapid atrial activity creates a repeating sawtooth pattern rather than discrete normal sinus P waves.

Only some atrial impulses conduct through the AV node, so the ventricular rate depends on the conduction ratio. A fixed ratio can produce a regular ventricular rhythm; a changing ratio can make it irregular.

Clarification: The 'single focus' description is a teaching simplification. Typical atrial flutter is an organized macroreentrant circuit; the central recognition feature here is the repeating flutter-wave pattern. [4]

Atrial Flutter

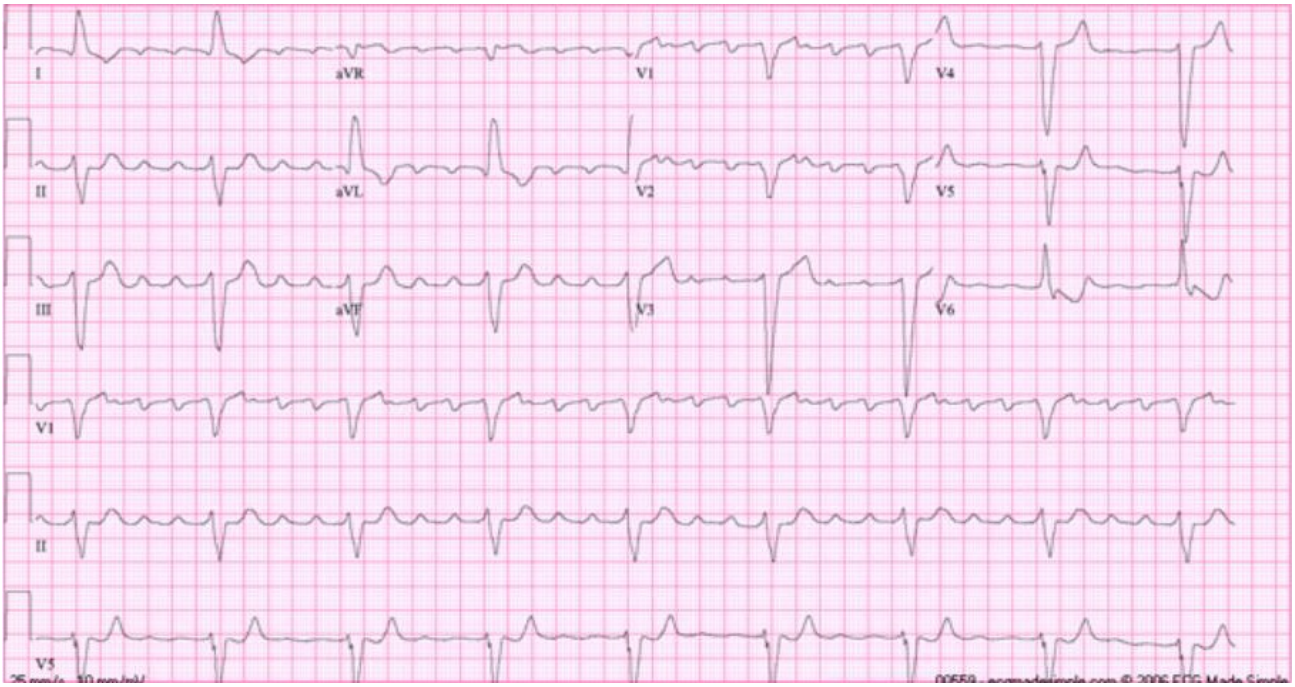


Lecture notes

The lecturer identified a 2:1 flutter example: two atrial flutter waves for each conducted ventricular complex. Count the atrial waves and QRS complexes rather than treating every visible deflection as a separate ventricular beat.

With a stable conduction ratio, the ventricular response can be quite regular despite the rapid atrial rate.

Atrial Flutter



Lecture notes

Variable conduction means the AV node does not pass the same proportion of flutter waves each time. The ventricular intervals can therefore vary.

The repeating organized flutter waves distinguish this example from the disorganized atrial baseline in AF. An irregular ventricular rhythm is not, by itself, enough to call the tracing AF.

Case Study

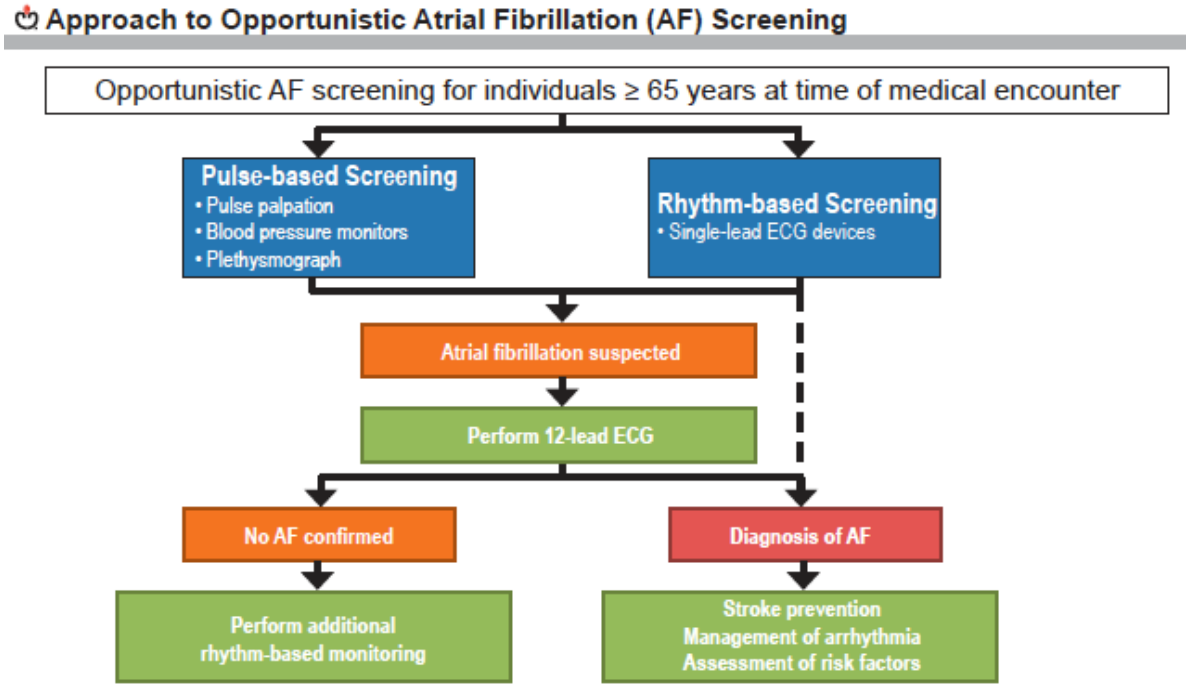
- A 56-year-old man presents to your clinic with atrial fibrillation
- He has been to ER 3 times in the last year with palpitations, requiring cardioversions
- He also has hypertension, dyslipidemia, and his BMI is 35
- Current medications include amlodipine, rosuvastatin, and bisoprolol 2.5 mg
- He says over the last 3 days he has been having palpitations again, but they aren't as severe as before.
- Will assume based on case this patient is ‘new to me/this clinic’

Lecture notes and case discussion

The case is a 56-year-old man with recurrent AF, three ED cardioversions in the past year, hypertension, dyslipidemia and BMI 35. He takes amlodipine, rosuvastatin and low-dose bisoprolol, and reports three days of recurrent palpitations.

Assume he is new to the clinic. Confirm what the current rhythm actually is, assess symptoms and stability, review previous investigations and treatment, and consider stroke prevention as well as rate or rhythm management.

Should We Routinely Screen for AF?



Lecture notes and discussion

Distinguish population screening from opportunistic screening at an existing medical encounter. The lecturer advocated checking for AF in people aged 65 and older when they are already being seen.

A pulse check, cardiac auscultation or an appropriate rhythm device may identify irregularity. If AF is suspected, obtain an ECG to confirm it.

Patients may bring watch or single-lead-device information. Treat it as information to assess, rather than assuming every device alert is a definitive diagnosis. Keep screening during future encounters when no AF is confirmed and the context calls for it.

Gathering More Information

- History
  - What does this patient understand about his atrial fibrillation?
  - OPQRST, symptom specific questions (cardiac and respiratory) and ROS of current episode
  - Atrial fibrillation history
    - date of first symptomatic attack and the date of first objective confirmation
    - duration and frequency of episodes (e.g. the dominant pattern of AF)
    - presence and nature of symptoms related to AF
    - symptom severity (including impact on quality of life)
    - treatment or procedures; details on previous cardioversions
  - Past med/surg Hx, medications + OTC, herbals/vitamins, allergies
  - Family Hx
  - Social Hx – work, home life, social supports
  - Lifestyle Hx – alcohol, drugs (stimulants specifically), smoking, diet, stress, exercise, mood

Lecture notes

Ask what the patient understands about AF, when it first occurred and how it has been managed. Establish episode frequency, duration, symptom severity and effect on work, exercise and daily life.

Use OPQRST plus directed cardiac and respiratory questions. Ask about prior cardioversions, medications, procedures, specialist assessment and what has or has not helped.

Review usual medical, surgical, medication, OTC, family and social history. Explore recent changes in alcohol, stimulants, smoking, stress, sleep, diet or exercise, and whether the patient recognizes triggers.

Gathering More Information

- Physical Exam
  - Vital signs and anthropometric measurements
    - Measure blood pressure and heart rate, height, weight, BMI and waist circumference
  - Cardiovascular, respiratory exam
    - Comprehensive cardiac and respiratory examination with a focus on determination of the causes of AF
- Differential Diagnoses
  - Arrythmia – AF, PVCs, PACs
  - Anxiety
  - Hyperthyroidism
  - Substance induced
- Diagnostic Investigations
  - ECG
    - Document presence of AF; PR, QRS, and QT intervals (e.g. baseline prior to therapy initiation or to monitor therapy)
    - Identify potential causes of AF (i.e. structural heart disease such as myocardial infarction, ventricular hypertrophy, atrial enlargement, congenital heart disease)
    - Identify factors that increase risk of adverse outcomes (i.e. conduction disturbances, sinus node dysfunction, or repolarization abnormalities)
  - Echocardiogram
    - Evaluate ventricular size, wall thickness, and function
    - Evaluate left atrial size and left atrial volume
    - Exclude significant valvular or congenital heart disease (e.g. atrial septal defect)
  - Labs
    - CBC, coagulation profile, electrolytes including calcium and magnesium, renal function, liver function, TSH, lipid profile, HbA1C

Lecture notes and clinical example

Do not assume every new palpitation is the patient's known AF. Consider PACs/PVCs, another arrhythmia, thyroid disease, substance effects and anxiety. Confirm the rhythm with ECG and assess with a cardiac and respiratory examination.

Echo assesses ventricular function and size, atrial size, and valve disease, particularly mitral stenosis. Obtain it when AF is newly diagnosed or structural information is needed.

The lecturer described a patient unaware of sustained rates around 140-160/min until pulmonary edema developed; EF had fallen from normal to about 20%. The example illustrates possible tachycardia-related dysfunction despite few preceding symptoms.

Tailor bloodwork to the assessment: CBC, electrolytes, renal function and TSH, with other baseline or risk-factor tests as indicated.

Common Signs and Symptoms of AF

- Cardiovascular-related symptoms
  - Palpitations – sensation of a rapid, fluttering, or pounding heartbeat
  - Irregular pulse – often detected on physical exam (irregularly irregular rhythm)
  - Tachycardia – resting heart rate >100 bpm is common
  - Chest discomfort or pain – may occur, especially in those with underlying heart disease
  - Fatigue or weakness – due to decreased cardiac output
  - Dizziness or lightheadedness – related to reduced cerebral perfusion
  - Syncope – rare, but possible if cardiac output drops significantly or during conversion pauses
- Respiratory symptoms
  - Shortness of breath (dyspnea) – especially with exertion
  - Exercise intolerance – reduced ability to perform usual activities
- Other possible findings
  - Anxiety – often associated with palpitations
  - Signs of heart failure – peripheral edema, orthopnea, or paroxysmal nocturnal dyspnea if AF is chronic or uncontrolled
  - *Asymptomatic – up to 30–40% of patients may have no obvious symptoms and are diagnosed incidentally (e.g., during a routine ECG)*

Lecture notes

Possible AF symptoms include palpitations, chest discomfort, fatigue, weakness, lightheadedness, dyspnea and reduced exercise tolerance. Syncope is uncommon and needs attention rather than being casually attributed to AF.

Look for heart-failure features when the rate is uncontrolled or cardiac function is impaired. Some patients have anxiety associated with the sensation; others have no symptoms at all.

The lecturer cited roughly 30-40% as potentially asymptomatic and stressed incidental detection during an examination or ECG for another reason.

Assessing Atrial Fibrillation Burden

- Severity of Atrial Fibrillation Score (SAF)
- Type of atrial fibrillation

Pattern	Definition
Paroxysmal AF	Continuous AF episode lasting longer than 30 seconds but terminating within 7 days of onset.
Persistent AF	Continuous AF episode lasting longer than 7 days but less than 1 year.
"Longstanding" persistent AF	Continuous AF >1 year in duration, in patients in whom rhythm control management is being pursued.
Permanent AF	Continuous AF for which a therapeutic decision has been made not to pursue sinus rhythm restoration.
Valvular AF	AF in the presence of any mechanical heart valve, or in the presence of moderate to severe mitral stenosis (rheumatic or nonrheumatic)

SAF Score	Symptoms Attributed to AF, Impact on QOL
Class 0	**Asymptomatic**
1	**Minimal effect:** • minimal and/or infrequent symptoms, or • single episode of AF without syncope or HF
2	**Minor effect:** • mild awareness of symptoms in patients with persistent/permanent AF, or • rare episodes (e.g., < a few per year) in patients with paroxysmal or intermittent AF
3	**Moderate effect:** • moderate awareness of symptoms on most days in patients with persistent/permanent AF, or • more common episodes (e.g., > every few months) or more severe symptoms, or both, in patients with paroxysmal or intermittent AF
4	**Severe effect:** • very unpleasant symptoms in patients with persistent/permanent AF and/or • frequent and highly symptomatic episodes in patients with paroxysmal or intermittent AF and/or • syncope thought to be due to AF and/or • congestive heart failure secondary to AF

- **Valvular AF** refers to patients with moderate to severe mitral stenosis; these patients have a higher risk of stroke than patients without this condition
- Everything else is called **Nonvalvular AF**
- After a diagnosis of atrial fibrillation, all patients should have an echocardiogram to determine if there are any underlying structural, valvular or functional reasons for the atrial fibrillation

Lecture notes and discussion

The lecturer corrected the headings above the two tables: the left table describes AF type, while the right is the symptom-severity scale. The SAF score ranges from 0, asymptomatic, to 4, severe symptoms or major impact.

Echo is important because significant mitral stenosis changes anticoagulant selection. The lecturer emphasized warfarin for AF with moderate-to-severe mitral stenosis; a DOAC may be used in eligible patients without that condition.

Her stated preference for apixaban was a clinical preference, with twice-daily adherence noted as a consideration.

Clarification: Mechanical heart valves also require warfarin; do not interpret 'everything else is nonvalvular' as permission to use a DOAC with a mechanical valve. [7]

Atrial Fibrillation
Epidemiological Considerations

- Risk of atrial fibrillation doubles with each decade of life
- Male sex is associated with 1.5-fold increase in risk of AF
- AF prevalence is 9.2% in females and 15% in males
- AF incidence in:
  - Females: 1.6 per 1000 person-years
  - Males: 1.8 per 1000 person-years
- Absolute number of patients with AF is higher in females than males

Canadian Cardiovascular Society

Lecture notes and related questions

The main epidemiological teaching was that AF becomes more likely with age and is more common in men. The lecturer described risk as increasing markedly with each decade.

Student question: does a single event count? A documented episode matters, even if brief. Record the event and its circumstances; distinguish an isolated episode during an acute reversible illness from an ongoing pattern, and reassess rather than assuming recurrence is impossible.

The cardioversion and anticoagulation questions raised during this interval are collected under slide 98, alongside the AF guideline reference.

Risk Factor Analysis

- Established Risk Factors:
  - Advancing age, **male sex**, **hypertension,** HFrEF, valvular heart disease (mitral stenosis), overt thyroid disease, OSA, **obesity**, excessive alcohol intake, congenital heart disease
- CCS guidelines also suggest new emerging and potential risk factors for atrial fibrillation
  - i.e. COPD, HFpEF, CAD, CKD, DM, tobacco use, genetics, inflammation
- Triggers for AF episodes
  - stimulants, alcohol, sleep deprivation, emotional stress, physical exertion, sleep/nocturnal, digestive
- Reversible causes or AF secondary to
  - cardiac or non-cardiac surgery, acute cardiac pathology or pulmonary pathology, acute infection, thyrotoxicosis, alcohol, pharmacologic agents (e.g. Ibrutinib), supraventricular tachycardia, ventricular pacing

Lecture notes

Established factors include age, hypertension, obesity, sleep apnea, heart failure, valvular disease, thyroid disease, congenital disease and excess alcohol. The lecturer highlighted obesity and OSA as particularly important in practice.

Ask about episode triggers such as stimulants, alcohol, sleep loss, stress and exertion. Consider AF secondary to surgery, acute infection, thyroid disease, acute cardiopulmonary illness or relevant drugs.

Treating a reversible contributor can remove the immediate trigger. Follow-up is still needed to establish whether AF has recurred and to revisit longer-term management.

Risk Markers and Co-morbid Conditions Associated with AF

5

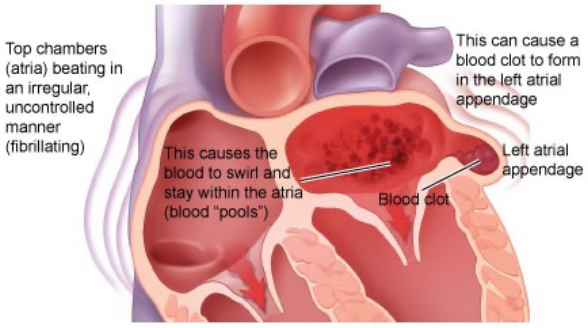
Established Risk Factors	Emerging Risk Factors	Potential Risk Factors
• Advancing age • Male sex • Hypertension • HF with reduced ejection fraction • Valvular heart disease • Overt thyroid disease • Obstructive sleep apnea • Obesity • Excessive alcohol intake • Congenital heart disease (e.g. early repair of atrial septal defect)	• Prehypertension and increased pulse pressure • Chronic obstructive pulmonary disease • HF with preserved ejection fraction • Subclinical hyperthyroidism • Coronary artery disease • Morphometric (increased height, increased birth weight)	• Familial/genetic factors • Tobacco use • Left atrial dilatation • LV hypertrophy • Inflammation • Diabetes • Pericardial fat • Subclinical atherosclerosis • Chronic kidney disease • Excessive endurance exercise • Electrocardiographic (atrial conduction delay, PR interval prolongation)

Lecture notes

This CCS table separates established, emerging and potential AF risk markers. The lecturer recommended the society's AF pocket guide as a useful clinical reference.

Use the table to broaden risk-factor review without treating every association as equally established or as a direct cause in an individual patient.

Atrial Fibrillation and Stroke



- left atrial appendage (LAA) is a unique substrate for stroke; elongated cul-de-sac lined with endothelium
- contractility of the LAA is reduced
- atrial thrombi almost always originate in the LAA
- thrombi have been strongly and consistently linked to reduced blood flow velocities (via echo doppler) in the LAA confirming the role of stasis in their genesis

Lecture notes

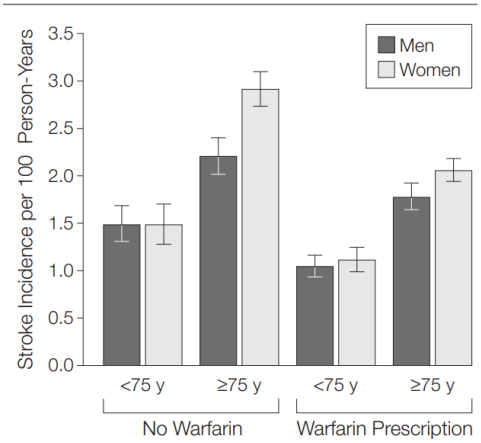
Stroke was the major preventable AF consequence emphasized in the lecture. The left atrial appendage is a pouch where reduced contraction and slower blood movement can permit thrombus formation.

If a thrombus embolizes, it can pass into the left ventricle, through the aorta and into the systemic circulation, including the cerebral circulation.

This is why stroke-prevention assessment remains necessary even when palpitations are mild or the patient feels well.

Sex Differences In Risk Of Stroke In AF Patients

- Females are at higher risk of AF-induced stroke
  - Likely to be under treated with respect to anticoagulation
  - Less likely to be cardioverted back to normal sinus rhythm
  - Less likely to undergo catheter ablation
- Severity of stroke is greater in females compared to males
  - Experiencing more functional disability
  - More likely to be institutionalized
- 12% increase in relative risk of all-cause mortality amongst females compared to males (relative risk, 1.12; 95% CI, 1.07-1.17)
- Female patients have greater residual risk of stroke even if anticoagulated properly



	No Warfarin		Warfarin Prescription	
Men				
No. of events	250	522	313	611
Total No.	5910	10549	9283	13656
Person-years	16864	23731	30091	34385
Women				
No. of events	194	972	288	1116
Total No.	3934	13466	7441	19274
Person-years	13154	33422	26136	54248

Canadian Cardiovascular Society

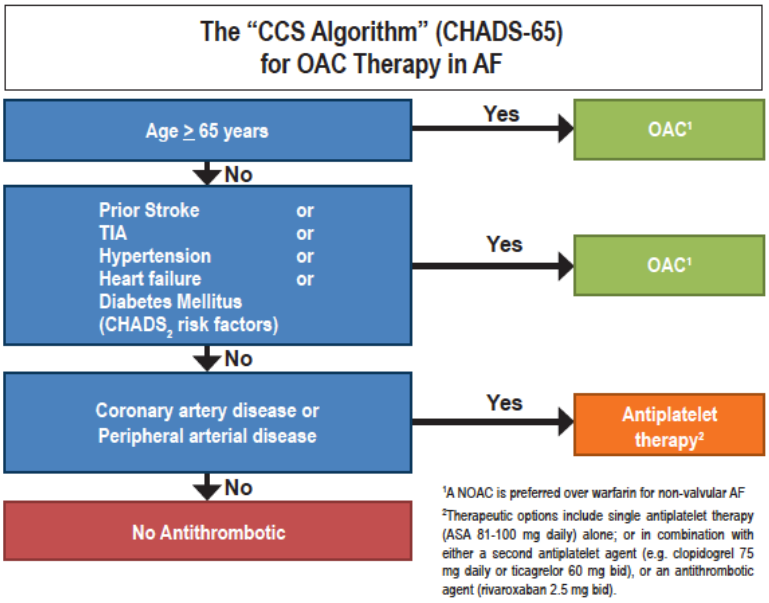
Lecture notes

The lecturer emphasized disparities affecting women with AF: undertreatment with anticoagulation, less access to cardioversion or ablation, and greater disability after AF-related stroke.

The practical point was to review whether an individual patient has received an appropriate stroke-prevention discussion and access to treatment options, rather than allowing sex or age-related assumptions to limit care.

The numerical comparisons on the slide were not explored in detail during the session.

Risk Scoring in Atrial Fibrillation



Lecture notes

The Canadian CHADS-65 pathway starts with age: age 65 or older supports anticoagulation. Below 65, assess CHADS2 factors: prior stroke/TIA, hypertension, heart failure or diabetes. Any such factor supports anticoagulation in the pathway discussed.

For younger patients without those factors, distinguish coexisting coronary or peripheral arterial disease from AF-related embolic risk. The diagram includes antiplatelet therapy in its vascular-disease branch.

Clarification: Antiplatelet therapy in that branch addresses the vascular indication; aspirin is not a substitute for anticoagulation for AF-related stroke prevention. Patients under 65 without CHADS2 factors generally do not need AF-directed antithrombotic treatment. [3]

Original $CHADS_2$ Score		
C	Congestive Heart Failure	1
H	Hypertension (>140/90mmHg)	1
A	Age ≥ 75	1
D	Diabetes Mellitus	1
S_2	Prior TIA or stroke	2

CHADS2 – VASc Score		
C	Congestive Heart Failure	1
H	Hypertension (>140/90 mmHg)	1
A	Age ≥ 75	2
D	Diabetes Mellitus	1
S_2	Prior TIA or stroke	2
V	Vascular disease (MI, aortic plaque etc)	1
A	Age 65-74	1
Sc	Sex category (Female = 1 pt)	1

Lecture notes and discussion

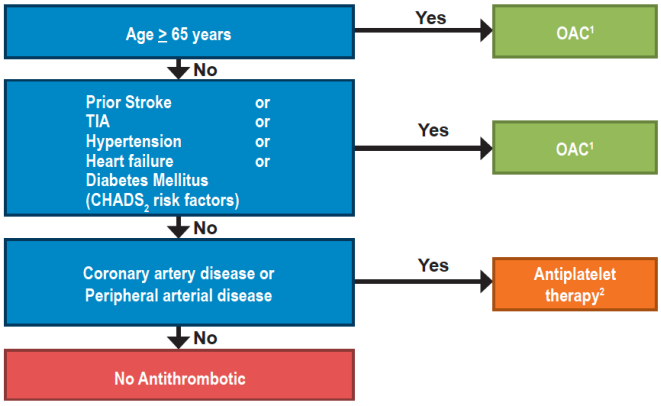
The lecturer uses CHADS-65 to decide whether anticoagulation is indicated and a validated numerical score when a patient wants a more detailed discussion of estimated annual stroke risk.

CHADS2 and CHA2DS2-VASc have different components and age weightings, as shown in the tables. Use the actual tool rather than mixing the two.

Risk estimates are aids to shared decision-making, alongside bleeding risk, comorbidities, preferences and treatment feasibility.

Stroke Risk

- Discussion with patient regarding what information has been provided to him with regards stroke risk and anticoagulation
- Calculating his risk score using validated tools
- Bleeding risk
- Other questions to ask: drug plan? drug cost?
- Calculating his risk score using validated tools
  - CHADS2 = 1 → OAC
  - CCS 65 → age less than 65 → HTN → OAC
  - Annual stroke risk 2.8%
- Bleeding risk
  - HAS-BLED
  - ATRIA Bleeding Risk Score
  - Low risk patient



	Condition	Points	$CHADS_2$ score	Stroke Risk %
C	Congestive heart failure	1	0	1.9
H	Hypertension	1	1	2.8
A	Age ≥ 75 years	1	2	4.0
D	Diabetes mellitus	1	3	5.9
S_2	Prior Stroke or TIA or Thromboembolism	2	4	8.5
			5	12.5
			6	18.2

$CHADS_2$ score = congestive heart failure, hypertension, age, diabetes mellitus, prior stroke or TIA or thromboembolism.

Lecture notes and case discussion

The 56-year-old case patient has hypertension, so the lecturer calculated CHADS2 = 1 and found anticoagulation indicated through CHADS-65 despite age below 65. The slide gives an annual stroke estimate of 2.8% for the referenced scoring table.

Discuss benefit, bleeding risk, drug cost, coverage and adherence. Present the information and recommendation, then establish what matters to the patient and what they will accept.

The discussion contrasted the possibility of bleeding with potentially disabling stroke; avoid assuming a patient's priorities. The quoted percentage is a score-based cohort estimate, not a precise prediction for this individual.

Bleeding Risk Scoring Tools

- HAS-BLED (Pisters et al, 2010)
  - Developed from the Euro Heart Survey on AF in 3,456 patients with atrial fibrillation followed for 12 months
  - 1 point for each of the following: hypertension, abnormal renal/liver function, stroke, bleeding history or predisposition, labile INR, elderly, or drug/alcohol use concomitantly
  - Score - Low (0), Intermediate (1–2), High (3+)
- ATRIA Risk Score (Fang et al, 2011) - Warfarin
  - Developed in 9186 atrial fibrillation patients enrolled in a large integrated healthcare system and followed for 6 years
  - Anemia (3 points), severe renal disease (3 points), age ≥ 75 yrs (2 points), prior bleed (1 point), hypertension (1 point)
  - Scoring - Low (0–3), Intermediate (4), High (5–10)

Lecture notes

HAS-BLED and ATRIA were presented as tools to structure bleeding-risk assessment before and during anticoagulation decisions. Consider the actual risk factors and address those that can be modified.

Discuss the difference between minor and major bleeding, relevant comorbidities, and the patient's concerns. A decision should integrate stroke risk, bleeding risk and preferences rather than rely on a score label alone.

The lecturer did not work through each component of these two tools in the live recording.

Rate Control Versus Rhythm Control Strategy

1. Rate Control:

- Primary goal: Control ventricular rate to relieve symptoms and prevent tachycardia-induced cardiomyopathy.
- Preferred when:
  - Patients are minimally symptomatic or asymptomatic.
  - Rhythm control is unlikely to maintain sinus rhythm (e.g., significant atrial remodeling or longstanding persistent AF).
  - Risks of rhythm control (antiarrhythmic drugs, procedures) outweigh benefits.
  - As a baseline strategy while assessing rhythm control candidacy.

2. Rhythm Control:

- Primary goal: Restore and maintain sinus rhythm to improve symptoms and quality of life.
- Preferred when:
  - Patients are highly symptomatic despite adequate rate control.
  - There is a need to improve left ventricular function (e.g., AF-induced cardiomyopathy).
  - In early AF (short duration) where maintenance of sinus rhythm is more likely to succeed.
  - Younger patients or those with paroxysmal AF who may benefit from long-term sinus rhythm.
  - As part of a broader strategy to reduce hospitalization and potentially improve outcomes, especially post-ablation.

Lecture notes

Rate control aims to control ventricular response, reduce symptoms and avoid tachycardia-related dysfunction. Rhythm control aims to restore and maintain sinus rhythm; the preferred approach depends on symptoms, comorbidities and the likelihood of success.

The lecturer described rate control and anticoagulation as work that can often be started and followed in primary care, within competence and the local clinical pathway.

Refer for a rhythm-control strategy because antiarrhythmic selection and procedures can be more complex. Stroke prevention still requires assessment whichever rhythm strategy is chosen.

Consultation or Referral

- Cardiology
  - If available Electrophysiology/Arrhythmia specialist
  - Rhythm control strategy
    - If fails medical management for rate control, is patient’s preference or other factors determine this is preferred
    - Pill in pocket antiarrhythmic drug therapy
    - Catheter ablation

Lecture notes

Refer to cardiology, or directly to electrophysiology/arrhythmia services where available, for rhythm-control planning, unsuccessful rate management or procedural options such as ablation.

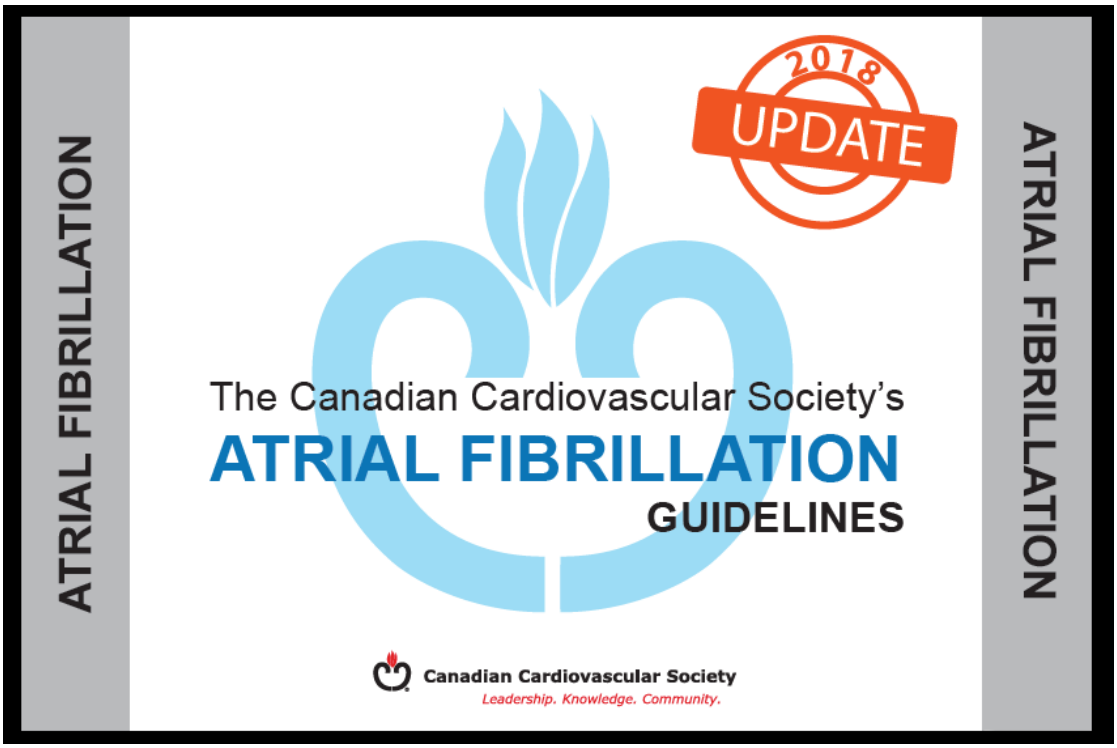
The patient's wish to pursue sinus rhythm and whether their other cardiac conditions would benefit from it also inform referral. Service availability and referral routes differ by location.



Slide reference

This CCS visual summarizes AF management themes. Use it as a map connecting risk-factor management, recognition, stroke prevention and specialist treatment options.

The lecturer did not separately explain every item in the infographic. The substantive discussion is captured under the corresponding slides above.



Related lecture questions

Before cardioversion, establish symptom onset, the reliability of that timing, and anticoagulation history. The lecturer warned against restoring rhythm when an atrial thrombus may have formed; some patients cannot reliably recognize when AF began.

For warfarin, she discussed obtaining a baseline INR and noted that mechanical-valve targets depend on valve type and position. Do not apply an AF INR target to every mechanical valve.

Clarification: The spoken '24 hours' and 'anytime if anticoagulated' rules need qualification. Cardioversion planning depends on stability, AF duration, stroke risk, adequate therapeutic anticoagulation and/or imaging to exclude thrombus. Canadian guidance uses risk-stratified timing; ESC 2024 uses a 24-hour threshold. Appropriate anticoagulation before and after the procedure remains important. [3,4]

Atrioventicular (AV) Blocks

Slide reference

The lecturer retained the AV-block section because conduction disease is relevant to syncope, but did not teach these slides in detail during this recording.

The following notes are concise summaries of the supplied slides, not reconstructed lecture dialogue.

Atrioventicular (AV) Blocks

- Result from interference in the conduction pathway between atria and ventricles
- AV Blocks are the most common cause of bradycardia
- Block can occur at the AV node, Bundle of His or bundle branches
- Block at the bundle branches may result in a wider QRS complex
- Can be transient or permanent
- Classifications:
  - First Degree
  - Second Degree (type I and type II)
  - Third Degree

Slide reference

AV block is impaired conduction between atria and ventricles. The site and severity of block affect which impulses conduct, the ventricular rate and sometimes QRS width.

Organize recognition into first degree, second degree Mobitz I or II, and complete block. Assess atrial activity separately from ventricular activity.

First Degree AV Block

- Delay of the electrical impulse between the atria and ventricle
- PRI is greater than 0.20 seconds
- First degree AV block is not considered a dysrhythmia
- Medications such as digitalis, beta-blockers and calcium channel blockers are usually the cause
- The sinus impulse is conducted normally to the AV node but is delayed longer than usual before being conducted to the ventricles

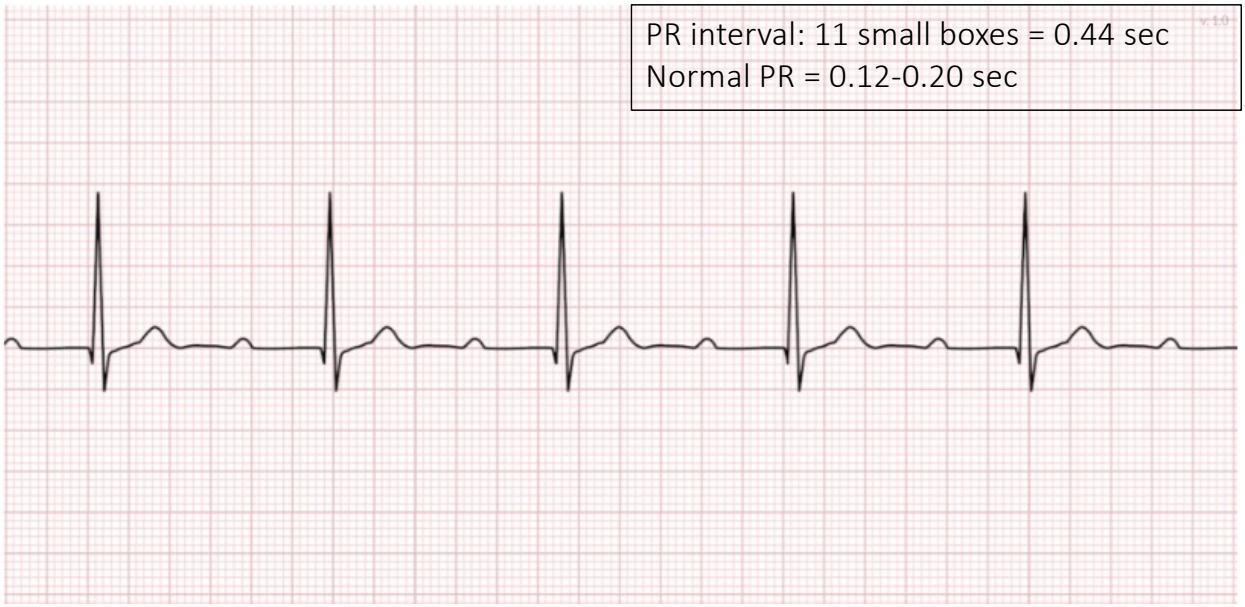
  - **Rate:** That of underlying rhythm, atrial = ventricular
  - **Rhythm:** Regular
  - **P Waves:** Upright and uniform
  - **P/QRS:** 1:1
  - **PRI:** Prolonged (> 0.20 seconds)
  - **QRS:** Normal, <0.12 seconds

Slide reference

First-degree AV block has a prolonged PR interval above 0.20 seconds with all atrial impulses conducted: P:QRS remains 1:1.

The slide lists AV-nodal-slowing medications among possible contributors. Identify the underlying rhythm and review the clinical and medication context rather than interpreting PR prolongation in isolation.

First Degree AV Block



Slide reference

The example labels PR as 11 small squares: 11 multiplied by 0.04 seconds equals 0.44 seconds. This is longer than the teaching range of 0.12-0.20 seconds.

The defining visual relationship is prolonged atrioventricular conduction with a conducted QRS after every P, rather than dropped ventricular complexes.

Second Degree AV Block – Type 1 and Type 2

- Occurs when the electrical impulse from the SA node is delayed or intermittently blocked
- ECG will show more P waves than QRS complexes
- Key to determining type of second degree block is the PR interval
  - Type 1 has **increasing** PR intervals, increasing until the QRS is "dropped" or missing
  - Type 2 has **constant** PR intervals, with randomly dropped QRS complexes
- Type 2 heart block is a permanent condition and can sometimes progress to third degree heart block
- This is an important contrast to Type 1 which is usually temporary and rarely progresses
- Type 2 to 3’ - patient has a high probability of needing a pacemaker
- Also referred to as Mobitz Type 1 or Wenkebach and Mobitz Type 2

Slide reference

In second-degree block, some P waves are not followed by a QRS. Use the PR pattern to help distinguish types.

Mobitz I: PR progressively lengthens before a nonconducted P. Mobitz II: conducted beats have a stable PR while some P waves do not conduct.

The slide highlights greater concern for progression in Mobitz II. Do not reduce assessment to 'temporary versus permanent'; identify the pattern and clinical risk.

Second Degree AV Block - Type 1

- **Rate:** atrial - that of underlying rhythm, ventricular - will depend on number of impulses conducted through AV node but will be less than atrial rate
- **Rhythm:** atrial - regular, ventricular - irregular
- **P Waves:** Sinus, normal size and shape
- **PRI:** Varies & progressively lengthens until a P wave occurs without a QRS
- **QRS:** Normal, <0.12 seconds

Slide reference

Mobitz I usually has a regular atrial rhythm with an irregular ventricular rhythm because an atrial impulse intermittently fails to conduct.

The key pattern is progressive PR lengthening followed by a P wave without a QRS. Ventricular rate is lower than atrial rate because fewer impulses reach the ventricles.

Second Degree AV Block - Type 1



Slide reference

On this tracing, mark each P wave, then compare the PR intervals of successive conducted beats. Find the nonconducted P and the repeating sequence.

This is a practice illustration; the lecturer did not give an individual walkthrough in the supplied recording.

Second Degree AV Block - Type I

Possible causes:

- Medications: digoxin, beta blockers, Ca channel blockers
- Increased vagal tone
- Myocarditis
- Ischemic heart disease
- Cardiac surgery

Signs & Symptoms:

- Presyncopal symptoms
  - Weakness
  - Dizziness
  - Lightneadedness
- Hemodynamic compromise

- Type 1 may occur in otherwise normal healthy children during sleep or in conditioned athletes

Slide reference

The slide lists medications, increased vagal tone, ischemia, inflammation and postoperative factors as possible contributors to Mobitz I. It can also occur during sleep or in trained athletes.

Assess symptoms and perfusion: weakness, dizziness or presyncope make the context different from an incidental asymptomatic finding.

Second Degree AV Block - Type 2

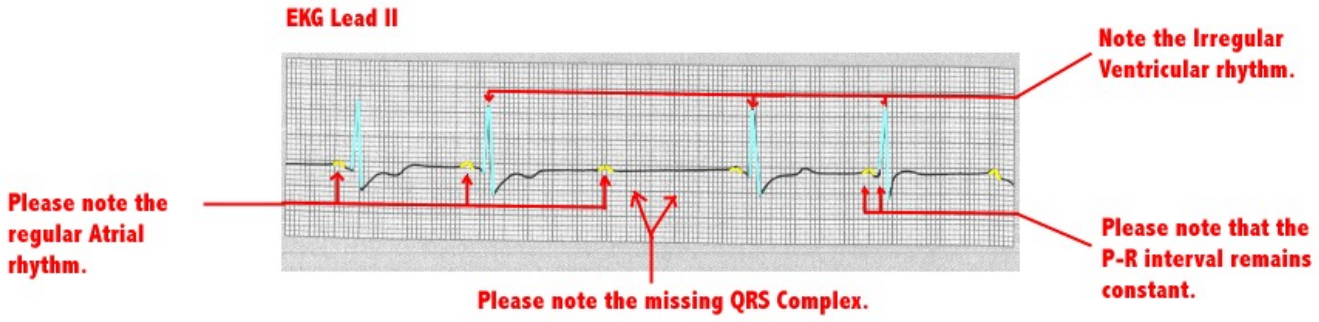
- A failure of some of the sinus impulses being conducted to the ventricles
- **Rate:** atrial - that of underlying rhythm (usually sinus), ventricular - depends on number of impulses conducted through AV node; less than atrial rate
- **Rhythm:** atrial - regular, ventricular - regular unless AV conduction ratio varies
- **P Waves:** Normal size and shape
- **P/QRS Ratio:** More P waves than QRS complexes
- **PRI:** May be normal or prolonged but remains constant
- **QRS:** Normal if block located in Bundle of His, wide if block located in bundle branches

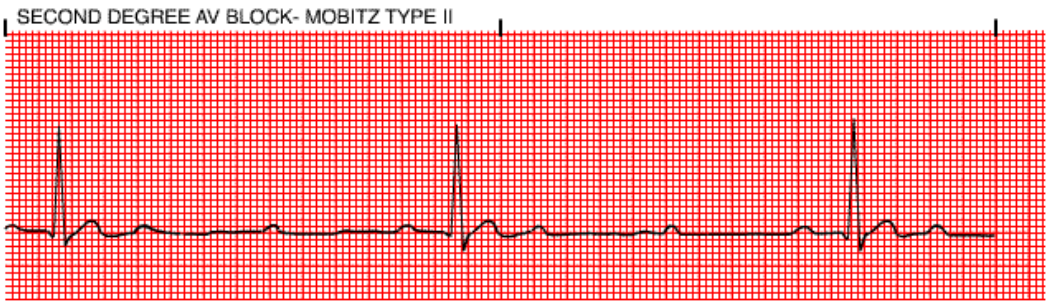
Slide reference

Mobitz II shows atrial impulses that fail to conduct despite a constant PR interval on the conducted beats. More P waves than QRS complexes are present.

The ventricular rhythm may be regular with a fixed conduction ratio or irregular with variable conduction. QRS width can vary with the level of conduction disease.

Second Degree AV Block - Type 2



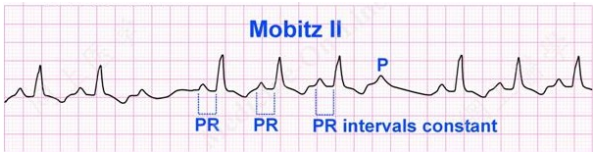


Slide reference

Practice recognizing nonconducted P waves while comparing the PR of the conducted beats. Unlike Mobitz I, do not look for a progressive PR staircase before the missed QRS.

No separate explanation of this particular tracing was recorded.

Second Degree AV Block - Type 2

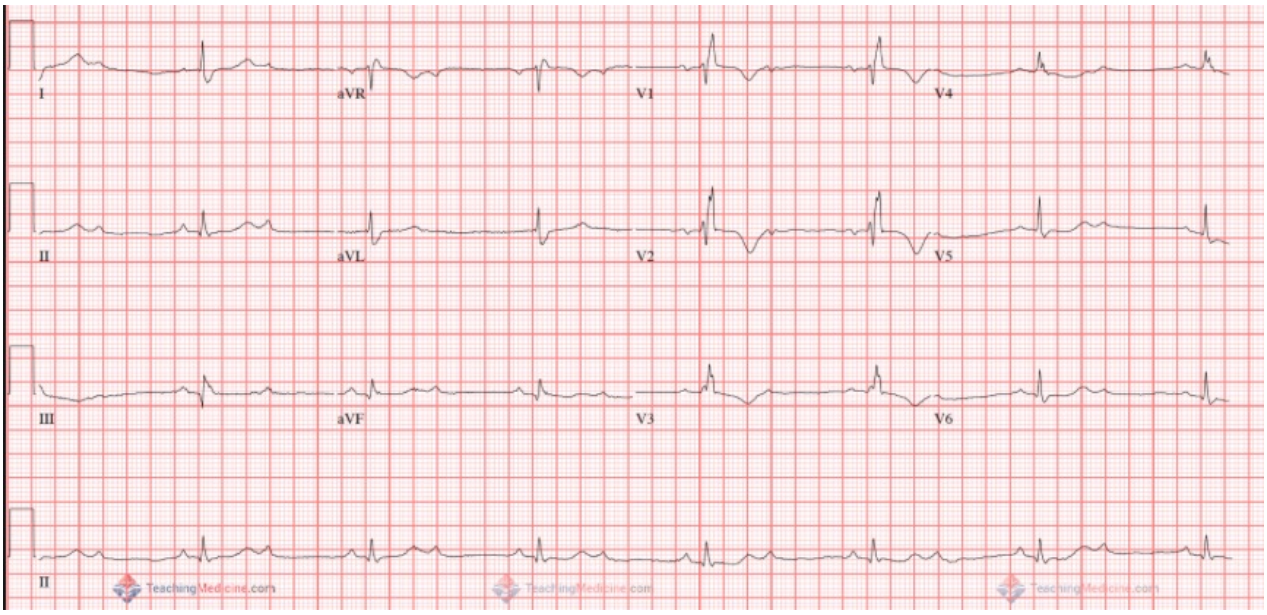


Slide reference

Use this additional example to count atrial and ventricular activity separately and examine the conduction ratio.

Interpret the entire pattern before assigning the block type; an isolated missed ventricular beat is not a complete description.

Second Degree AV Block - Type 2



Slide reference

This is another conduction-block practice tracing. Identify P waves, conducted QRS complexes and whether the conducted PR intervals are stable.

The supplied audio does not include the lecturer's answer-by-answer analysis of this example.

Second Degree AV Block, Type II

Possible causes:

- Post open-heart surgery where there is trauma at or below the Bundle of His
- Medications: digoxin, beta blockers, Ca channel blockers
- Increased vagal tone
- Myocarditis
- Ischemic heart disease
- Cardiac surgery
- Conduction system disease

Signs & Symptoms:

- Weakness
- Dizziness
- Syncope
- Hemodynamic compromise

Slide reference

Review the slide's possible contributors, including ischemia, conduction-system disease, medications and cardiac procedures, in the patient's actual context.

Weakness, dizziness, syncope and hemodynamic compromise are important symptoms. The later syncope discussion specifically identifies Mobitz II as a high-risk ECG finding requiring urgent assessment.

Third Degree or Complete Heart Block

- All electrical impulses from the atria are blocked before they reach the ventricle
- Atrial block could occur at the AV junction, Bundle of His, or the bundle branches
- P waves will be present and are consistent in shape and are regular in rhythm but will have no relationship to the QRS complexes
- Complete blocks that occur at the AV junction will be faster in rate and have a normal QRS duration
- Complete blocks that occur at the Bundle of His or the bundle branches will have a slower rate and a QRS duration >0.12 seconds

Slide reference

Complete heart block means atrial impulses do not conduct to the ventricles. P waves and QRS complexes continue independently, without a consistent PR relationship.

The escape rhythm's origin influences its rate and QRS width. A more distal ventricular escape is generally slower and broader than a junctional escape.

Third Degree or Complete Heart Block

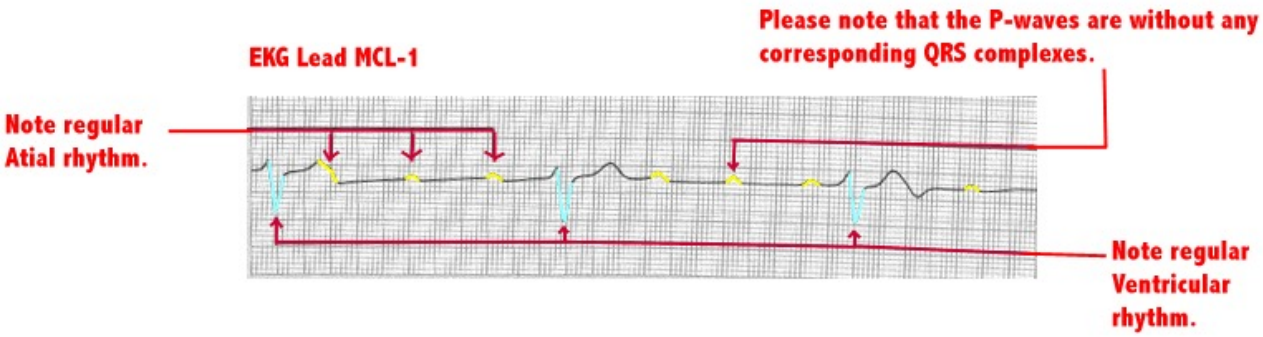
- A complete absence of conduction between the atria and ventricles
- The atria and ventricles beat independently of each other
- **Rate:** atrial - that of underlying rhythm (usually sinus), ventricular - usually slower than normal depending on rhythm
- **Rhythm:** atrial – regular, ventricular - regular
- **P Waves:** Sinus P waves
- **PRI:** P waves have no relationship to QRS
- **QRS:** Depends on origin
    - <0.12 seconds for junctional
    - >0.12 seconds for ventricular

Slide reference

Assess P-P regularity and R-R regularity separately: each may be regular while the two are unrelated. This atrioventricular dissociation is the central teaching feature here.

Do not force a PR measurement when there is no consistent relationship. Determine the ventricular rate and assess the patient's perfusion.

Third Degree or Complete Heart Block



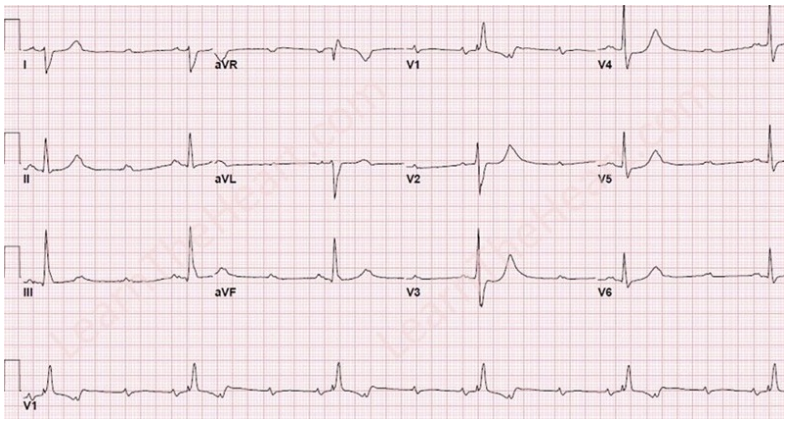


Slide reference

On the tracing, follow the P waves across the strip without using QRS position as their reference, then follow the ventricular complexes separately.

Look for the lack of a repeated P-to-QRS relationship. No separate live interpretation of this image was recorded.

Third Degree or Complete Heart Block



Slide reference

This additional complete-block example allows practice identifying atrial and ventricular independence and the escape rhythm's appearance.

Compare it with the first- and second-degree examples rather than treating all slow rhythms as sinus bradycardia.

Third Degree AV Block or Complete Heart Block

Possible causes:

- Congenital
- CV surgery
- Medications: digitalis, tricyclic antidepressants
- Acute coronary syndrome
- Post MI

Signs & Symptoms:

- Asymptomatic to severe hemodynamic compromise
- Dizziness
- Syncope
- Weakness
- Chest pain

Slide reference

The slide lists congenital disease, cardiac surgery, medication effects and ischemic events among possible settings for complete block. Presentation can range from few symptoms to marked hemodynamic compromise.

Dizziness, syncope, weakness and chest pain require attention. In the later syncope section, complete block is explicitly treated as a high-risk finding.

Syncope… Not For The Faint of Heart

- clinical syndrome in which transient loss of consciousness (TLOC) is caused by a period of inadequate cerebral blood flow and oxygenation, most often the result of an abrupt drop of systemic blood pressure
  - the inadequate cerebral nutrient flow is of relatively brief duration, and, by definition, syncope is self-limited
- term "syncope" is often misapplied to encompass other forms of abrupt collapse such as seizures or accidents, which may or may not be accompanied by TLOC
- syncope is a transient, self-limited loss of consciousness caused by transient, self-terminating, inadequate nutrient flow to the brain
  - syncopal episodes may or may not be preceded by prodromal symptoms
  - presyncope (or near-syncope) is a clinical manifestation that is suggestive of an impending syncope often occurring in conjunction with a similar set of prodromal symptoms reflecting the same conditions that might lead to syncope in the same individual at other times



Lecture notes

Syncope is a transient, self-limited loss of consciousness caused by inadequate cerebral perfusion, often with a fall in systemic blood pressure. Consciousness returns when adequate perfusion is restored.

Patients may use 'passed out' to describe a seizure, fall or other collapse, so establish whether there was an actual transient loss of consciousness and what happened around it.

Presyncope refers to warning symptoms suggesting an impending faint, without necessarily losing consciousness. Syncope may occur with or without a prodrome; that distinction helps assess cause.

Clinical features of syncope that suggest a cause

Neurally mediated syncope:
Absence of heart disease
Long history of recurrent syncope
After sudden unexpected unpleasant sight, sound, smell, or pain
Prolonged standing or crowded, hot places
Nausea, vomiting associated with syncope
During a meal or postprandial
With head rotation or pressure on carotid sinus (as in tumors, shaving, tight collars)
After exertion
Syncope due to OH:
After standing up
Temporal relationship with start or changes of dose of vasodepressive drugs leading to hypotension
Prolonged standing, especially in crowded, hot places
Presence of autonomic neuropathy or Parkinsonism
Standing after exertion
Cardiovascular syncope:
Presence of definite structural heart disease
Family history of unexplained sudden death or channelopathy
During exertion or supine
Abnormal ECG
Sudden onset palpitation immediately followed by syncope
ECG findings suggesting arrhythmic syncope:
▪ Bifascicular block (defined as either LBBB or RBBB combined with left anterior or left posterior fascicular block)
▪ Other intraventricluar conduction abnormalities (QRS duration ≥0.12 s)
▪ Mobitz I second-degree AV block
▪ Asymptomatic inappropriate sinus bradycardia (<50 bpm), sinoatrial block or sinus pause ≥3 s in the absence of negatively chronotropic medications
▪ Nonsustained VT
▪ Preexcited QRS complexes
▪ Long or short QT intervals
▪ Early repolarization
▪ RBBB pattern with ST elevation in leads V1 to V3 (Brugada syndrome)
▪ Negative T waves in right precordial leads, epsilon waves and ventricular late potentials suggestive of ARVC
▪ Q waves suggesting myocardial infarction

OH: orthostatic hypotension; ECG: electrocardiogram; LBBB: left bundle branch block; RBBB: right bundle branch block; AV: atrioventricular; bpm: beats per minute; VT: ventricular tachycardia; ARVC: arrhythmogenic right ventricular cardiomyopathy.

Reproduced with permission from: European Heart Rhythm Association (EHRA), Heart Failure Association (HFA), Heart Rhythm Society (HRS), et al. Guidelines for the diagnosis and management of syncope (version 2009): the Task Force for the Diagnosis and Management of Syncope of the European Society of Cardiology (ESC). Eur Heart J 2009; 30:2631. Copyright © 2009 Oxford University Press.

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Major cardiovascular causes of syncope

Reflex-mediated*
▪ Vasovagal • Orthostatic vasovagal syncope: usually after prolonged standing, frequently in a warm environment, etc • Emotional vasovagal syncope: secondary to fear, pain, medical procedure, etc • Unknown trigger
▪ Situational • Micturition, defecation • Swallowing • Coughing/sneezing
▪ Carotid sinus syndrome
Orthostatic hypotension*
▪ Medication-related • Diuretics (eg, thiazide or loop diuretics) • Vasodilators (eg, dihydropyridine calcium channel blockers, nitrates, alpha blockers, etc) • Antidepressants (eg, tricyclic drugs, SSRIs, etc)
▪ Volume depletion • Hemorrhage • Gastrointestinal losses (ie, vomiting or diarrhea) • Diminished thirst drive (primarily in older patients)
▪ Autonomic failure • Primary: pure autonomic failure, Parkinson disease, multiple system atrophy, Lewy body dementia • Secondary: diabetes mellitus, amyloidosis, spinal cord injuries, autoimmune neuropathy (eg, Guillain-Barré), paraneoplastic neuropathy
Cardiac
▪ Tachyarrhythmias • Ventricular tachycardia • Supraventricular tachycardias
▪ Bradyarrhythmias (with inadequate ventricular response) • Sinus node dysfunction • Atrioventricular block
▪ Structural disease • Severe aortic stenosis • Hypertrophic cardiomyopathy • Cardiac tamponade • Prosthetic valve dysfunction • Congenital coronary anomalies • Cardiac masses and tumors (eg, atrial myxoma)
▪ Cardiopulmonary/vascular • Pulmonary embolus • Severe pulmonary hypertension • Aortic dissection

SSRI: selective serotonin reuptake inhibitor.

* Reflex-mediated syncope and syncope due to orthostatic hypotension are more likely to occur, or are more severe, when other factors may also be contributing, such as medication(s) causing low blood pressure, volume depletion, pulmonary diseases causing reduction in brain oxygen supply, alcohol use, and/or environmental factors (excessive heat or humidity).

Adapted from: Brignole M, Moya A, de Lange FJ, et al. 2018 ESC Guidelines for the diagnosis and management of syncope. Eur Heart J 2018; 39:1883.

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Lecture notes

Use the figure to organize possible causes and the clinical features that point toward cardiac syncope. The lecturer's focus was recognizing cardiovascular causes within the much wider differential.

The detailed history on the next slides matters because the patient can have an entirely normal examination after the event has resolved.

Syncope
Key History Taking Considerations

- history taking is paramount in presyncope/syncope because physical exam findings are often benign
- **frequency and duration**
  - number and frequency of episodes - benign causes (such as vasovagal faint) is variable; single or very rare episodes, others will have multiple episodes over many years
  - patient with multiple episodes occurring over a short period of time (i.e. several days or weeks) is more likely to be suffering from a serious underlying disorder (likely cardiac) and requires more aggressive evaluation
  - span of time marked by episodes - the longer the period of time over which episodes have occurred (i.e., many years versus very recent onset) and the younger the patient's age at onset (particularly < 35 years), the less likely that the cause of syncope is life-threatening
- **triggers**
  - reflex syncope may be triggered by emotional or orthostatic stress, fear, or intense pain, or by a warm and/or crowded environment; may not be possible to identify the trigger
  - situational syncope (a type of reflex syncope) occurs during or immediately after certain apparent triggers such as urination, defecation, coughing, swallowing, or after eating a meal

Lecture notes

Establish the number of episodes, their frequency, the time span over which they have occurred and age at first onset. New clustered events over days or weeks are more concerning than a long, stable pattern beginning when young.

Ask about warning symptoms and triggers. Heat, prolonged standing, emotional distress, pain, urination, defecation or coughing can support reflex or situational syncope in context.

Abrupt collapse without warning raises suspicion for arrhythmia, especially with an injury suggesting no attempt to protect against the fall. The lecturer contrasted a gradual hot, nauseated, dizzy faint with being well one moment and on the ground the next.

Syncope

- **patient position**
  - assessing the patient's position (i.e., supine, sitting, or standing) at the time of syncope, along with any recent changes in position during the preceding few minutes prior to syncope, provides clues to the etiology
  - **while in an unprotected position** (i.e., prolonged standing) - reflex syncope most commonly occurs when the patient is upright (standing or seated) and almost never when supine
  - syncope resulting from orthostatic hypotension is frequently associated with a change from a supine to erect posture, although several minutes may pass between the patient arising and the subsequent collapse.
  - **while supine** - syncope that occurs when the patient is supine or recumbent suggests a cardiac arrhythmia
    - even if the arrhythmia persists, the TLOC is often brief given the body's adaptive hemodynamic compensatory responses (i.e., vascular constriction).
  - **relation to exercise** – timing of syncope in relation to exercise is very important, as syncope during full-flight exercise may be indicative of a serious condition (i.e., exercise-triggered tachyarrhythmia or hypotension, aortic stenosis, hypertrophic cardiomyopathy)
    - syncope immediately after exercise tends to be more innocent and reflex in origin

Lecture notes

Clarify posture and recent position changes: prolonged standing, rising from bed, sitting, or lying down can suggest different mechanisms. The lecturer was especially concerned by abrupt unexplained loss of consciousness while seated; supine events also raise concern for arrhythmia.

Distinguish syncope during exercise from symptoms immediately after stopping. Collapse during exertion raises concern for a serious cardiac cause, including arrhythmia or structural obstruction.

No single position establishes the diagnosis. Combine it with prodrome, trigger, associated symptoms and cardiac background.

Syncope

- **onset/prodrome**
  - symptoms preceding syncope can point toward a specific cause
  - warning premonition period of at least a few seconds or longer prior to losing consciousness
- **witnessed signs**
  - manner in which the collapse occurred, appearance of the patient, duration of the loss of consciousness, alterations in the patient's breathing pattern, any physical movements
- **associated symptoms**
  - patients should be asked about experiencing any of the following during the event or shortly after recovery: confusion, fatigue, injury, nausea, vomiting, feeling cold or clammy, palpitations, shortness of breath, chest pain, bladder or bowel incontinence
  - persistence of nausea, pallor, and diaphoresis in addition to prolonged fatigue (lasting minutes to hours) after a syncope episode suggest a reflex event (particularly a vasovagal episode); these findings are helpful in distinguishing reflex syncope from syncope due to an arrhythmia.
  - neurologic changes or confusion during the recovery period may suggest a stroke or seizure
- **past medical/preexisting conditions**
- **drugs and alcohol**
- **medications** – classes/categories and why?
- **family history** - sudden cardiac death, drowning

Lecture notes

Obtain a witness account when possible: the person who lost consciousness cannot describe the whole event. Ask about the collapse, duration, movements, breathing and recovery, as well as symptoms immediately before and after.

Nausea, pallor, sweating and prolonged fatigue can support a reflex episode. Prolonged confusion or focal neurological changes suggest a different process and need assessment.

The lecturer contrasted a prolonged washed-out feeling after a vasovagal faint with rapid recovery after an arrhythmia terminates. Recovery pattern is a clue, not a diagnosis by itself.

Review medical conditions, medications, alcohol and other substances, and family sudden death or unexplained drowning.

Syncope

- Physical Exam
  - Full cardiac and neurological
  - Orthostatic blood pressure
  - Often will have normal physical exam findings
- Diagnostics
  - ECG
  - Echocardiogram
  - Holter monitor
  - CT – head
  - EEG
- Focused Clinical Practice Update
  - Canadian Cardiovascular Society Clinical Practice Update on the Assessment and Management of Syncope (2020)
  - https://onlinecjc.ca/action/showPdf?pii=S0828-282X%2819%2931549-1

Lecture notes

Perform cardiac and neurological examinations and orthostatic vital signs. A normal examination after recovery does not exclude an important cause.

Choose investigations from the history and suspected mechanism. ECG and, when indicated, echo or ambulatory monitoring address possible cardiac causes. The lecturer pointed to the CCS syncope practice update as a risk-assessment reference.

The slide lists CT and EEG among possible investigations; it should not be read as a routine panel for every faint. Their role depends on evidence of neurological disease, seizure or injury.

Syncope

Table 2. High-risk electrocardiogram features

Feature	Description
Bradyarrhythmia	
Sinus node dysfunction	Asymptomatic inappropriate sinus rate < 50 bpm or slow AF (40-50 bpm), sinus block, sinus pause > 3 seconds in the absence of negatively chronotropic medications
Conduction disease	Bifascicular block
	Intraventricular conduction delay (QRS 120 ms)
	Second-degree AV block type 1 with prolonged PR interval
	Second-degree AV block type 2
	Third-degree AV block
Tachyarrhythmia	
Supraventricular	Ventricular pre-excitation
	Supraventricular tachycardia or AF
Ventricular tachycardia	Nonsustained ventricular tachycardia
	Evidence of acute ischemia or previous myocardial infarction
	Long (> 460 ms) QT on repetitive ECGs or short (< 340 ms) QT interval
	Type 1 Brugada
	Brugada pattern (RBBB with ST elevation V_1-V_3)
	Arrhythmogenic right ventricular cardiomyopathy features (negative T waves in right precordial leads, epsilon wave, ventricular late potentials)
	Ventricular hypertrophy

Table 1. Low- and high-risk predictors for risk stratification

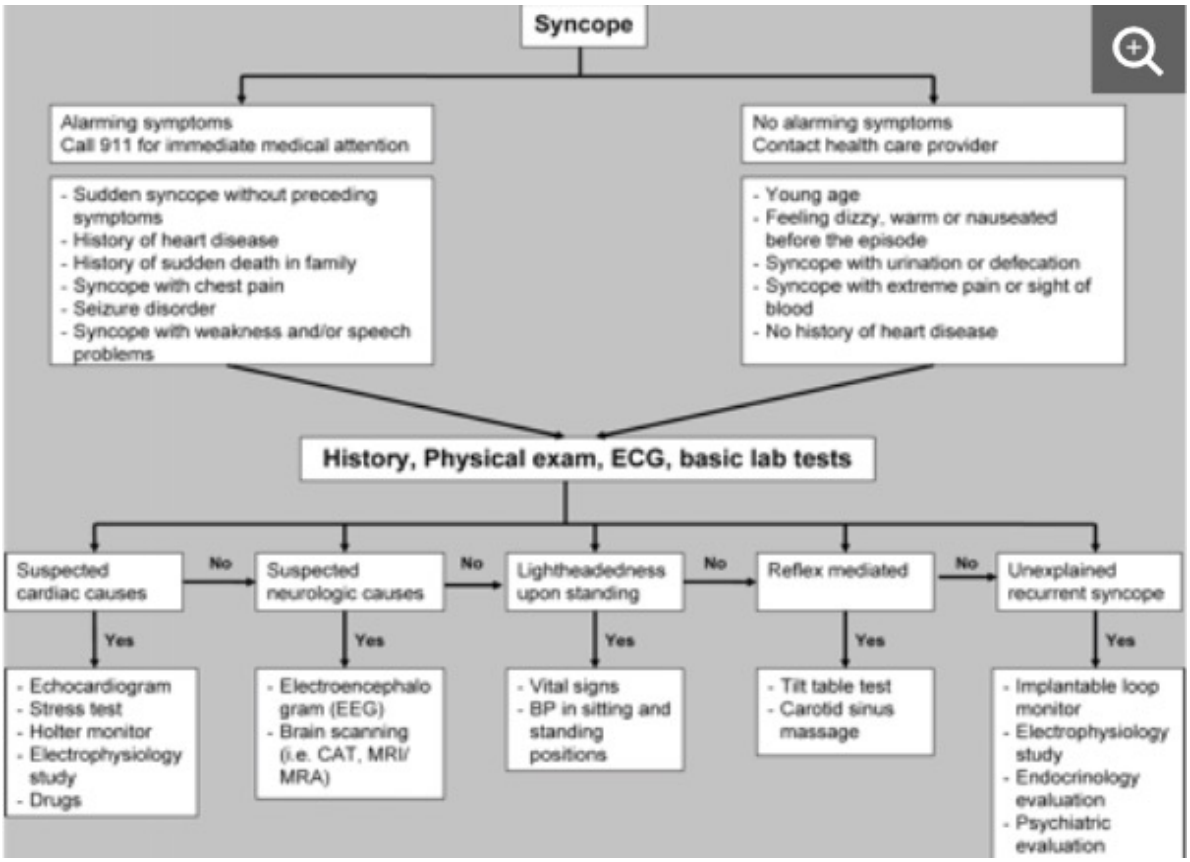
	Low-risk features	High-risk features
History[17-21,28-35]	Prodrome typical of reflex syncope	Symptoms suggestive of cardiac disease
	Triggers/specific situations typical of reflex syncope	Syncope during exertion or supine or without prodrome
	Positional syncope	History of cardiovascular disease (ischemia, arrhythmic, obstructive, valvular)
	Absence of cardiovascular disease	Concomitant trauma
		Family history of sudden cardiac death (age younger than 50 years)
Physical examination[17,30,36]	Normal	Abnormal vital signs
		Abnormal cardiac exam
12-Lead ECG[17,21,29,36]	Normal	Any bradyarrhythmia, tachyarrhythmia, or conduction disease
Laboratory values[37-40]	Normal	Elevated cardiac biomarkers or other relevant abnormal blood tests with a suspected related diagnosis

Lecture notes

The lecturer highlighted high-risk ECG findings in a patient with syncope: VT or runs of VT, significant rapid supraventricular rhythms, Mobitz II and complete heart block. These warrant urgent assessment rather than routine outpatient follow-up.

For a patient presenting with a dangerous rhythm and concerning symptoms in clinic, she described calling an ambulance for emergency transfer.

Combine the ECG with history and examination: known heart disease, exertional or supine syncope, absent prodrome, significant injury, abnormal vital signs and a concerning family history increase suspicion. A normal examination alone is not sufficient reassurance.



Lecture notes

Use the pathway to connect risk stratification with the next action. Low-risk features allow a more staged assessment for reflex or other benign causes; alarming features require a different urgency.

Order tests to answer the suspected mechanism rather than working through every box. The lecturer emphasized that primary care can perform substantial initial assessment when the presentation is genuinely low risk.

Ventricular Rhythms

Lecture notes

Ventricular tachycardia was identified as a rhythm students need to recognize. The following examples connect ventricular ectopy with potentially sustained ventricular arrhythmias.

Separate rhythm recognition and emergency escalation from the specialist's longer-term arrhythmia treatment plan.

Ventricular Tachycardia

- three or more PVCs occur in a row and the ventricular rate exceeds 100 beats/minute
- extremely unstable rhythm when prolonged
- can occur in short, paroxysmal bursts lasting fewer than 30 seconds and causing few or no symptoms
- can be sustained, requiring immediate treatment to prevent death, even in patients initially able to maintain adequate cardiac output
- results from increased myocardial irritability
- conditions that can cause ventricular tachycardia include:
  - myocardial ischemia, infarction, coronary artery disease
  - valvular heart disease
  - heart failure (poor LV function)
  - Cardiomyopathy
  - electrolyte imbalances
  - drug toxicity
  - pro-arrhythmic effects of some anti-arrhythymics

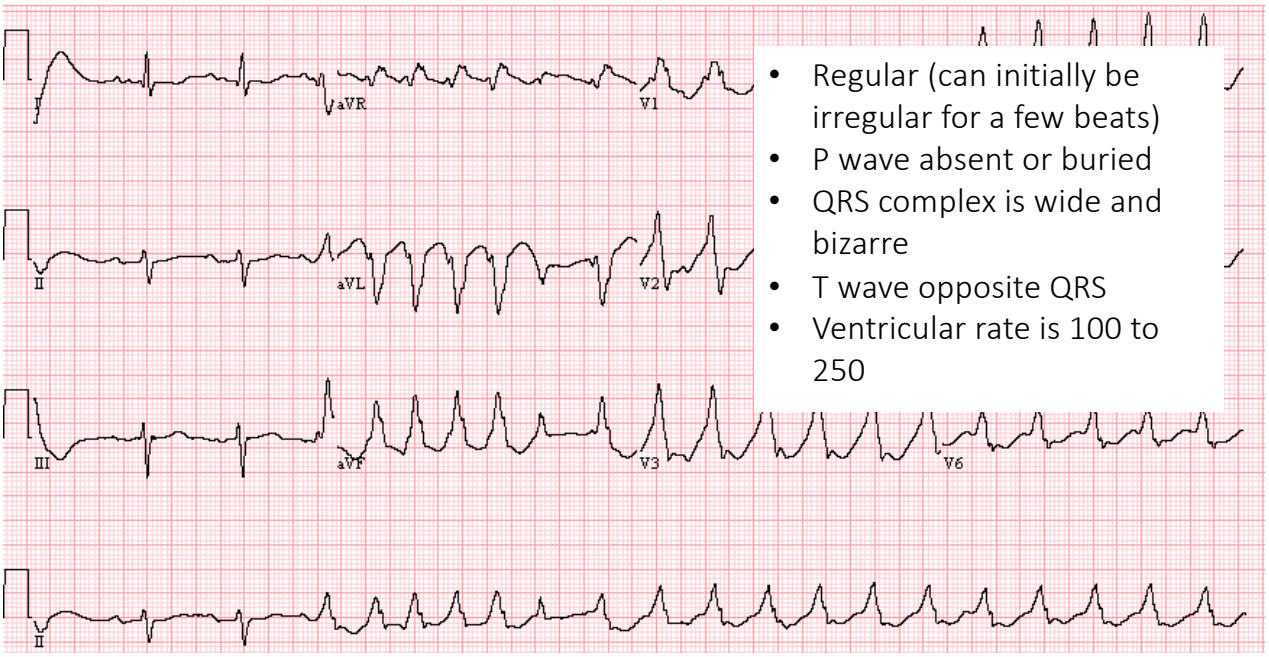
Lecture notes

The lecturer clarified the definition: three or more consecutive ventricular beats at a rate above 100/min are VT. This supersedes the earlier four-beat wording during the PVC discussion.

Nonsustained runs may last less than 30 seconds and produce few symptoms, particularly when cardiac function is otherwise good. Symptoms and risk are greater with underlying structural disease or reduced LV function; absence of symptoms does not make VT harmless.

Prolonged VT can become pulseless or deteriorate into VF. Consider ischemia, infarction, cardiomyopathy, valvular disease, impaired LV function, electrolyte disturbance, toxicity and proarrhythmic drugs as possible settings.

Ventricular Tachycardia



Lecture notes

Recognize the rapid, broad ventricular complexes. Monomorphic VT has a similar QRS appearance from beat to beat; polymorphic VT has changing morphology.

For an approximately regular example, the lecturer estimated rate using 300 divided by large boxes between complexes: two boxes is about 150/min; shorter intervals imply a higher rate.

Rate calculation supports the description but does not replace assessing the rhythm and the patient's circulation.

Ventricular Fibrillation

- Chaotic pattern of electrical activity in the ventricles in which electrical impulses arise from many different foci
- Produces no effective muscular contraction and no cardiac output
- Untreated ventricular fibrillation causes most cases of sudden cardiac death in people outside of a hospital
- Causes of ventricular fibrillation include:
  - myocardial ischemia
  - MI
  - untreated ventricular tachycardia
  - underlying heart disease
  - acid-base imbalance
  - electric shock
  - severe hypothermia
  - electrolyte imbalances, such as hypokalemia, hyperkalemia, and hypercalcemia
  - drug toxicity, including digoxin
  - severe hypoxia.
- Ventricles quiver instead of contract, so cardiac output falls to zero
  - if fibrillation continues, it leads to ventricular standstill and death

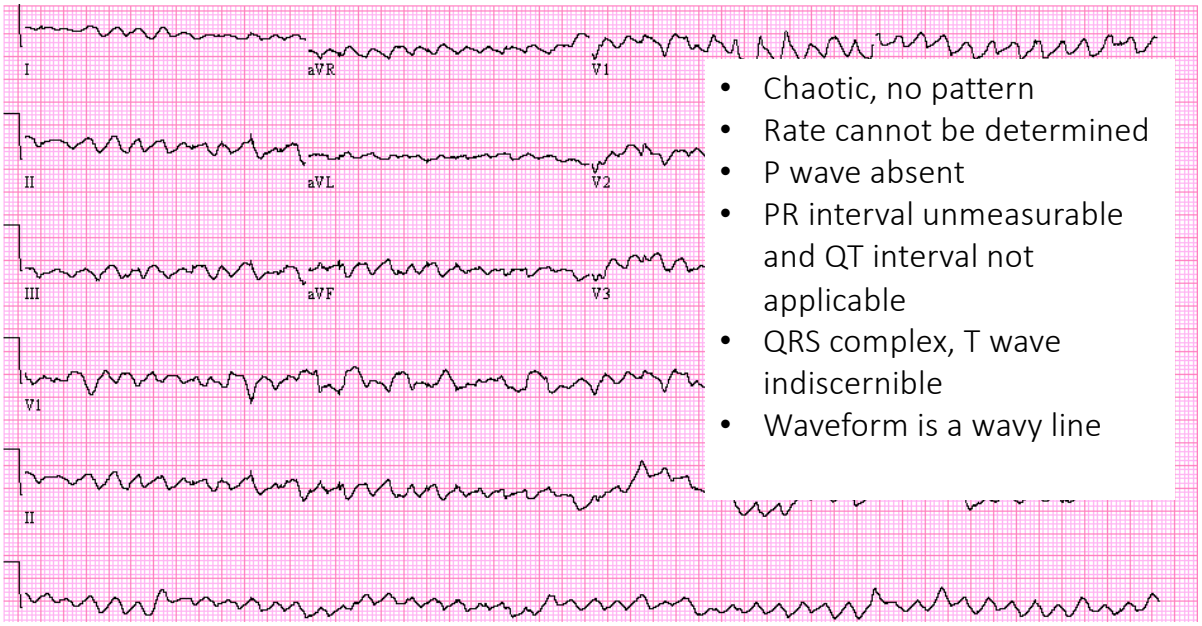
Lecture notes

VF is disorganized ventricular electrical activity without effective ventricular contraction or cardiac output. It is a cardiac-arrest rhythm.

The lecturer linked it to the same serious substrates discussed for VT, including ischemia, damaged myocardium, electrolyte or metabolic problems and toxicity. Prolonged VT can deteriorate into VF.

The key distinction from a perfusing rhythm is the absence of effective circulation, not simply a very fast rate.

Ventricular Fibrillation



Lecture notes and practice context

The waveform has no organized, identifiable P-QRS-T sequence and no meaningful ventricular rate to count.

The lecturer used a discharge-summary example: a familiar primary-care patient survives an out-of-hospital VF arrest and returns with new diagnoses, specialists and a recovery plan.

The NP may not manage the specialist arrhythmia treatment, but needs to understand what happened, its implications and their role in recovery and comorbidity management. Recognizing the terminology makes the discharge plan clinically meaningful.

Case Study 1

- Kenneth (he/him) is a 59-year-old nurse, who has had several episodes of chest pain for the past four months. When asked, he estimates having had five to six episodes of moderate chest tightness with exertion (e.g., while playing football with his sons). He states he usually rests when having an episode of chest pain and the pain goes away after about ten minutes. He denies any sharp or stabbing chest pain during exertion or chest pain aggravated by movement. He also denies having increased shortness of breath, palpitations or nausea.
  - PMH: Asthma x 10 years; GERD x 5 years
  - Medications: Budesonide 2 puffs BID, Salbutamol MDI 2 puffs QID PRN, Ranitidine 150 mg po BID
  - FMH: MI @ age 55 for his father; mother diabetes mellitus type 2 @ 63 yrs
  - Social History: Married, father of four, shift worker

    Lifestyle: Smoked ½ ppd x 20 years (quit when diagnosed with asthma), but recently restarted smoking 3-4 cigarettes per day, ETOH 3-4 beers/day
- The physical examination reveals:
  - Tall, pleasant man with truncal obesity
  - Ht 180 cm (5’10”); Wt 98 kg (215 lbs); BMI 30; waist circumference 118 cm
  - BP 142/85 (sitting), 148/86 (supine), HR 72 (reg), RR 18
  - CVS: S1 S2 normal, no S3 or S4, no murmurs, no peripheral edema, JVP 2 cm at 45°, no chest pain reproduced by palpation
  - Resp: General decreased air entry, no wheezes, no crackles

Lecture notes and case discussion

Kenneth's chest tightness occurs with substantial exertion, such as playing football, and resolves after rest. He is not describing sharp, movement-provoked pain or ongoing symptoms at the office visit.

Assess his risk background: paternal MI at 55, smoking resumed after quitting, approximately 10 prior pack-years, obesity, raised blood pressure and regular alcohol intake. His otherwise unremarkable examination does not establish whether coronary disease is present.

The lecturer used this history to introduce stable exertional symptoms, risk factors and the distinction from an active acute coronary presentation.

Case Study 1

- List Kenneth’s modifiable and non-modifiable risk factors for atherosclerosis.
- Which symptoms significantly increase the probability of an acute MI?
- What is the likelihood that Kenneth is having an MI based on his symptoms? Provide rationale.
- Would this be an appropriate case to consult? Explain.

Lecture notes and case discussion

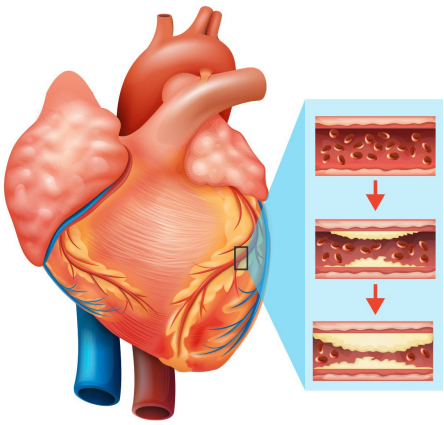
Classify Kenneth's risk factors into those he can modify and those he cannot. His history supports investigating coronary disease even though he currently has no chest pain.

The lecturer considered the predictable exertional-and-rest pattern more consistent with stable angina than an active MI at that visit. She contrasted this with a patient experiencing ongoing chest pain in the office, which changes urgency.

This is a case-based assessment, not a rule that being pain-free excludes ACS. Clarify recent changes, symptoms at rest, duration and associated features before deciding the pathway.

Coronary Artery Disease

- most common cause of cardiovascular morbidity and mortality
- in recent years, heart disease (which includes CAD/ ischemic heart disease among other cardiac conditions) has been the 2nd leading cause of death in Canada
- in 2023, there were 57,890 deaths due to “diseases of heart” in Canada; the percentage of all deaths due to heart disease was about 17.7%
- atherosclerosis is a systemic disease characterized by lipid deposition and smooth muscle cell migration in the intima of the larger arteries
  - starts in childhood with the development of fatty streaks
- risk factors
  - non-modifiable: age, male sex, family history, ethnicity (African, Hispanic, Latino, Southeast Asian)
  - modifiable: smoking, HTN, diabetes, dyslipidemia, obesity, sedentary lifestyle, diets high in saturated fats, drug/alcohol use



Lecture notes

Atherosclerosis is systemic: lipid deposition and inflammation affect arterial walls, and risk develops over time. Coronary disease was presented as a major cause of cardiovascular illness and death.

Nonmodifiable factors include age, family history and sex; the lecturer discussed ethnicity-related differences in risk. Premature coronary disease in a first-degree relative was described using male onset before 55 and female onset before 65.

Modifiable factors include smoking, hypertension, diabetes, dyslipidemia, obesity, inactivity, diet and substance use. Assess them because they inform both diagnostic probability and subsequent prevention.

Chronic Stable Angina

- typical
  1. retrosternal chest pain, tightness or discomfort radiating to left (± right) shoulder/arm/neck/jaw, associated with diaphoresis, nausea, anxiety
  2. predictably precipitated by the “3 Es”: exertion, emotion, eating
  3. brief duration, lasting <10-15 min and typically relieved by rest and nitrates
- atypical/probable angina (meets 2 of the above)
- non-cardiac chest pain (meets none or 1 of the above)
- anginal equivalents: dyspnea, acute LV failure, flash pulmonary edema

Lecture notes

Typical stable angina has a reproducible pattern: retrosternal tightness or discomfort, provoked by exertion, emotion or eating, and relieved with rest or prescribed nitrate. Episodes are generally brief, resolving within about 10-15 minutes in the teaching description.

Radiation can involve the arm, shoulder, neck or jaw; sweating, nausea or anxiety may accompany it. The lecturer emphasized predictability and the level of exertion needed to provoke symptoms.

Some patients have anginal equivalents rather than classic chest pain. Reproducible exertional dyspnea that resolves with rest may be the presenting complaint. Ask directly about associated tightness rather than assuming the problem is solely respiratory.

Angina/Chest Pain Differential Diagnoses

- Chest pain is a localized sensation of distress or discomfort that may or may not be associated with actual tissue damage
- Cardiac - Ischemia, atherosclerosis, MI, angina, myocarditis, pericarditis, dissecting aortic aneurysm
- Musculoskeletal - Muscle strain and inflammation caused by costochondritis, chest wall syndrome, cervicodorsal arthritis or intercostal myositis
- Neurologic – Nerve inflammation 2’ herpes zoster and nerve root compression
- Pulmonary - PE, pneumothorax/hemothorax, tension pneumothorax, pneumonia, empyema, pulmonary neoplasm, bronchiectasis, pleuritis, pulmonary hypertension, and TB
- Gastrointestinal - GERD, esophageal rupture, spasm, esophagitis, ulceration, achalasia, neoplasm, PUD, gastritis, pancreatitis, and biliary colic
- Mediastinal - Lymphoma, thymoma
- Vascular - Dissecting aortic aneurysm and aortic rupture
- Psychogenic – Stress caused by anxiety, depression, or panic disorder

Lecture notes

Chest discomfort has a wide differential: cardiac, musculoskeletal, neurological, pulmonary, gastrointestinal, vascular and psychological causes.

The lecturer emphasized the history as the main tool for narrowing these possibilities. Establish location, quality, duration, triggers, relief, associated symptoms and risk factors before anchoring on a familiar diagnosis such as reflux or anxiety.

The slide includes serious alternatives, including aortic disease, pulmonary embolism and pneumothorax; the clinical presentation determines which require urgent exclusion.

Canadian Cardiovascular Society (CCS) Functional Classification of Angina

Canadian Cardiovascular Society grading of angina pectoris

Grade	Description
Grade I	Ordinary physical activity does not cause angina, such as walking and climbing stairs. Angina with strenuous or rapid or prolonged exertion at work or recreation
Grade II	Slight limitation of ordinary activity. Walking or climbing stairs rapidly, walking uphill, walking or stair climbing after meals, or in cold, or in wind, or under emotional stress, or only during the few hours after awakening. Walking more than two blocks on the level and climbing more than one flight of ordinary stairs at a normal pace and in normal conditions
Grade III	Marked limitation of ordinary physical activity. Walking one or two blocks on the level and climbing one flight of stairs in normal conditions and at normal pace
Grade IV	Inability to carry on any physical activity without discomfort, anginal syndrome may be present at rest
References ***Campeau Lucien. Grading of angina pectoris. Circulation 1976;54:522-3*** ***Available on the Canadian Cardiovascular Society Website at www.ccs.ca***	

Lecture notes

Use the CCS angina class to communicate functional limitation. Class I means symptoms only with strenuous, rapid or prolonged exertion; II is slight limitation of ordinary activity; III is marked limitation; IV is inability to do physical activity without discomfort, with possible symptoms at rest.

In a consult letter, 'CCS class III' immediately conveys substantially more limitation than class I. The lecturer compared this shared language with the NYHA functional classification used for heart-failure symptoms.

Describe the actual activity threshold as well as the class, especially if it is changing.

Clinical Assessment & Diagnostics

- history including directed risk factor assessment and physical exam
  - physical exam will often be non-contributory
- labs: Hb, HbA1C/fasting glucose, fasting lipid profile
- ECG (at rest and during episode of chest pain if possible)
- CXR (suspected heart failure, valvular disease, pericardial disease, aortic dissection/aneurysm, or signs or symptoms of pulmonary disease)
- stress testing, nuclear medicine imaging (MIBI scan) or coronary angiography
- echocardiogram to assess for wall motion abnormalities, valvular disease, LV function in patients with Hx of prior MI, pathological Q waves, signs or symptoms of CHF

Lecture notes

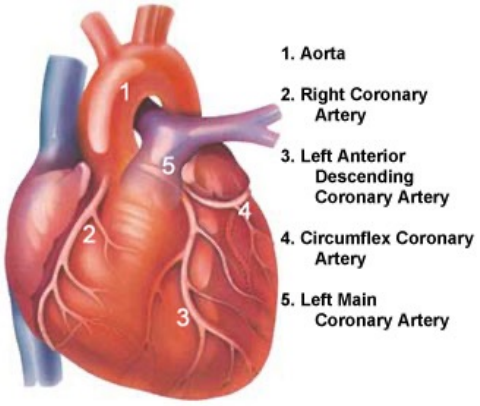
Take a detailed history and assess vascular risk factors; the physical examination may be normal. ECG is important, with additional tests selected for the question being asked.

The lecturer discussed bloodwork for risk factors, CXR when dyspnea or examination findings suggest heart failure or pulmonary disease, and echo for cardiac structure, function and regional movement.

Coronary investigations may include exercise testing, perfusion imaging or angiography, depending on the presentation. She stated that her NP ordering pathway required a physician for nuclear perfusion imaging; treat that as the local lecture context rather than a universal current ordering rule.

Acute Coronary Syndromes (ACS)

- Used to refer to the pathophysiologic continuum that begins with plaque rupture in a coronary artery and ultimately results in permanent cell damage (infarction) if the process is not arrested
- ACS continuum
  - Unstable angina (UA)
  - Non-ST elevation myocardial infarction (NSTEMI)
  - ST elevation myocardial infarction (STEMI)
- ECG changes consistent with ischemia
  - T wave flattening, inversion or peaked
    - can also be non-specific
  - ST depression
  - ST elevation



Lecture notes

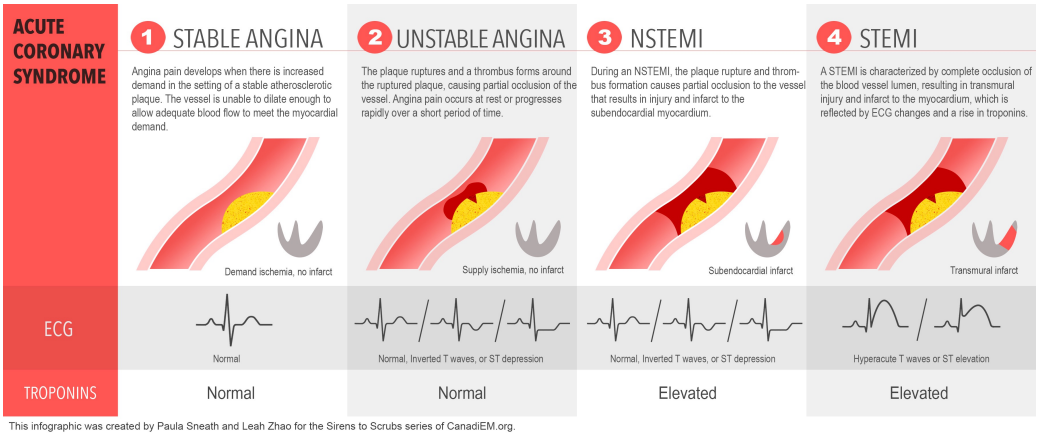
ACS comprises unstable angina, NSTEMI and STEMI. The lecturer described plaque rupture and thrombus formation leading to ischemia and, when injury is sufficient, myocardial necrosis.

Review ST segments and T waves across the relevant leads. The symptoms, ECG pattern, biomarkers and time course together determine the diagnosis and pathway.

Do not think of NSTEMI as automatically a minor event. A delayed presentation after days of symptoms can involve substantial injury. Prompt STEMI recognition and reperfusion may limit damage; both forms need timely attention.

The practical message was 'time is muscle': the eventual injury depends partly on how long tissue is deprived of adequate flow.

- Unstable angina
  - Change in pattern of angina symptoms, including prolonged or worsening angina and new onset of severe angina symptoms
  - non occlusive thrombus
  - non-specific on 12-Lead ECG, normal cardiac enzymes
- Myocardial infarction: occurs where a portion of heart muscle is deprived of oxygen due to blockage of a coronary artery
  - STEMI: Complete thrombus occlusion, ST elevation on 12-Lead ECG or new LBBB, elevated cardiac enzymes, high risk for cardiac arrest
  - NSTEMI: Non-occlusive thrombus sufficient to cause tissue damage and myocardial necrosis, ST depression +/- T wave, inversion on 12-Lead ECG, elevated cardiac enzymes



Lecture notes

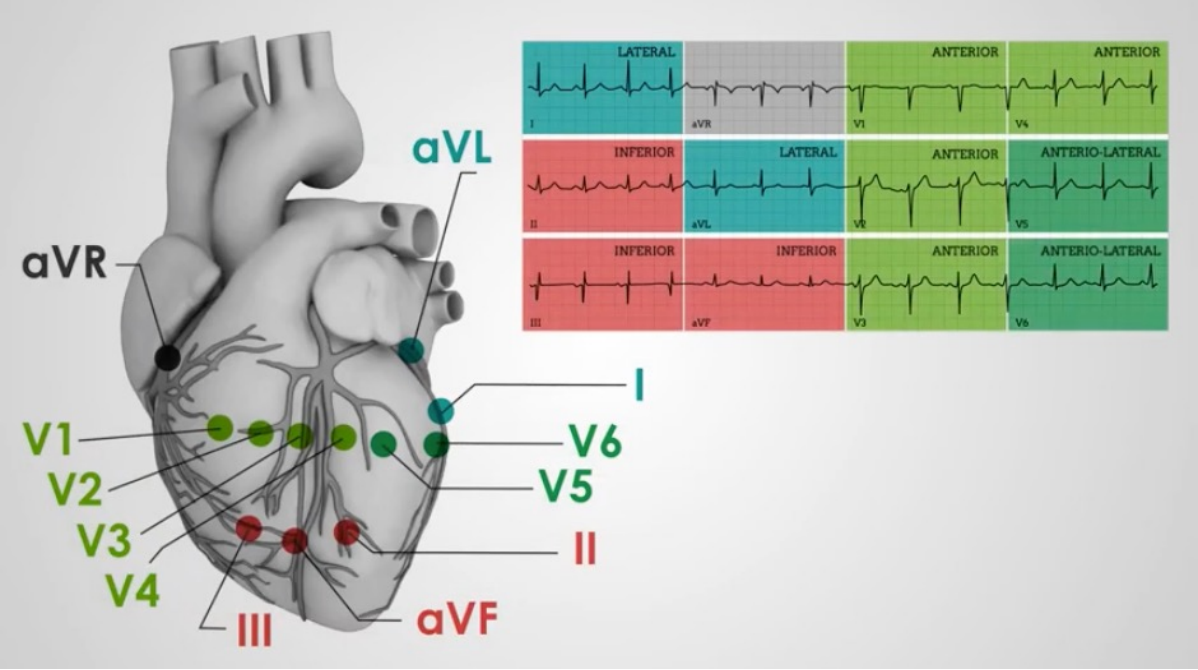
Stable angina is predictable with a consistent exertional threshold. Increasing duration, a lower activity threshold, slower relief or symptoms at rest suggest a changing pattern and possible instability.

The lecture distinguishes unstable angina, ischemia without infarction, from MI with myocardial injury. STEMI and NSTEMI describe important clinical ECG/biomarker presentations; prolonged occlusion can cause more extensive wall injury.

Clarification: The slide's 'new LBBB' wording should not be used as an automatic STEMI criterion. New LBBB alone is not a STEMI equivalent; assess for ischemia using the full presentation and appropriate ECG criteria. [5]

Clarification: Complete versus partial blockage is a simplified teaching model, not an infallible distinction between STEMI and NSTEMI. [5,6]

12 Lead ECG and Coronary Artery Territories

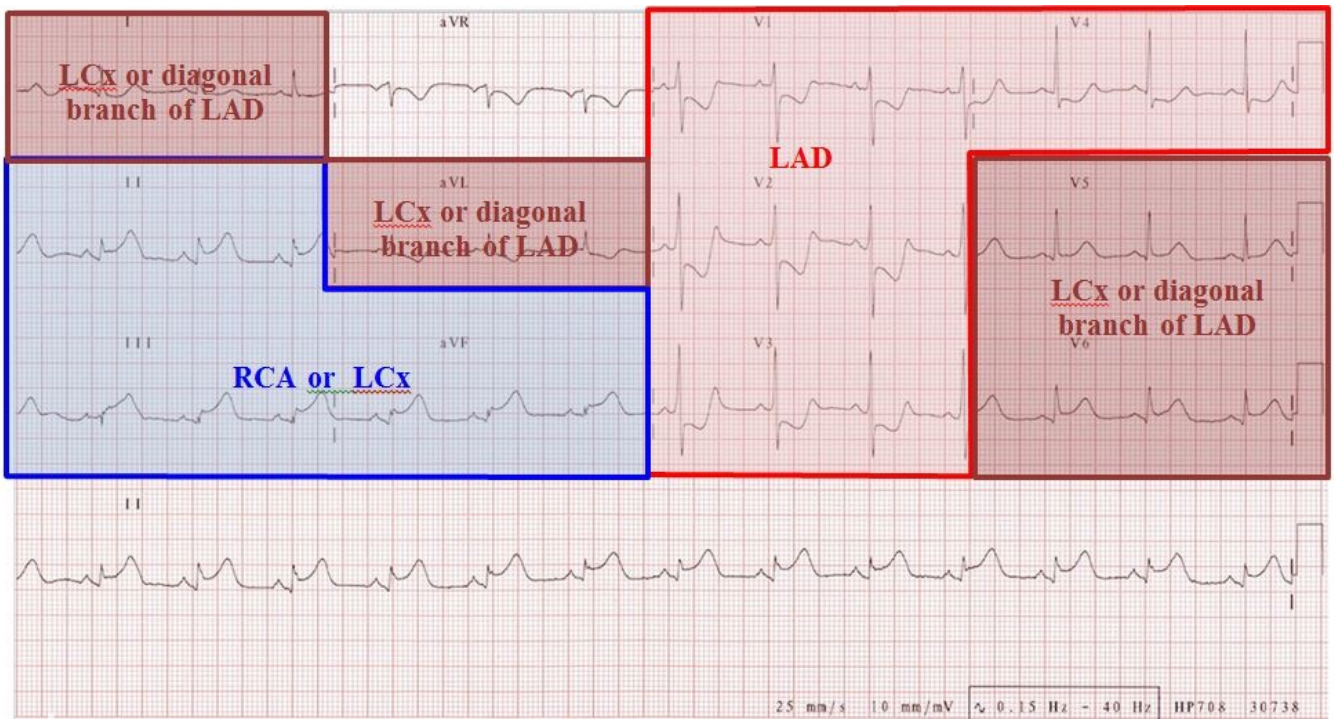


Lecture notes

Use groups of leads to localize the affected myocardial territory. The lecturer highlighted inferior leads II, III and aVF; lateral I and aVL; lateral chest leads V5-V6; and anterior/septal V1-V4.

Changes in V1-V4 raise concern for the LAD territory in the appropriate presentation. Inferior changes commonly relate to the RCA, although coronary anatomy varies, as the next diagram indicates.

Relate the lead pattern to symptoms and the whole ECG. The territory map supports interpretation; it is not a definitive angiogram of the patient's coronary anatomy.



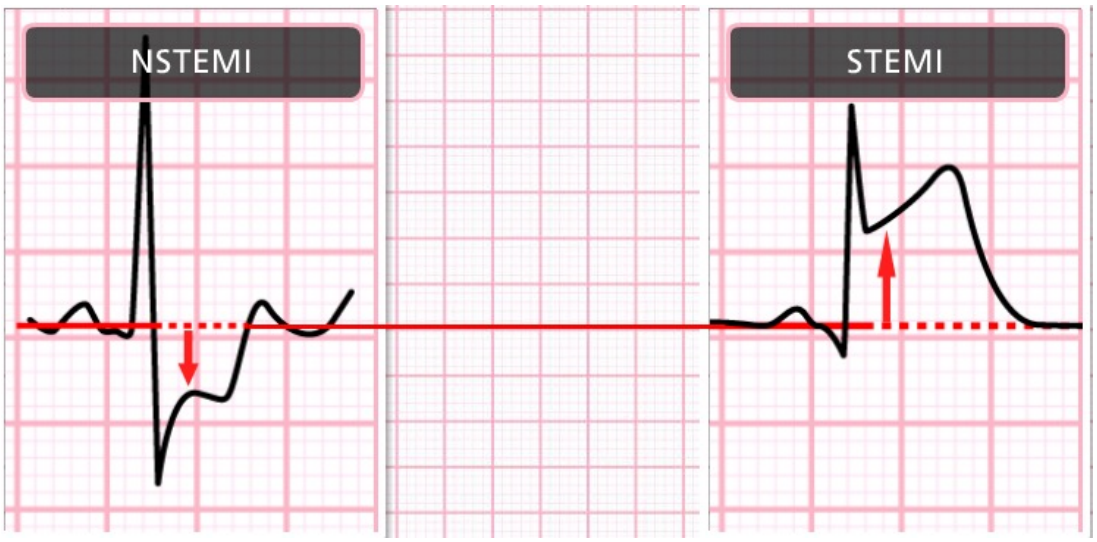
- ❖ T wave flattening, inversion or peaked
- ❖ ST depression
- ❖ ST elevation

Lecture notes

Look for ST elevation, ST depression and T-wave abnormalities within meaningful lead groups. The annotated map connects those groups to likely coronary supply.

The lecturer's example was ST elevation across V1-V4 suggesting an LAD-related anterior event. Use the figure to revise contiguous lead groups rather than memorizing an isolated lead change as a diagnosis.

STEMI – ST elevation myocardial infarction
NSTEMI – non-ST elevation myocardial infarction



Related lecture explanation

The illustration contrasts common ST-segment patterns used in the STEMI/NSTEMI discussion. It was not given a separate detailed walkthrough.

The lecturer emphasized that the amount of damage is affected by the duration and extent of impaired flow and how quickly treatment occurs. The absence of ST elevation does not imply that myocardial injury is small.

Q waves in ACS

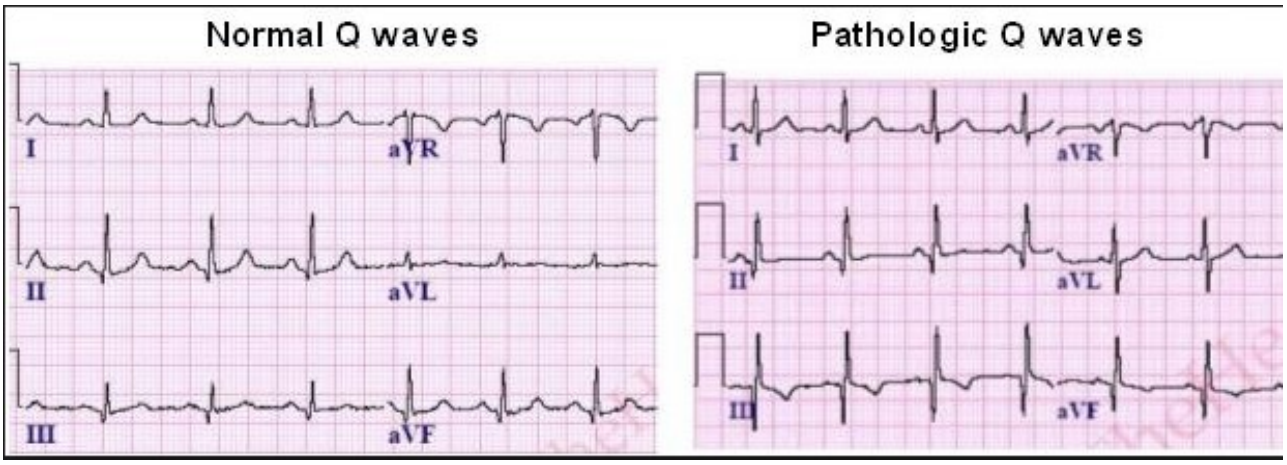
- Pathologic **Q waves** are a sign of previous **myocardial infarction**
- They are the result of absence of electrical activity
- A **myocardial infarction** can be thought of as an electrical 'hole' as scar tissue is electrically dead and therefore results in pathologic **Q waves**

Lecture notes

The lecturer described pathological Q waves as a possible clue to current or previous myocardial damage, including an infarction the patient may not recall.

If the ECG suggests a prior event, revisit the history and cardiovascular risk and consider further cardiac assessment rather than accepting an unfamiliar finding without explanation.

Clarification: Not every Q wave is abnormal, and even pathological Q waves are not exclusive to infarction. Small septal Q waves can be normal; other conditions can produce infarct-like patterns. Interpret morphology, lead distribution and the clinical context. [6]



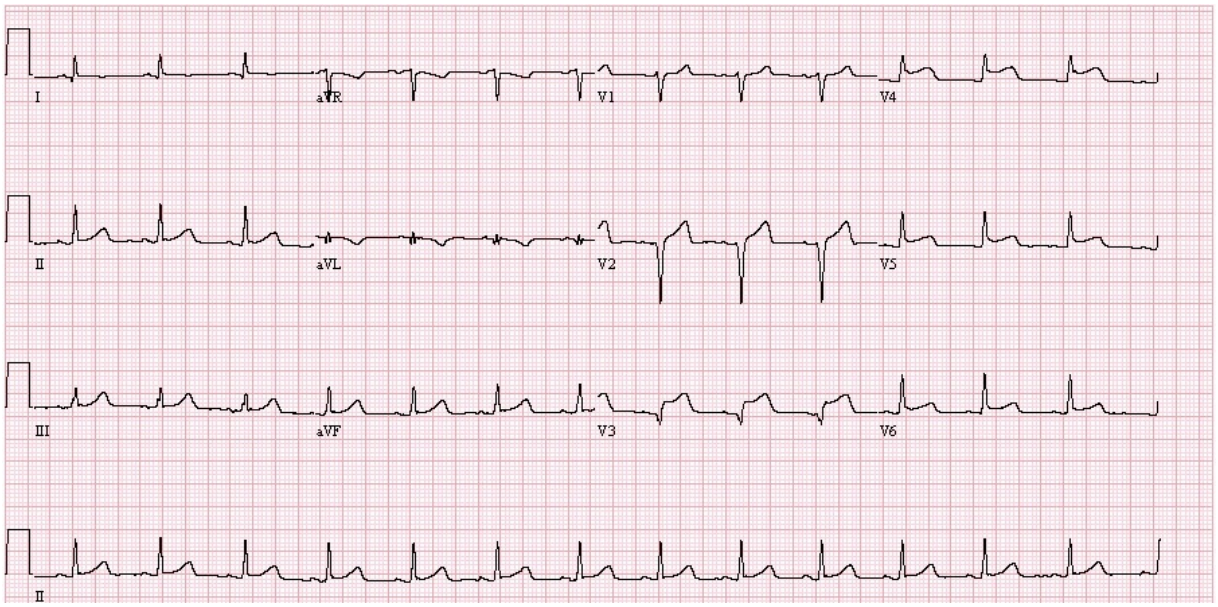
- Q waves are considered pathological if:
  - > 40 ms (1 mm) wide
  - > 2 mm deep
  - > 25% of depth of QRS complex
  - Seen in leads V1-3
- Pathological Q waves usually indicate current or prior myocardial infarction

Lecture notes

The slide lists width, depth and distribution features that raise concern for pathological Q waves. The lecturer's practical point was to recognize an unexpected potential marker of myocardial damage and investigate its context.

Clarification: The listed measurements are a simplified screen. Q-wave interpretation is lead-dependent and cannot be reduced to 'any Q wave means MI.' Use the qualification on slide 142 when reviewing this table. [6]

Where is the ischemia?

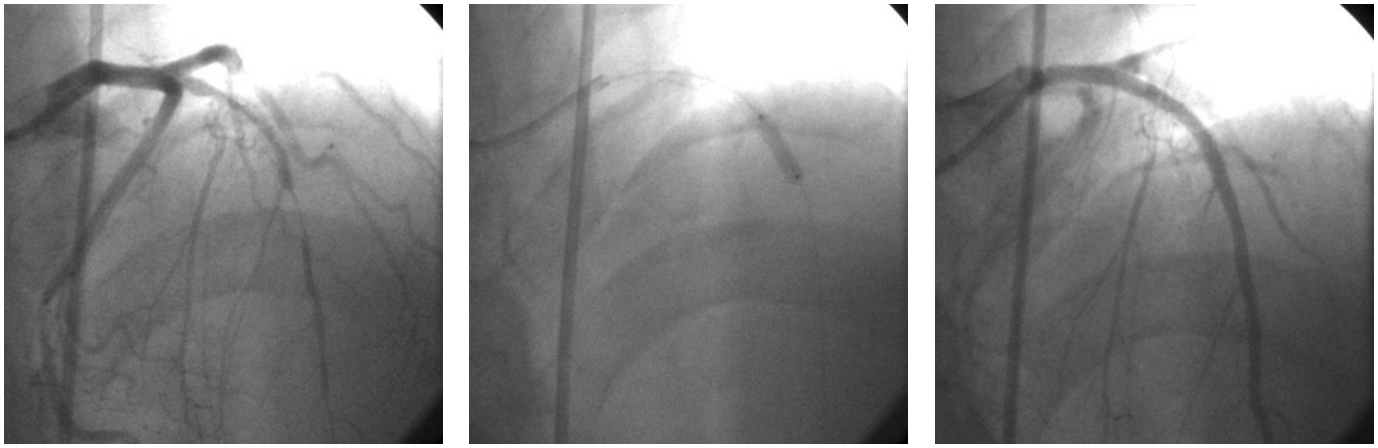


Slide reference

This ECG was left as a localization exercise rather than answered in detail in the recording. Identify the leads with ST-T abnormalities, group them anatomically and compare with slides 139-140.

An instructor-confirmed answer for this specific tracing is not available in the supplied audio.

Anterior STEMI
PCI to the LAD

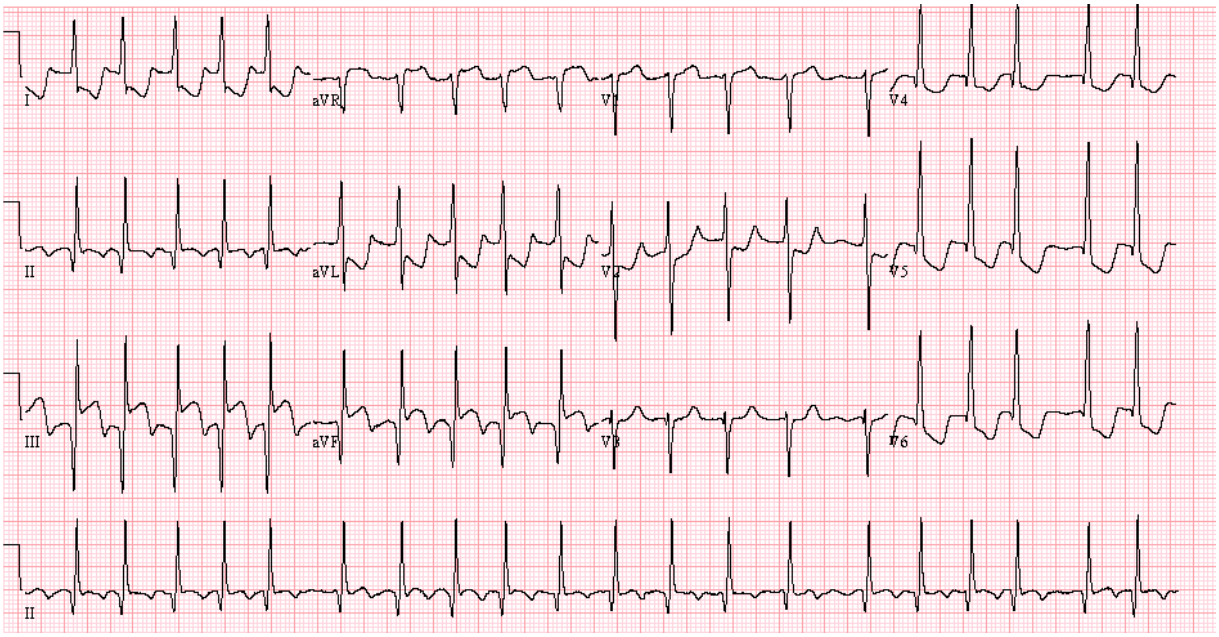


Related lecture explanation

The lecturer returned to angiographic images while discussing Mr. Hart. Compare coronary flow before intervention, the wire and stent placement, and flow after the stent is deployed.

She pointed out a large territory that was not filling before intervention and became visible afterward. The slide is labelled anterior STEMI with PCI to the LAD; it illustrates the anatomical target of reperfusion, rather than proving that Mr. Hart's own presentation was a STEMI.

Where is the ischemia?

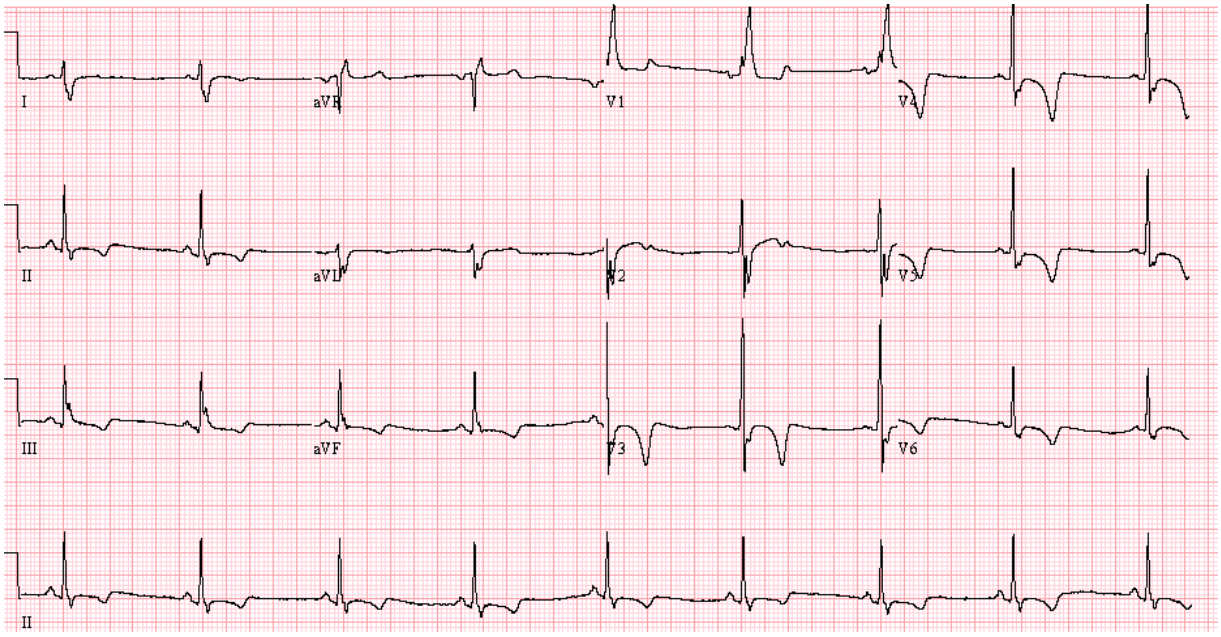


Slide reference

Independent ECG localization exercise. Assess the whole tracing, identify the main abnormal lead groups and describe the associated territory using the map on slides 139-140.

No specific verbal answer was given for this image in the recording.

Where is the ischemia?

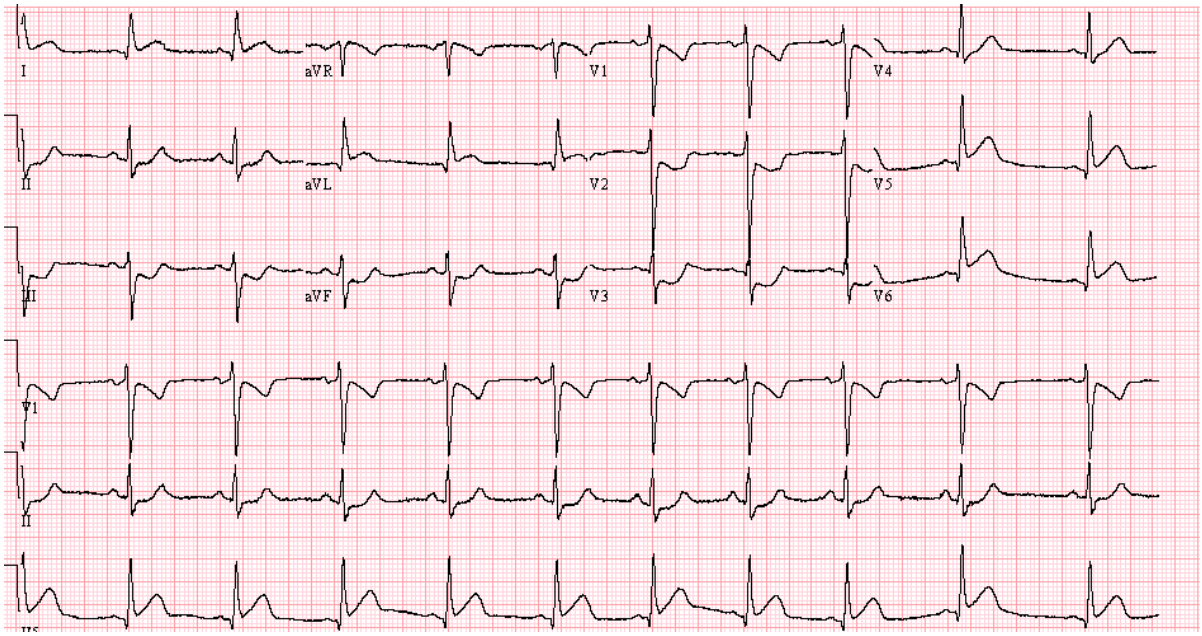


Slide reference

Use the same approach for this additional ECG: rhythm and rate first, then QRS, ST segments and T waves in contiguous leads.

This was supplied for review. The recording does not provide a separate worked answer for this tracing.

Where is the ischemia?



Slide reference

Another localization practice tracing. Describe the findings before naming a likely territory, and compare with the lead map rather than relying on one conspicuous waveform.

The lecturer moved to Mr. Hart's case without a recorded answer-by-answer analysis of this example.

Julie's Cardiology Case Study 1

- Mr. Hart (he/him) is a 58-year-old male who presents to his NP-PHC with complaints of new shortness of breath on exertion
- He exercises regularly walking 45-60 min 3-4 times per week
- He has a PMHx of hypertension which is well controlled on Micardis Plus (80 mg-25 mg); HbA1C in early 2022 was 5.4% and LDL was 3.6mmol/L
- On further history taking, he endorses a 'chest tightness' that is always present with the shortness of breath and he thought it was related to his breathing
- No relevant family history
- The NP-PHC ordered an ECG

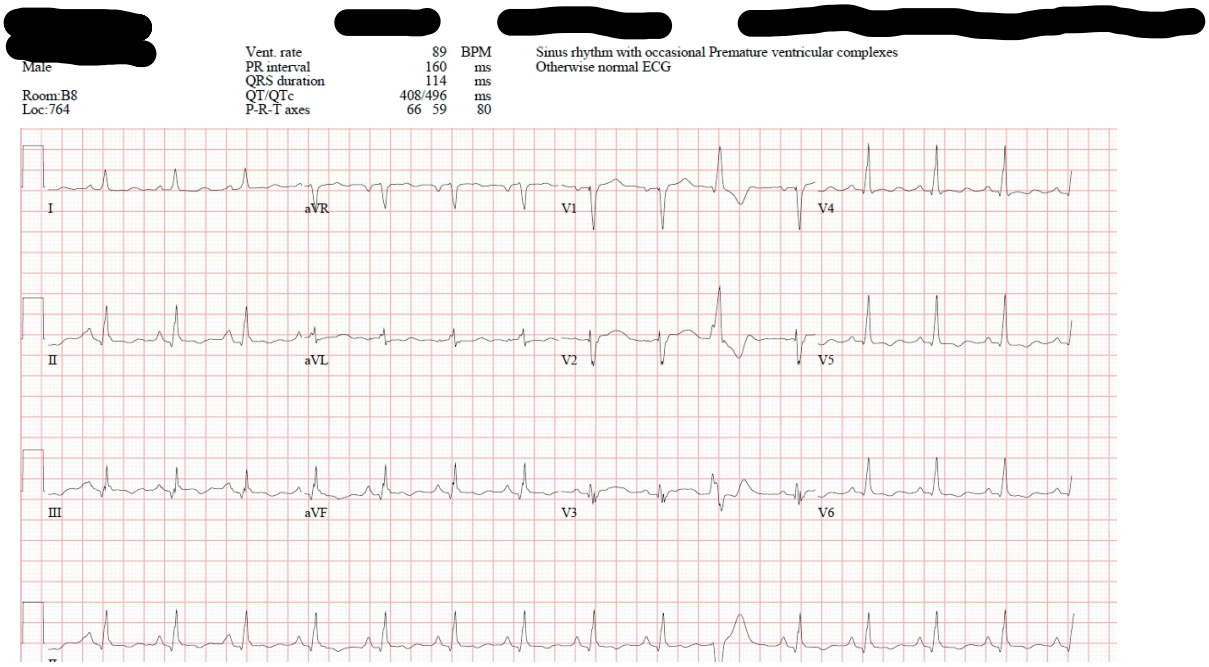
Lecture notes and case discussion

Mr. Hart is 58 and reports new exertional dyspnea despite regular walking. Directed questioning reveals accompanying chest tightness that he had attributed to his breathing.

His hypertension is treated, and the case provides previous HbA1c and LDL results. No relevant family history is reported. The NP orders an ECG.

The teaching point is to unpack dyspnea: the patient may not volunteer exertional chest discomfort unless asked, and a respiratory description may conceal a cardiac symptom pattern.

Julie’s Cardiology Case Study 1



Lecture notes

The ECG is reported as sinus rhythm with occasional PVCs. The lecturer checked the lower rhythm strip and agreed with that description, then reviewed the remaining leads.

Do not stop after reading the machine label. Confirm the rhythm yourself and assess whether the tracing explains the symptoms.

The subsequent case demonstrates that a relatively unremarkable resting ECG does not settle the cause of progressive exertional limitation.

Julie’s Cardiology Case Study 1

- Mr. Hart returns for a follow-up appointment after his ECG
- He has cut back on his walking; he has slowed down his pace, now walking only 30 min; his shortness of breath is mild when walking; he can still perform his ADLs with no symptoms
- Based on the ECG, what next steps would the NP-PHC take?
  - Echocardiogram
  - PFTs
  - CXR

Lecture notes and discussion

At follow-up, Mr. Hart has slowed his pace and reduced walks from 45-60 minutes to about 30 minutes. He remains comfortable with ordinary activities at home. The change from his own baseline is clinically important.

Students suggested CBC and electrolytes; the lecturer also discussed BNP or NT-proBNP for dyspnea, CXR and echo. PFTs were not considered unreasonable if respiratory disease remained in the differential.

Select tests according to what the history and examination suggest. A test that identifies a different cause may help, but progressive exertional dyspnea with tightness still requires cardiac consideration.

Julie’s Cardiology Case Study 1
Echo Report

Adult Echocardiography Report

Name: [redacted]	**Study Date:** [redacted]	**BP:** 95/64 mmHg
MRN: [redacted]	**Patient Location:** [redacted]	**HR:** 87
DOB: [redacted]	**Gender:** Male	**Height:** 165 cm
Age: 58 yrs	**Study Location:** [redacted]	**Weight:** 68 kg
Image Quality Index: Fair		
Reason For Study: NSTEMI .Chest pain		**BSA**: 1.7 m^2
		BMI: 25.0 kilograms/m^2
Contrast Used: Definity Contrast; uneventful		

Interpretation Summary

An image enhancing agent (Definity ultrasonic contrast) was used in the acquisition of images for this study. Administration of the agent was uneventful. 1. Normal left ventricular size. LVEF 35-40%. Akinesis of the anteroseptum, inferoseptum, inferior wall and apex. No left ventricular thrombus.
2. Normal right ventricular size and systolic function.
3. Thickened mitral valve leaflets with mild mitral regurgitation.
4. Insufficient tricuspid regurgitation Doppler signal to estimate right ventricular systolic pressure.

No prior study for comparison.

Lecture notes

The echo summary reports normal LV size but reduced LVEF of 35-40%, with akinesis involving the anteroseptum, inferoseptum, inferior wall and apex. No LV thrombus is reported; RV size and systolic function are normal.

The lecturer emphasized that chamber size and pumping function are different findings: normal size does not mean normal LV function.

Read beyond the first sentence. Connect the regional movement abnormalities and reduced EF with the patient's exertional symptoms, then review the valves and the rest of the report.

Julie's Cardiology Case Study 1

Let's review the details of this report

- What findings are most concerning to the NP-PHC on this report?
- What is the most likely cause of this reduced LV function? Why?
- What would be the appropriate next steps for the NP-PHC in this case?
- What investigation could the NP-PHC expect would be next for this patient?
- Any guesses what that investigation would demonstrate?
- Hint… what walls are not moving well?
- What supplies those walls?

Left Ventricle
The BSA indexed LV end diastolic volume is 81.0 ml./sq.m.. Normal left ventricular volumes when compared to reference ranges for contrast enhanced images. There is normal left ventricular wall thickness. Left ventricular systolic function is moderately reduced. Ejection Fraction = 35-40%. The left ventricular ejection fraction was calculated using contrast enhanced biplane method of disks. There is mild diastolic dysfunction (normal left atrial pressure - Grade 1). There are regional wall motion abnormalities. There is apical akinesis. There is anteroseptal wall akinesis. There is inferior wall hypokinesis. There is inferoseptal wall hypokinesis. There is no thrombus.

Right Ventricle
The right ventricle is normal size. The right ventricular systolic function is normal.

Atria
The left atrial size is normal. Right atrial size is normal. No interatrial shunt or patent foramen ovale is visualized by color Doppler.

Mitral Valve
The mitral valve leaflets appear thickened, but open well. There is mild mitral regurgitation.

Tricuspid Valve
The tricuspid valve leaflets are thin and pliable. There is trace tricuspid regurgitation. The right atrial pressure is estimated at 3mmHg. Insufficient tricuspid regurgitation Doppler signal to estimate right ventricular systolic pressure.

Aortic Valve
The aortic valve opens well. The aortic valve is tricuspid. No hemodynamically significant valvular aortic stenosis. No aortic regurgitation is present.

Pulmonic Valve
The pulmonic valve cusps are thin and pliable; valve motion is normal. There is trace pulmonic valvular regurgitation.

Great Vessels
The aortic root is normal size. The proximal ascending aorta is normal size.

Pericardium/Pleural
There is no pericardial effusion.

Cardiac Rhythm
The baseline ECG displays normal sinus rhythm.

Lecture notes and case reasoning

The most concerning findings are reduced EF and regional wall-motion abnormalities. Akinesis means absent motion; hypokinesis means reduced motion. The lecturer linked the regional pattern to possible impaired coronary blood supply and suspected ischemic heart disease.

These findings can explain dyspnea, reduced exercise tolerance and exertional discomfort even without a dramatic acute presentation. They require timely cardiology assessment rather than attributing symptoms to occasional PVCs.

The expected next specialist investigation discussed was coronary angiography to define anatomy and obstruction. Review lipids, diabetes status and other vascular risks so that primary care can address its part of risk reduction.

The case supports suspected coronary disease; echo alone does not establish the precise coronary lesion.

Julie’s Cardiology Case Study 2

- Mr. Aorta (he/him) is a 51-year-old male who had an inferior STEMI while visiting his daughter out of province
- He presented to hospital within 60 minutes of the onset of his chest pain and underwent successful rescue PCI to the RCA
- His hospitalization was uneventful, and he was started on goal directedmedical therapy as per guidelines post STEMI
- The NP-PHC received a discharge summary with documentation of the event, an echocardiogram report showing an EF of 45% with inferior wallmotion abnormalities and a post STEMI ECG that was NSR with no ST or T wave changes
- The discharge summary requests primary care follow-up with a repeat echocardiogram to reassess LV function

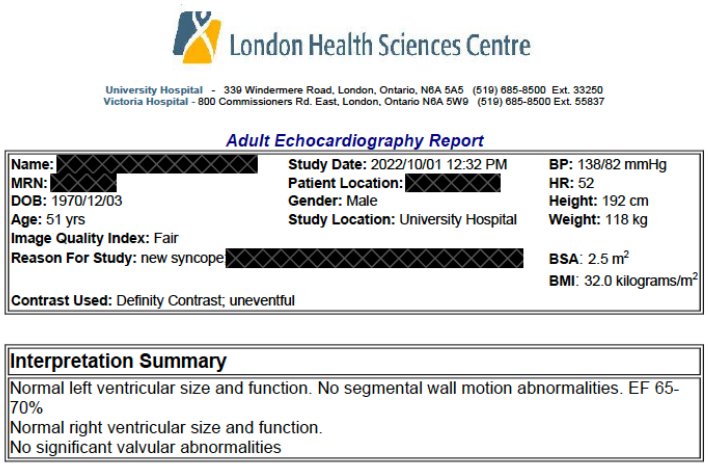
Slide reference

This case was left for independent review. Mr. Aorta had an inferior STEMI treated with RCA PCI; the discharge echo showed EF 45% and inferior regional wall-motion abnormalities. Follow-up reassessment of LV function was requested.

Use the case to distinguish the acute discharge findings from later recovery and to identify the follow-up responsibilities communicated to primary care.

Julie’s Cardiology Case Study 2

- What time frame post STEMI would be appropriate to reassess the LV function?
  - Why?



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University Hospital - 339 Windermere Road, London, Ontario, N6A 5A5 (519) 685-8500 Ext. 33250
Victoria Hospital - 800 Commissioners Rd. East, London, Ontario N6A 5W9 (519) 685-8500 Ext. 55837

Adult Echocardiography Report

Name: [redacted]	**Study Date:** 2022/10/01 12:32 PM	**BP:** 138/82 mmHg
MRN: [redacted]	**Patient Location:** [redacted]	**HR:** 52
DOB: 1970/12/03	**Gender:** Male	**Height:** 192 cm
Age: 51 yrs	**Study Location:** University Hospital	**Weight:** 118 kg
Image Quality Index: Fair		
Reason For Study: new syncope [redacted]		**BSA**: 2.5 m^2
		BMI: 32.0 kilograms/m^2
Contrast Used: Definity Contrast; uneventful		

Interpretation Summary
Normal left ventricular size and function. No segmental wall motion abnormalities. EF 65-70% Normal right ventricular size and function. No significant valvular abnormalities

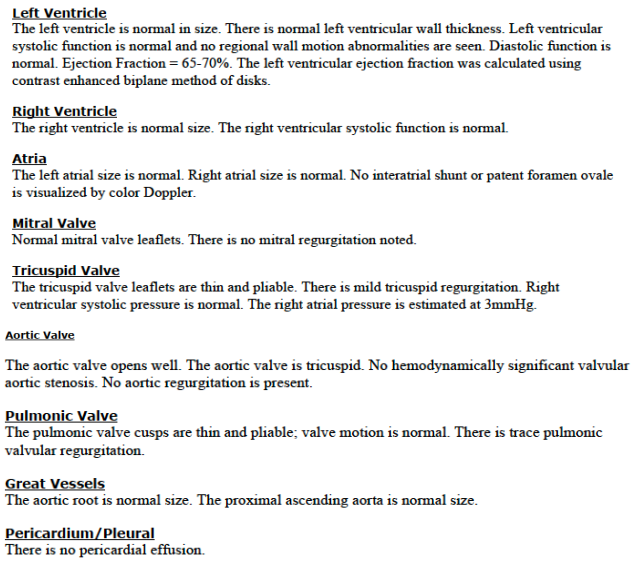
Slide reference

Self-study question: when should LV function be reassessed after STEMI, and why? The supplied recording does not contain the lecturer's specific timing recommendation or a worked answer.

Retain this as a question to resolve against the course material and the patient's discharge or cardiology plan rather than inserting an unspoken interval into the lecture notes.

Julie's Cardiology Case Study 2

- How does the NP-PHC interpret this echocardiogram report?
- What other referral would be appropriate at this time?
- What other diagnostic testing will be appropriate for Mr. Aorta? Now or within the next 12 months?



Left Ventricle
The left ventricle is normal in size. There is normal left ventricular wall thickness. Left ventricular systolic function is normal and no regional wall motion abnormalities are seen. Diastolic function is normal. Ejection Fraction = 65-70%. The left ventricular ejection fraction was calculated using contrast enhanced biplane method of disks.

Right Ventricle
The right ventricle is normal size. The right ventricular systolic function is normal.

Atria
The left atrial size is normal. Right atrial size is normal. No interatrial shunt or patent foramen ovale is visualized by color Doppler.

Mitral Valve
Normal mitral valve leaflets. There is no mitral regurgitation noted.

Tricuspid Valve
The tricuspid valve leaflets are thin and pliable. There is mild tricuspid regurgitation. Right ventricular systolic pressure is normal. The right atrial pressure is estimated at 3mmHg.

Aortic Valve

The aortic valve opens well. The aortic valve is tricuspid. No hemodynamically significant valvular aortic stenosis. No aortic regurgitation is present.

Pulmonic Valve
The pulmonic valve cusps are thin and pliable; valve motion is normal. There is trace pulmonic valvular regurgitation.

Great Vessels
The aortic root is normal size. The proximal ascending aorta is normal size.

Pericardium/Pleural
There is no pericardial effusion.

Slide reference

The displayed follow-up report describes normal LV size and wall thickness, normal systolic function, EF 65-70% and no regional wall-motion abnormality. RV function is normal, and no major valve abnormality is described.

Compare these findings with the initial EF 45% and inferior movement abnormality in slide 154: the supplied example shows recovery of function.

The lecturer did not give recorded answers to the additional referral and future-testing questions on this slide.

Julie’s Cardiology Case Study 3

- Mrs. Ventricle (she/her) is a 91-year-old woman who presents to the NP-PHC with complaints of exertional chest discomfort and shortness of breath
- PMHx: HTN well controlled on Perindopril and Amlodipine, osteoarthritis, Type II DM on Metformin and Sitagliptin (last HbA1C 7.3%), CKD Stage II
- The physical exam is unremarkable except for a new murmur not previously heard on auscultation
- How are murmurs described or classified?

Slide reference

Mrs. Ventricle is 91 with exertional chest discomfort, dyspnea and a newly recognized murmur. The case includes hypertension, diabetes and CKD among her comorbidities.

This case was left for independent review. Describe the murmur carefully and relate it to the symptoms before reviewing the echo findings on subsequent slides.

Julie’s Cardiology Case Study 3

- character of a murmur is described by intensity (grade), pitch (frequency), configuration, timing, quality, location, and radiation
- intensity - determined by the quantity and velocity of blood flow at the site of its origin, the transmission characteristic of the tissues between the blood flow and stethoscope, the site of auscultation or recording, and the distance of transmission
- pitch - frequency of the murmur determines the pitch, which may be high or low; can be described as harsh, rumbling, scratchy, grunting, blowing, squeaky, vibratory, and musical
- location and radiation - location on the patient's chest where the murmur is loudest is typically described as apical, and parasternal or over the base of the heart
- parasternal murmurs are further characterized by the intercostal space and right or left side of the sternum
- assess the area over which the murmur is audible (radiation)
- classification by timing - duration of a murmur is assessed by determining the length of systole or diastole that the murmur occupies; murmur can be long (i.e. it occupies most of systole or diastole), or it can be short

Slide reference

Use a consistent murmur description: timing, site of maximal intensity, radiation, grade, pitch, quality and configuration.

Distinguish how loud the murmur is from its other characteristics. A clear description allows another clinician to understand what was heard and why a particular valve lesion is being considered.

The lecturer did not separately teach each descriptor during this recording.

Grading of the intensity of heart murmurs

Grade 1	Faintest sound that can be detected; often detected by cardiologists but not by general clinicians
Grade 2	Soft murmur that is readily detectable
Grade 3	Louder than grade 2 but not associated with a palpable thrill
Grade 4	Easily detected murmur associated with a palpable thrill*
Grade 5	Very loud murmur audible with the stethoscope placed lightly on the chest
Grade 6	Extremely loud murmur audible with the stethoscope off of the chest

* A grade 4 murmur requires the presence of a precordial thrill. A thrill located only in the suprasternal notch or neck and not on the precordium does not fulfill this criterion.

References:

1. *Marino BS, Goldblatt A. Heart sounds and murmurs. In: Lilly LS, ed. Pathophysiology of heart disease. Philadelphia: Lea & Febiger; 1992. p.18.*
2. *Freeman AR, Levine SA. The clinical significance of the systolic murmur. A study of 1,000 consecutive "non-cardiac" cases. Ann Intern Med 6:1371, 1933.*

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Slide reference

The table reviews grades 1-6. Grades 1-3 describe increasing audibility without a thrill; a palpable thrill is part of the higher-grade descriptions.

Use the original table for the precise grade definitions. This grading exercise was supplied for review, without an additional spoken walkthrough.

Julie’s Cardiology Case Study 3

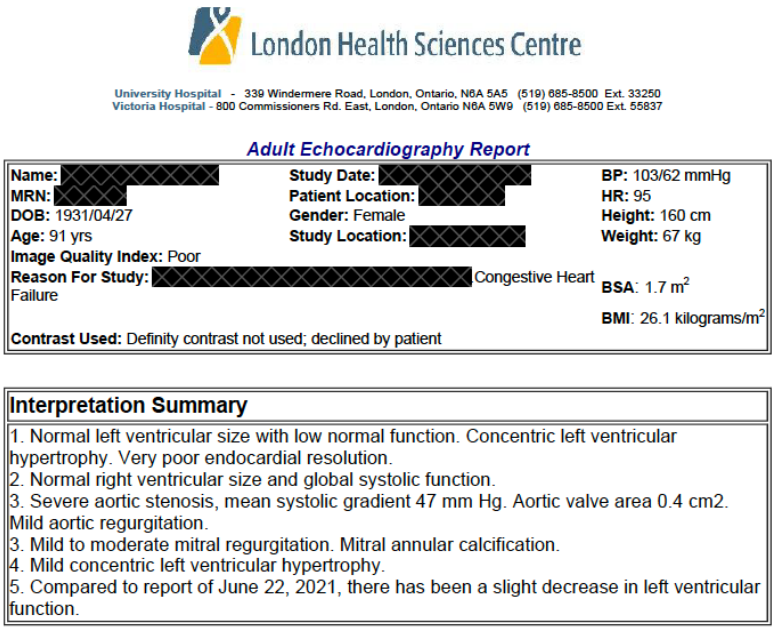
- Mrs. Ventricle’s murmur is characterized as the following:
  - Systolic ejection murmur, crescendo decrescendo
  - Grade 3
  - Radiates across the precordium and to the carotid arteries
- What differential diagnosis would the NP-PHC make at this point?
  - Aortic stenosis, aortic sclerosis, heart murmur NYD
- What would be appropriate next step for Mrs. Ventricle?
  - Echocardiogram
  - Referral to Cardiology or depending on severity Cardiac Surgery

Slide reference

The murmur is described as grade 3, systolic ejection, crescendo-decrescendo, radiating across the precordium and to the carotids.

The slide's differential includes aortic stenosis and aortic sclerosis, with echo as the next investigation and referral based on severity. This is the course's supplied case progression, not an independently discussed live answer.

Julie’s Cardiology Case Study 3



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Adult Echocardiography Report

Name: [redacted]	**Study Date:** [redacted]	**BP:** 103/62 mmHg
MRN: [redacted]	**Patient Location:** [redacted]	**HR:** 95
DOB: 1931/04/27	**Gender:** Female	**Height:** 160 cm
Age: 91 yrs	**Study Location:** [redacted]	**Weight:** 67 kg
Image Quality Index: Poor		
Reason For Study: [redacted] Congestive Heart Failure		**BSA**: 1.7 m^2
		BMI: 26.1 kilograms/m^2
Contrast Used: Definity contrast not used; declined by patient		

Interpretation Summary

1. Normal left ventricular size with low normal function. Concentric left ventricular hypertrophy. Very poor endocardial resolution.
2. Normal right ventricular size and global systolic function.
3. Severe aortic stenosis, mean systolic gradient 47 mm Hg. Aortic valve area 0.4 cm2. Mild aortic regurgitation.
3. Mild to moderate mitral regurgitation. Mitral annular calcification.
4. Mild concentric left ventricular hypertrophy.
5. Compared to report of June 22, 2021, there has been a slight decrease in left ventricular function.

Slide reference

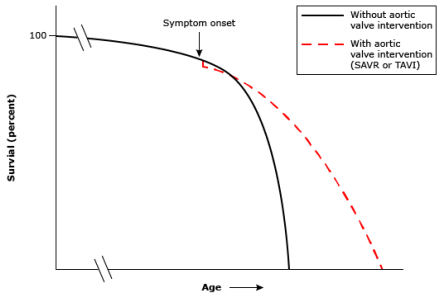
The echo summary identifies severe aortic stenosis, mean systolic gradient 47 mmHg and valve area 0.4 cm2. LV size is normal, with low-normal function and concentric hypertrophy; mild aortic regurgitation and mild-to-moderate mitral regurgitation are also described.

Use these actual report findings to connect the murmur and exertional symptoms with a structural diagnosis. The original report is reproduced above for review of image limitations and other details.

Julie’s Cardiology Case Study 3

- What symptoms are classic of aortic stenosis?
- What is the natural course of severe aortic stenosis?
- Prevalence of AS increases with age; prospective population-based studies prevalence of AS varied from 0.2% 50-59 years, 1.3% at ages 60-69, 3.9% 70-79 years, and 9.8% 80 to 89 years

Survival in patients who develop symptomatic severe aortic stenosis



This graph demonstrates the survival of patients with untreated and treated calcific AS. Prior to the onset of symptoms, patients with calcific aortic valve disease are at risk of cardiovascular morbidity and mortality given associated comorbidities (eg, risk of coronary artery disease). Hence, mortality for patients with asymptomatic calcific AS is higher than for the general population. For patients with severe AS, the outlook dramatically worsens when symptoms develop (eg, dyspnea of cardiac origin, syncope, or angina pectoris) and untreated patients exhibit mortality rates that approach 50% in 2 years. In patients with symptomatic severe AS, SAVR or TAVI are associated with short-term procedural risk but provide improved longer-term survival.

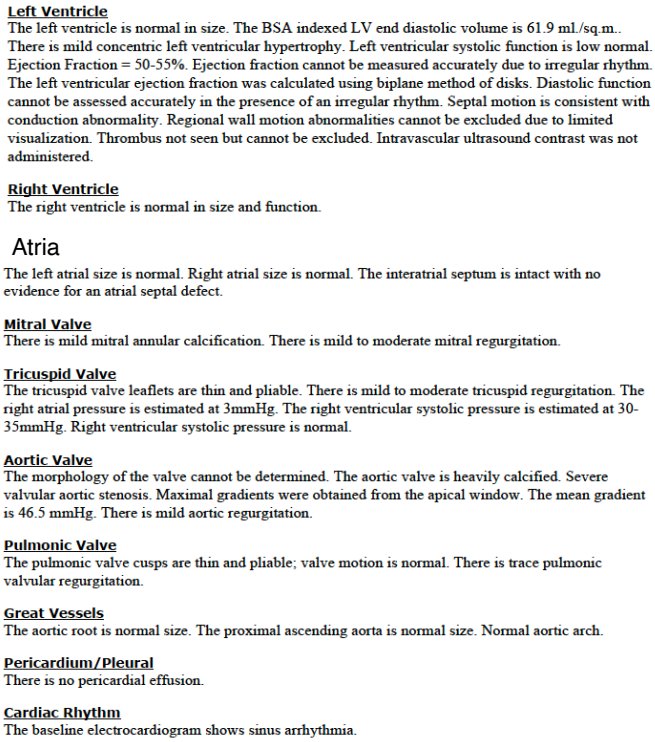
SAVR: surgical aortic valve replacement; TAVI: transcatheter aortic valve implantation; AS: aortic stenosis.

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Slide reference

The slide asks about classic aortic-stenosis symptoms and the natural history of severe disease and gives age-related prevalence figures.

These were self-study questions in the supplied recording. Review them with the symptomatic case and the severe echo findings rather than assuming the lecturer provided a recorded answer.



Left Ventricle
The left ventricle is normal in size. The BSA indexed LV end diastolic volume is 61.9 ml./sq.m.. There is mild concentric left ventricular hypertrophy. Left ventricular systolic function is low normal. Ejection Fraction = 50-55%. Ejection fraction cannot be measured accurately due to irregular rhythm. The left ventricular ejection fraction was calculated using biplane method of disks. Diastolic function cannot be assessed accurately in the presence of an irregular rhythm. Septal motion is consistent with conduction abnormality. Regional wall motion abnormalities cannot be excluded due to limited visualization. Thrombus not seen but cannot be excluded. Intravascular ultrasound contrast was not administered.

Right Ventricle
The right ventricle is normal in size and function.

Atria

The left atrial size is normal. Right atrial size is normal. The interatrial septum is intact with no evidence for an atrial septal defect.

Mitral Valve
There is mild mitral annular calcification. There is mild to moderate mitral regurgitation.

Tricuspid Valve
The tricuspid valve leaflets are thin and pliable. There is mild to moderate tricuspid regurgitation. The right atrial pressure is estimated at 3mmHg. The right ventricular systolic pressure is estimated at 30-35mmHg. Right ventricular systolic pressure is normal.

Aortic Valve
The morphology of the valve cannot be determined. The aortic valve is heavily calcified. Severe valvular aortic stenosis. Maximal gradients were obtained from the apical window. The mean gradient is 46.5 mmHg. There is mild aortic regurgitation.

Pulmonic Valve
The pulmonic valve cusps are thin and pliable; valve motion is normal. There is trace pulmonic valvular regurgitation.

Great Vessels
The aortic root is normal size. The proximal ascending aorta is normal size. Normal aortic arch.

Pericardium/Pleural
There is no pericardial effusion.

Cardiac Rhythm
The baseline electrocardiogram shows sinus arrhythmia.

Slide reference

This is the detailed report accompanying the aortic-stenosis case. It describes a heavily calcified valve, severe stenosis, an estimated EF around 50-55%, concentric LV hypertrophy and limited image quality.

Read the full report alongside the summary on slide 161, including the ventricular findings, valves and limitations. No separate oral interpretation was recorded.

Commonly Ordered Tests in Cardiology Transthoracic Echocardiography (TTE)

- ultrasound beams are directed across the chest wall to obtain images of the heart
- evaluation of chamber size, LVEF, wall motion abnormalities, myocardial ischemia and complications of MI, evaluation of wall thickness, valve morphology, proximal great vessel morphology, pericardial effusion
- evaluation of unexplained hypotension, murmurs, syncope, and congenital heart disease or other cardiac symptoms and unexplained signs on physical exam (i.e. leg edema)
- bubble study improves visualization and provides real-time assessment of intracardiac blood flow using conventional agent is agitated saline
- visualization of right heart and intracardiac shunts, most commonly patent foramen ovale (PFO) and intrapulmonary shunt
- contrast (no renal effects) study helps visualize blood flow and apex of the heart

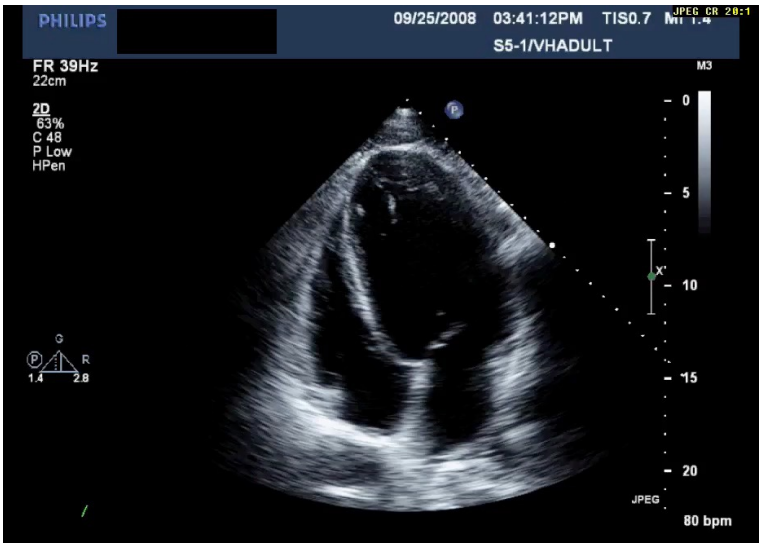
Slide reference

The remaining investigation slides were provided as a reference: what a test shows, when it is useful, and how it helps answer a diagnostic question.

TTE assesses chamber size, ventricular function, regional wall motion, valves and pericardial findings. The slide distinguishes agitated-saline studies for shunt assessment from image-enhancing contrast used to improve endocardial visualization.

The lecturer did not separately expand every listed indication in this closing section.

Commonly Ordered Tests in Cardiology
Transthoracic Echocardiography (TTE)



Slide reference

The image illustrates a transthoracic echo view. Use it with the previous slide to distinguish an imaging study of cardiac structure and movement from an ECG recording of electrical activity.

No individual interpretation of this echo image was given in the audio.

Commonly Ordered Tests in Cardiology Transesophageal Echocardiography (TEE)

- ultrasound probe inserted into the esophagus to allow for better resolution of the heart and structures
- better visualization of posterior structures, including left atrium, mitral, and aortic valves, interatrial septum
- indications
  - TTE is used to evaluate valvular morphology and function (particularly the aortic and mitral valves)
  - intracardiac thrombi, tumors, valvular vegetations (infective endocarditis), aortic dissection, aortic atheromas, prosthetic valve function, shunt, technically inadequate transthoracic study
  - valuation for left atrial thrombus/left atrial appendage thrombus in a patient with atrial fibrillation/atrial flutter to facilitate clinical decision making regarding electrical cardioversion or ablation
- risks
  - serious complications are extremely rare (<1 in 5000)
  - esophageal perforation
  - gastrointestinal bleeding
  - pharyngeal hematoma

Slide reference

TEE places an ultrasound probe in the esophagus to obtain closer views of selected structures. The slide lists valve assessment, intracardiac thrombus, endocarditis and other structural questions among its uses.

An AF-related use is assessment for left atrial or appendage thrombus when planning cardioversion. This is a different test from a routine chest-wall TTE and has procedural risks.

The bullet beginning 'TTE is used' appears within the TEE section; read the test name in context rather than treating it as a separate lecture instruction.

Commonly Ordered Tests in Cardiology
Exercise Stress Testing

- treadmill or bicycle exercise with electrocardiographic and blood pressure monitoring
- indications
  - patients with intermediate (10-90%) pretest probability of CAD based on age, gender, and symptoms
  - ST depression <1 mm at rest, no left bundle branch block, no digoxin
- exercise test results stratify patients into 3 risk groups
  - 1. low risk patients can be treated medically without invasive testing
  - 2. intermediate risk patients may need additional testing in the form of exercise imaging studies or cardiac catheterization
  - 3. high risk patients should be referred for cardiac catheterization

Slide reference

Exercise testing combines exertion with ECG and blood-pressure monitoring. The slide describes using the result to inform risk assessment and whether additional imaging or invasive testing is needed.

Suitability depends on the clinical question, the patient's ability to exercise and whether the baseline ECG permits useful interpretation. No detailed live selection algorithm was taught in the closing comments.

Commonly Ordered Tests in Cardiology
Nuclear Stress Testing - MIBI

- myocardial perfusion imaging (MPI) with ECG-gated single photon emission computed tomography (SPECT), using radiolabelled tracer
- evaluates myocardial viability, detects ischemia, and assesses perfusion and LV function simultaneously
- predicts the likelihood of further cardiac event rates independent of the patient's history, examination, resting ECG, and stress ECG
- often denoted as MIBI scan with reference to radiolabelled tracer (sestamibi)
- stress with either treadmill or IV vasodilator stress (dipyridamole, adenosine)
- images of the heart obtained during stress and at rest 3-4 h later

Slide reference

Nuclear myocardial perfusion imaging compares tracer distribution under stress and at rest to evaluate perfusion. The deck describes exercise or pharmacological stress with ECG-gated SPECT and the tracer sestamibi, often called a MIBI scan.

Use this as a reference for the kind of information the test provides; the lecturer did not walk through the acquisition process or individual results in the recording.

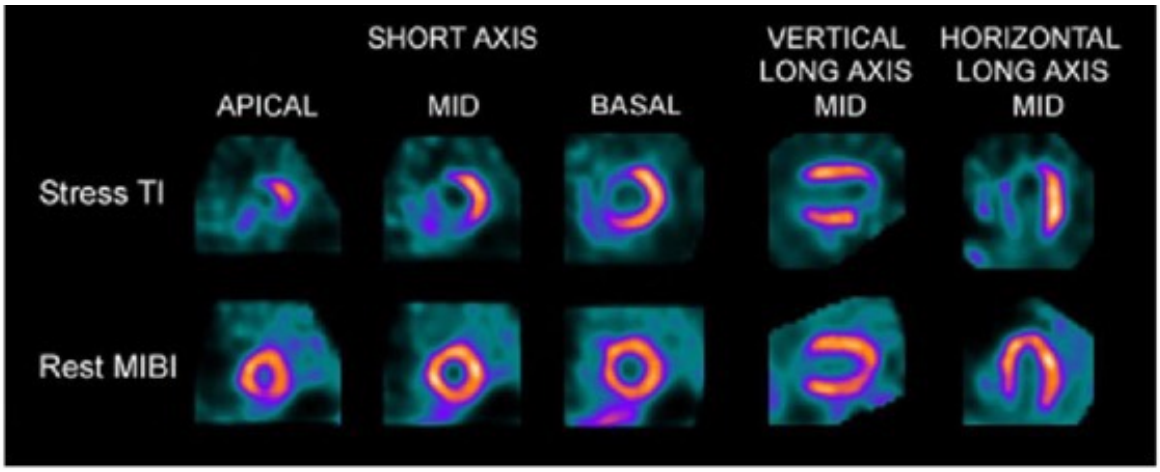
Commonly Ordered Tests in Cardiology Nuclear Stress Testing - MIBI

- to diagnose CAD in possible ACS patients with non-diagnostic ECG and negative serum biomarker
- exercise
  - when ECG cannot be interpreted appropriately due to LBBB or abnormal baseline ECG
  - intermediate pre-test probability with normal/equivocal exercise ECG
  - in patients with previous imaging whose symptoms have changed
  - to diagnose ischemia
- dipyridamole/adenosine MPI
  - exercise testing is always preferred
  - pharmacological stress imaging test for patients who cannot exercise or do not want to hold cardiac meds

Slide reference

The slide lists circumstances in which stress imaging may add information beyond a standard exercise ECG, including baseline ECG limitations, changed symptoms or inability to exercise adequately.

Match the modality to the diagnostic question and patient factors. The lecturer's local ordering-pathway comment is recorded under slide 136; it should not be generalized across jurisdictions or dates.



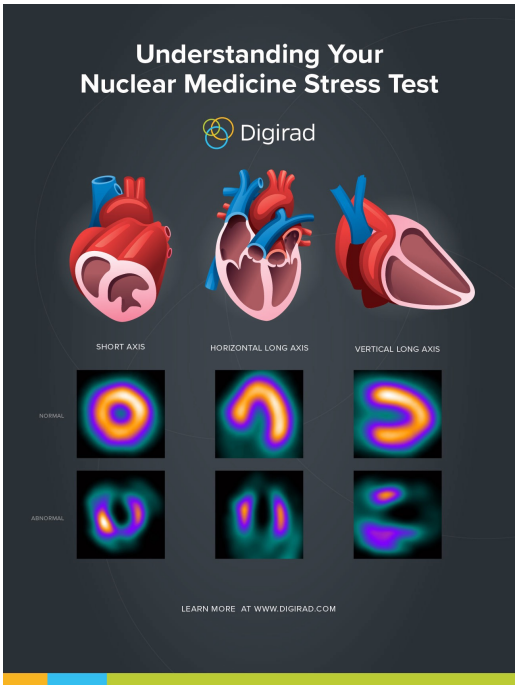
Slide reference

This image places stress and rest perfusion views side by side in several cardiac planes. Use it to understand the comparison underlying the test.

No case-specific interpretation or verbal answer for this image was provided in the supplied audio.

Nuclear Stress Testing – MIBI Red Flags

- Ischemia involving > 10% of the LV mass
- Resting ischemia
- Always be concerned ‘balanced ischemia’… where all 3 main vessel are diseased so no obvious ischemia appears
- If history and symptoms fit the bill… then move on to next level of investigation



Slide reference

The slide highlights substantial ischemic burden and warns about balanced ischemia: relative perfusion imaging can appear less striking when multiple territories are similarly affected.

Its main teaching point is to revisit the clinical story when symptoms and risk remain concerning despite a reassuring-looking investigation. This was supplied as reference content, without further live explanation.

Commonly Ordered Tests in Cardiology
Left Heart Catheterization- Coronary Angiography

- accomplished by introducing a catheter into the brachial or femoral artery and advancing it through the aorta, across the aortic valve, and into the left ventricle
- evaluates mitral and aortic valvular defects and myocardial disease
- systolic and end-diastolic pressure tracings recorded
- LV size, wall motion and ejection fraction can be assessed by injecting contrast into the LV (left ventriculography) via femoral/radial artery catheterization
- cardiac output and index (measured by the Fick oxygen method or the indicator dilution method)
- indications
  - identification of the extent and severity of CAD and evaluation of left ventricular function
  - assessment of the severity of valvular or myocardial disorders (e.g. aortic stenosis or insufficiency, mitral stenosis or insufficiency, and various cardiomyopathies) to determine the need for surgical correction
  - collection of data to confirm and complement noninvasive studies
  - investigating CAD in patients with confusing clinical features or chest pain of uncertain origin

Slide reference

The deck introduces invasive catheter-based assessment of coronary anatomy, LV function and hemodynamics. Define which question the proposed procedure is intended to answer.

Read this overview alongside the next slide's more specific discussion of coronary angiography. The lecturer's live example focused on coronary flow and stenting rather than teaching every listed measurement or access route.

Commonly Ordered Tests in Cardiology
Left Heart Catheterization- Coronary Angiography

- radiographic visualization of the coronary vessels after injection of radiopaque contrast media
- coronary vasculature accessed via the coronary ostia
- indications
  - to define the coronary anatomy and the degree of luminal obstruction of the coronary arteries
  - to determine the presence and extent of obstructive CAD
  - to assess the feasibility and appropriateness of various forms of therapy, such as revascularization by percutaneous or surgical interventions (PCI)
  - can be used when the diagnosis of CAD is uncertain and cannot be excluded by non-invasive techniques
- contraindications
  - severe renal failure due to contrast agent toxicity (must check patient's renal status due to contrast agent toxicity)
- risks
  - major complications for diagnostic catheterization (death, MI, stroke) <3 in 1000
  - minor complications e.g. vascular assess issue, kidney damage <1/100

Slide reference

Coronary angiography visualizes contrast within coronary vessels to define anatomy and obstruction and help plan revascularization when appropriate.

The slide emphasizes contrast-related renal considerations and procedural risks. Integrate renal status, clinical urgency and the specialist plan rather than using a single listed contraindication without context.

The live angiographic explanation is captured under slide 145.

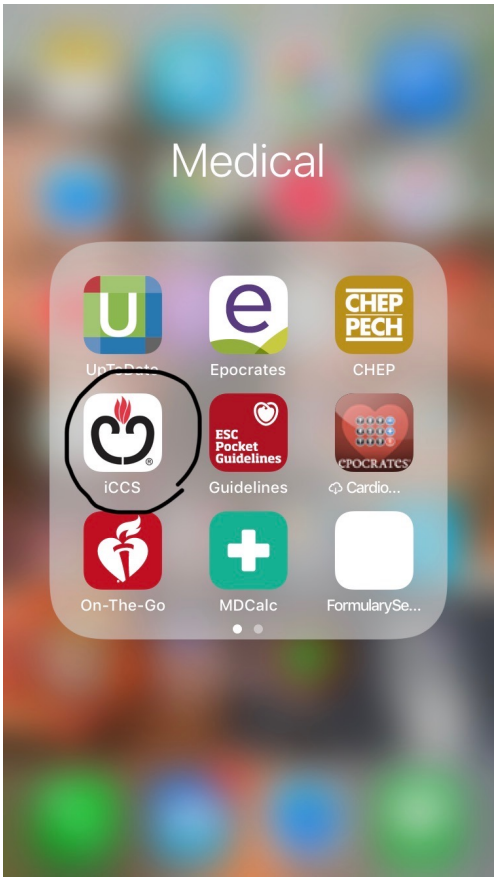
Cardiac Investigations… Clinical Pearls

- Remember…
  - They are not the “silver bullets” of diagnostic cardiology
  - It gives you a “picture” at a moment in time
  - It’s “all in the history”
    - Detailed history and physical can give you many of the answers you need
    - Think about what question you are asking before ordering a specific diagnostic test

Slide reference

The closing reference reinforces a theme repeated throughout the lecture: decide what clinical question a test is meant to answer, and interpret it together with the history and examination.

A test samples particular information at a particular time. A normal brief recording or otherwise reassuring result may not explain intermittent or progressive symptoms.

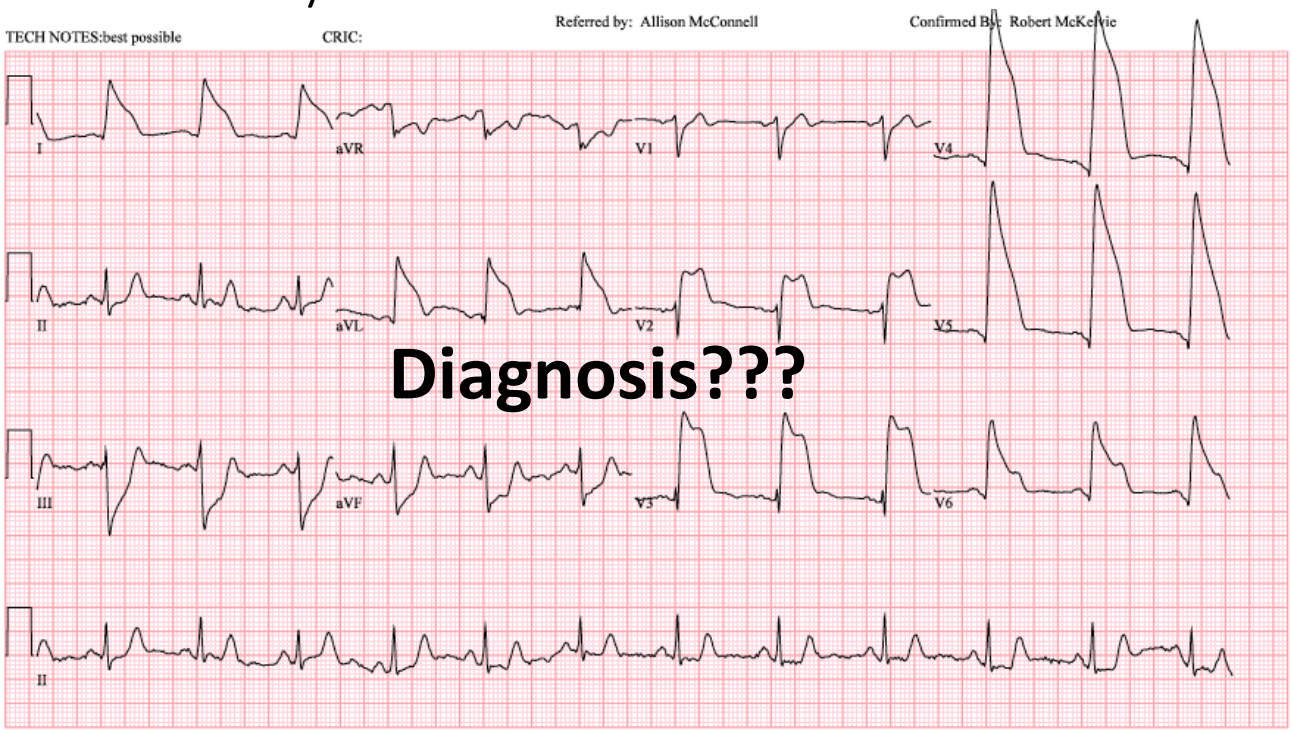


Slide reference

This slide shows clinical reference apps rather than additional case teaching. No separate spoken recommendations about the individual apps were captured.

The substantive AF and syncope references discussed in the lecture appear in the relevant sections above.

Julie's WTW (What The Wow!!!) Case Study

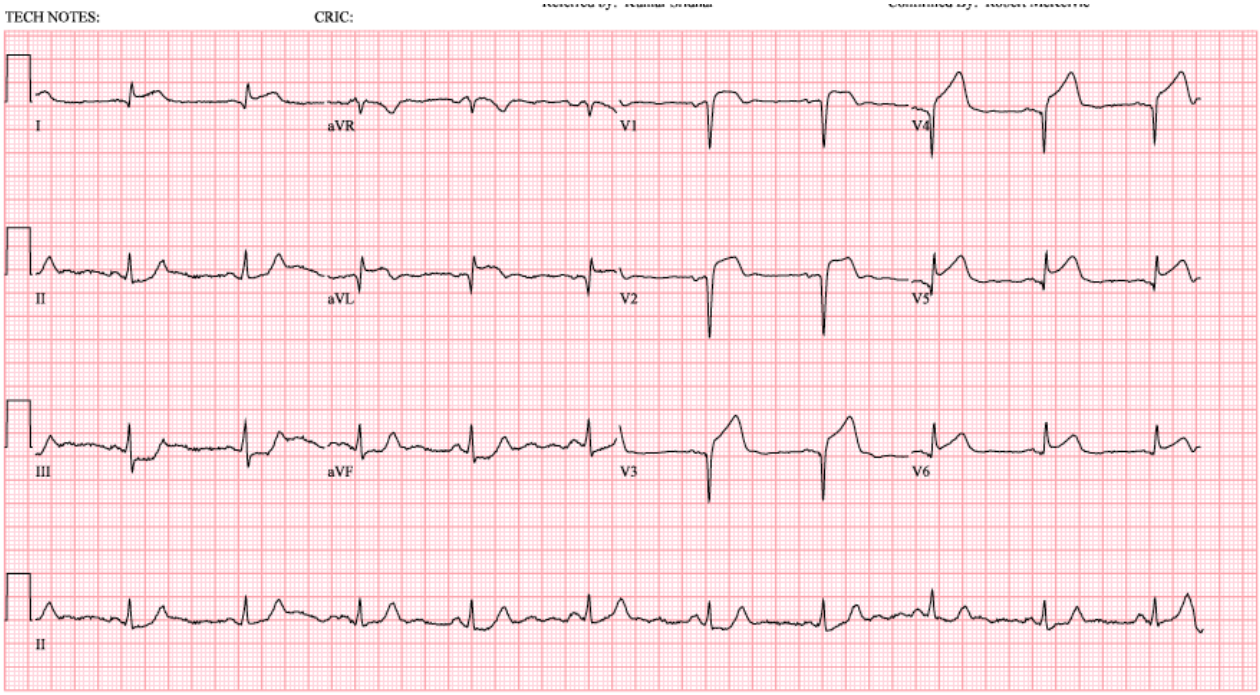


Slide reference

An additional ECG challenge is supplied at the end of the deck. It was not discussed in the recording.

The following slides identify the case as an anterior STEMI treated with LAD PCI. Compare the before-and-after tracings as a self-study exercise; do not interpret this as a separate spoken case presentation.

Code STEMI
Post Percutaneous Coronary Intervention



Slide reference

The deck labels this tracing as post-PCI following a code STEMI. Compare it with the preceding tracing to examine the evolution of the ECG after intervention.

No detailed verbal interpretation or timing of this follow-up tracing is available in the recording.

Anterior STEMI – large infarct
PCI to LAD

- Heart function post MI?
  - Guesses?

Interpretation Summary
Mildly dilated left ventricle with severely reduced systolic function. Ejection fraction = 20-25%. Regional wall motion abnormalities as specified. Normal right ventricular size and systolic function. Mild mitral regurgitation. Mild tricuspid regurgitation. The right ventricular systolic pressure is estimated at 30-35mmHg.

Grade 1). The left ventricle is globally hypokinetic with regional variation. There is apical akinesis. There is anterior wall akinesis. There is anteroseptal wall akinesis. The basal lateral walls are best preserved. Intravascular ultrasound contrast was administered for left ventricular opacification to improve visualization. There is no thrombus.

Slide reference

The final supplied case summary identifies a large anterior infarct treated with LAD PCI. The echo shows severely reduced LV systolic function, EF 20-25%, with regional abnormalities including apical, anterior and anteroseptal akinesis.

This answers the slide's question about post-MI function using the printed report itself. The lecturer did not separately discuss the case in the supplied recording.

References for clarifications

Lecture notes and slide summaries are based on the supplied October 22 recording and student PDF. The following sources support the separately labelled clarifications.

[1] ACC AHA HRS 2018 Bradycardia and Conduction Delay Guideline

Open source

[2] Merck Manual Professional Electrocardiography

Open source

[3] Canadian Cardiovascular Society 2020 AF Guideline Chapter 8 Stroke Prevention

Open source

[4] ESC 2024 Atrial Fibrillation Guideline

Open source

[5] ACC 2022 Expert Consensus Decision Pathway on Acute Chest Pain

Open source

[6] Fourth Universal Definition of Myocardial Infarction 2018

Open source

[7] British Columbia Atrial Fibrillation Diagnosis and Management

Open source